PrepFusion

Electronic Devices and Circuits

Detailed Solutions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electronic Devices and Circuits (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Contents

SOLUTIONS

I

Basic Semiconductor Physics

Topics

1. Physics and Fundamentals
2. Carrier Transport Phenomena
3. Semiconductor Under Non-Equilibrium
4. Fermi Dirac Distribution and Energy Band Diagrams

Q1.26.1 NAT | 2026 | 2M | Ans: -36.50 to -34.50 OR 34.50 to 36.50 | ● ● ○

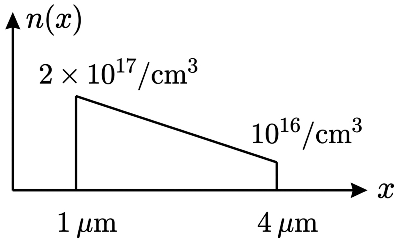

Fig. Solution Q1.26.1

The diffusion constant D_n is calculated using Einstein's relation:

$$D_n = \mu_n V_T = 1400 \times 25 \times 10^{-3} = 35\,\text{cm}^2/\text{s}$$

The electron concentration $n(x)$ varies linearly from $x = 1\,\mu$m to $x = 4\,\mu$m. The concentration gradient is:

$$\frac{dn(x)}{dx} = \frac{n(4\,\mu\text{m}) - n(1\,\mu\text{m})}{4\,\mu\text{m} - 1\,\mu\text{m}}$$

$$\frac{dn(x)}{dx} = \frac{10^{16} - 2 \times 10^{17}}{3 \times 10^{-4}\,\text{cm}} = \frac{-1.9 \times 10^{17}}{3 \times 10^{-4}} \approx -6.33 \times 10^{20}\,\text{cm}^{-4}$$

Now, it is observed that electron concentration decreases along positive-X axis, which implies the diffusion current flows along the negative-X axis direction. Here, since the direction of diffusion current has not been specified, both positive and negative values are correct.

The diffusion current density J_{diff} is given by:

$$J_{\text{diff}} = qD_n \frac{dn(x)}{dx}$$

Taking the magnitude:

$$|J_{\text{diff}}| = \left|1.6 \times 10^{-19} \times 35 \times (-6.33 \times 10^{20})\right|$$

$$|J_{\text{diff}}| = 3544.8\,\text{A/cm}^2 \approx 35.45\,\text{A/mm}^2$$

As the direction of current is not specified, the answer is:

$$\boxed{35.45 \text{ OR } -35.45}$$

Mistakes to be avoided

- Ensure consistent units; convert the distance x from μm to cm ($1\,\mu\text{m} = 10^{-4}$ cm) before calculating the gradient.
- Do not confuse the sign convention; electron diffusion current flows opposite to the direction of electron diffusion.

Q1.25.1 MSQ | 2025 | 1M | Ans: A, D | ● ○ ○

Silicon is a group-IV element. To create an n-type semiconductor, we must dope silicon with a group-V element (pentavalent), which donates an extra electron to serve as a free carrier.

Evaluating each dopant option:

- **(A) Arsenic:** Group-V element ⇒ n-type dopant.
- **(B) Boron:** Group-III element ⇒ p-type dopant.

- **(C) Gallium:** Group-III element $\Rightarrow$ p-type dopant.
- **(D) Phosphorous:** Group-V element $\Rightarrow$ n-type dopant.

$$\boxed{(A)\text{ Arsenic , }(D)\text{ Phosphorous}}$$

Mistakes to be avoided

- Do not confuse the groups; remember that "n" stands for negative (extra electrons from Group-V) and "p" stands for positive (extra holes from Group-III).

Q1.25.2 MCQ 2025 2M Ans: B

We are given the following parameters for the semiconductor:

$$n_i = 2.5 \times 10^{16}\,\text{m}^{-3}$$
$$\mu_n = 0.15\,\text{m}^2/\text{Vs}$$
$$\mu_p = 0.05\,\text{m}^2/\text{Vs}$$

The intrinsic conductivity σ is given by the formula:

$$\sigma = qn_i(\mu_n + \mu_p)$$

Substituting the given values:

$$\sigma = 1.6 \times 10^{-19} \times 2.5 \times 10^{16} \times (0.15 + 0.05)$$
$$\sigma = 1.6 \times 10^{-19} \times 2.5 \times 10^{16} \times 0.20$$

Thus, the conductivity is:

$$\sigma = 0.8 \times 10^{-3} = 8 \times 10^{-4}\,\text{S/m}$$

The resistivity ρ is the reciprocal of conductivity:

$$\rho = \frac{1}{\sigma} = \frac{1}{8 \times 10^{-4}}$$
$$\rho = 1250\,\Omega\text{m} = 1.25\,\text{k}\Omega\text{m}$$

$$\boxed{(B)1.25}$$

Q1.25.3 MCQ 2025 2M Ans: B

Given: Electron mobility, $\mu_n = 0.38\,\text{m}^2/\text{Vs}$ at $T = 300\,\text{K}$

The diffusion constant D_n is related to mobility by the Einstein relation:

$$D_n = \mu_n V_T$$

Where the thermal voltage V_T at temperature T is given by:

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 0.025875$$

At $T = 300\,\text{K}$:

$$D_n = 0.38 \times 0.025875 \approx 0.009833\,\text{m}^2/\text{s}$$

Converting the unit to cm^2/s:

$$D_n = 0.009833 \times 10^4\,\text{cm}^2/\text{s} = 98.33\,\text{cm}^2/\text{s}$$

$$\boxed{(B)\ 98}$$

Q1.24.1 MSQ 2024 1M Ans: A, C, D ● ● ○

Based on the equilibrium condition, the net current density must be zero, implying that drift and diffusion currents must cancel each other out ($J_{\text{drift}} + J_{\text{diff}} = 0$).

Evaluating the logic for each statement based on the carrier profile :

- **Statement (a) is true:** Electrons diffuse from regions of high concentration to low concentration (from B to C). Since electrons are negatively charged, the conventional diffusion current flows in the opposite direction, from **C to B**.
- **Statement (c) is true:** If electrons diffuse from B to A, equilibrium requires an opposing drift current directed from **B to A** to maintain a net current of zero.
- **Statement (d) is true:** A drift current flowing from B to C (to oppose diffusion) requires an electric field directed from **B to C**. This field provides the necessary force to push the negative electrons in the direction of the required drift.
- **Statement (b) is false:** The drift current from B to A requires an electric field directed from **B to A**, not A to B.

The system remains in equilibrium as the internal electric field perfectly balances the carrier concentration gradient across the entire profile.

$$\boxed{(A)\text{For } x \text{ between B and C, the electron diffusion current is directed from C to B.}}$$

$$\boxed{(C)\text{ For } x \text{ between B and A, the electron drift current is directed from B to A.}}$$

$$\boxed{(D)\text{For} x \text{between B and C, the electric field is directed from B to C .}}$$

Q1.24.2 MCQ 2024 1M Ans: A ● ● ○

Plot (a) is the only correct representation because it illustrates the three fundamental regimes of carrier concentration in an extrinsic semiconductor as a function of temperature:

- **Intrinsic (High T , Low $1/T$):** Thermal energy is so high that $n \approx n_i$, causing a steep exponential rise in concentration.
- **Saturation (Mid T , Mid $1/T$):** All donor atoms are ionized, but the temperature is too low for significant intrinsic excitation, keeping $n \approx N_d$ (forming a flat plateau).
- **Freeze-out (Low T , High $1/T$):** Thermal energy is insufficient to ionize the dopants, causing electrons to "freeze" back into donor states, leading to a drop in carrier concentration.

$$\boxed{A}$$

Mistakes to be avoided

- Check the axes of the given graph; the horizontal axis is $1/T$, not T.

Q1.24.3 NAT 2024 2M Ans: 0.17 to 0.19 ● ● ○

Given data: Energy gap from conduction band to Fermi level, $E_c - E_f = 0.25\,\text{eV}$ Thermal voltage, $V_T = 20\,\text{mV} = 0.02\,\text{V}$

It is mentioned that 95% of dopants are ionized, which means 5% of the electrons stay back at the donor energy level E_d. The degeneracy factor is 2, meaning E_d has 2 available states.

The probability of finding an electron at the donor level E_d is 0.05. Since there are 2 states, the probability of finding an electron in one specific state at E_d is:

$$P(E_d) = \frac{0.05}{2} = 0.025$$

Using the Maxwell-Boltzmann approximation for the probability function:

$$f(E_d) \approx e^{-(E_d-E_f)/kT} = 0.025$$

Note that $kT = qV_T$, so:

$$\frac{E_d - E_f}{kT} = -\ln(0.025)$$

$$E_d - E_f = -V_T \ln(0.025)$$

$$E_d - E_f = -0.02 \times (-3.6888) \approx 0.0738\,\text{eV}$$

To find the ionization energy $E_c - E_d$:

$$E_c - E_d = (E_c - E_f) - (E_d - E_f)$$

$$E_c - E_d = 0.25 - 0.0738 = 0.1762\,\text{eV}$$

$$\boxed{0.176}$$

Key Points

- The probability of an electron occupying a an energy state is given by $f(E_d) = \frac{1}{1+e^{(E_d-E_f)/kT}}$.
- For $f(E_d) \ll 1$, this simplifies to the Boltzmann approximation used above.

Mistakes to be avoided

- Do not forget to account for the degeneracy factor 2 when calculating the occupation probability per state. Degeneracy factor of an energy level gives the number of energy states present on that level.

Q1.23.1 MCQ 2023 1M Ans: A ● ○ ○

The correct answer is (A) **degenerate n-type**. A semiconductor is said to be degenerately doped when its Fermi Level enters into the Conduction Band (in case of n-type doping) or the Valence Band (in case of p-type doping). This occurs due to doping concentration exceeding the density of states.

$$\boxed{(A)\ \text{degenerate } n\text{-type.}}$$

Q1.23.2 MCQ 2023 1M Ans: A ● ○ ○

The correct answer is **(A)**, as an intrinsic semiconductor at absolute zero has a completely filled valence band and a completely empty conduction band due to the lack of thermal energy.

$$\boxed{(A)}$$

All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.

Q1.23.3 NAT 2023 2M Ans: 2.20 to 2.30 ● ○ ○

Given data: Intrinsic carrier concentration, $n_i = 1 \times 10^{10}\,\text{cm}^{-3}$ Hole concentration is given as:

$$p = 1.5 \times n_i = 1.5 \times 10^{10}\,\text{cm}^{-3}$$

According to the Mass Action Law:

$$np = n_i^2$$

$$n = \frac{n_i^2}{p} = \frac{n_i^2}{1.5n_i} = \frac{n_i}{1.5}$$

$$n = \frac{10^{10}}{1.5}\,\text{cm}^{-3}$$

It is given that the electron and hole drift currents are equal:

$$I_{p,\text{drift}} = I_{n,\text{drift}}$$

The drift current components are defined as $pq\mu_p EA$ and $nq\mu_n EA$ respectively. Equating them:

$$pq\mu_p EA = nq\mu_n EA$$

$$p\mu_p = n\mu_n$$

Substituting the value of p and n:

$$1.5 \times 10^{10}\mu_p = \frac{10^{10}}{1.5} \times \mu_n$$

$$\frac{\mu_n}{\mu_p} = 1.5 \times 1.5$$

$$\frac{\mu_n}{\mu_p} = 2.25$$

$$\boxed{2.25}$$

Key Points

- **Mass Action Law:** Under thermal equilibrium, the product of free electron and hole concentrations is constant, $np = n_i^2$.
- **Drift Current Density:** $J = \sigma E = (nq\mu_n + pq\mu_p)E$.

Mistakes to be avoided

- Ensure you do not confuse the drift current with diffusion current; drift depends on the electric field E.
- Double-check the algebraic manipulation of the ratio to avoid inversion errors (e.g., calculating μ_p/μ_n instead of μ_n/μ_p).

Q1.23.4 NAT 2023 2M Ans: 60.00 to 70.00 ● ● ○

Given data: Energy difference, $E_F - E_V = 0.35\,\text{eV}$ (at $T_2 = 400\,\text{K}$), Thermal voltage at $T_1 = 300\,\text{K}$ is $V_{T1} = 0.026\,\text{V}$ (or $kT_1 = 0.026\,\text{eV}$).

First, we calculate the thermal voltage V_{T2} at $T_2 = 400\,\text{K}$ using the relation $V_T \propto T$:

$$\frac{V_{T1}}{V_{T2}} = \frac{T_1}{T_2} \implies V_{T2} = V_{T1} \times \frac{T_2}{T_1}$$

$$V_{T2} = 0.026 \times \frac{400}{300} \approx 0.03466\,\text{eV}$$

Next, we find the effective density of states in the valence band N_{v2} at 400 K. Given $N_{v1} = 1 \times 10^{19}/\text{cm}^3$ at 300 K and knowing $N_v \propto T^{3/2}$:

$$\frac{N_{v2}}{N_{v1}} = \left(\frac{T_2}{T_1}\right)^{3/2}$$

$$N_{v2} = 1 \times 10^{19} \times \left(\frac{400}{300}\right)^{3/2} \approx 1.5396 \times 10^{19}/\text{cm}^3$$

The hole concentration p at 400 K is calculated using the Fermi-Dirac distribution (Boltzmann approximation):

$$p = N_v e^{-(E_F - E_V)/kT_2}$$

$$p = 1.5396 \times 10^{19} \times e^{-0.35/0.03466}$$

$$p = 1.5396 \times 10^{19} \times e^{-10.098}$$

$$p \approx 63.36 \times 10^{13}$$

$$\boxed{63.36}$$

Key Points

- $V_T \propto T$ and $N_c, N_v \propto T^{3/2}$.
- Hole Concentration: $p = N_v \exp\left(-\frac{(E_F - E_V)}{kT}\right)$.

Mistakes to be avoided

- Do not forget to scale N_v with temperature; using the 300 K value at 400 K is a common error.
- Ensure the exponent is dimensionless by using consistent units for $(E_F - E_V)$ and kT (both in eV).

Q1.22.1

MSQ | 2022 | 2M | Ans: A , C | ● ○ ○

A. In intrinsic semiconductor at equilibrium, $n = p = ni$.
B. Collector region is generally lightly doped than base region in a BJT.
C. Total current is spatially constant in a two terminal electronic device, however individual currents vary spatially under dark and steady-state condition.
D. Beyond 300 K, mobility of electron decreases with increase in temperature.
Hence, statement(s): (A) and (C) are correct.

$$\boxed{(A)\text{ Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium}}$$

$$\boxed{(C)\text{Total current is spatially constant in a two terminal electronic device in dark under steadystate condition.}}$$

Q1.22.2

MCQ | 2022 | 1M | Ans: A | ● ● ○

The continuity equation is given as:

$$\frac{dn}{dt} = \frac{1}{q}\frac{dJ_n(x)}{dx} + G_n(x,t) - R_n(x,t)$$

Since the question mentions steady-state condition, no electric field, and carrier generation only at $x = 0$, we conclude that for $x > 0$:

- $Gn = 0$, because excess carriers are generated only at the illuminated surface.

- $E = 0$, since electric field effects are ignored.
- $\frac{dn}{dt} = 0$ due to steady-state condition.

Thus the continuity equation becomes:

$$0 = \frac{1}{q}\frac{dJ_n(x)}{dx} + G_n(x) - R_n(x)$$

The electron current density $J_n(x)$ is given as:

$$J_n(x) = q\mu_n E + qD_n\frac{dn}{dx}$$

Substituting $J_n(x)$ into the continuity equation, we get:

$$0 = \mu_n n\frac{dE}{dx} + \mu_n E\frac{dn}{dx} + G_n(x) - Rn(x) + Dn\frac{d^2n}{dx^2}$$

Given : Electric field is ignored, therefore $E = 0$ and $\frac{dE}{dx} = 0$.

$$0 = 0 + 0 + 0 - R_n + Dn\frac{d^2n}{dx^2}$$

$$R_n = Dn\frac{d^2n}{dx^2}$$

Recombination relation for excess minority carriers is given as:

$$R_n = \frac{\Delta n}{\tau_n}$$

Equating both equations, we get:

$$\frac{\Delta n}{\tau_n} = Dn\frac{d^2n}{dx^2}$$

Using $L_n = \sqrt{D_n\tau_n}$,

$$\frac{d^2n}{dx^2} = \frac{\Delta n}{L_n^2}$$

Solving this differential equation gives:

$$\Delta n(x) = \Delta n(0)exp\left(-\frac{x}{L_n}\right)$$

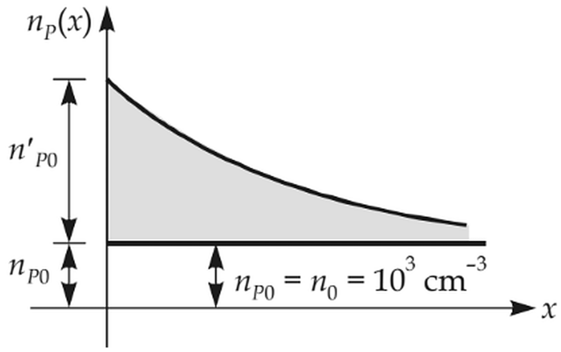

Fig. Solution Q1.22.2

Substituting at $x = 2\mu m$:

$$\Delta n(2\mu m) = 10^{14} exp(-1) = 0.37 \times 10^{14}\ \text{cm}^{-3}$$

$$\boxed{(A)\ 0.37 \times 10^{14}\ \text{cm}^{-3}}$$

Q1.22.3 MCQ 2022 1M Ans: C ● ○ ○

The objective is to find the energy difference $E_c - E_{fn}$. The relationship between the conduction band energy level E_C, Fermi level E_F, and electron concentration n is given by:

$$E_C - E_F = kT \ln\left(\frac{N_c}{n}\right)$$

For two different doping concentrations n_1 and n_2, we have:

$$E_C - E_{F1} = kT \ln\left(\frac{N_c}{n_1}\right) \tag{i}$$

$$E_C - E_{F2} = kT \ln\left(\frac{N_c}{n_2}\right) \tag{ii}$$

Subtracting equation (ii) from equation (i):

$$(E_C - E_{F1}) - (E_C - E_{F2}) = kT \ln\left(\frac{\frac{N_c}{n_1}}{\frac{N_c}{n_2}}\right) = kT \ln\left(\frac{n_2}{n_1}\right)$$

Substituting the given values ($E_C - E_{F1} = 200\,\text{meV}$, $kT = 26\,\text{meV}$, $n_1 = 1 \times 10^{16}$, and $n_2 = 0.5 \times 10^{16}$):

$$200\,\text{meV} - (E_C - E_{F2}) = 26\,\text{meV} \times \ln\left(\frac{0.5 \times 10^{16}}{1 \times 10^{16}}\right)$$

$$200\,\text{meV} - (E_C - E_{F2}) = 26\,\text{meV} \times \ln(0.5) \approx -18\,\text{meV}$$

$$E_C - E_{F2} = 200 + 18 = 218\,\text{meV}$$

$$\boxed{(C)\ 218\ \text{meV}}$$

Key Points

- As the electron concentration n decreases, the Fermi level moves further away from the conduction band edge.

Q1.22.4 NAT 2022 2M Ans: 0.57 to 0.61 ● ● ●

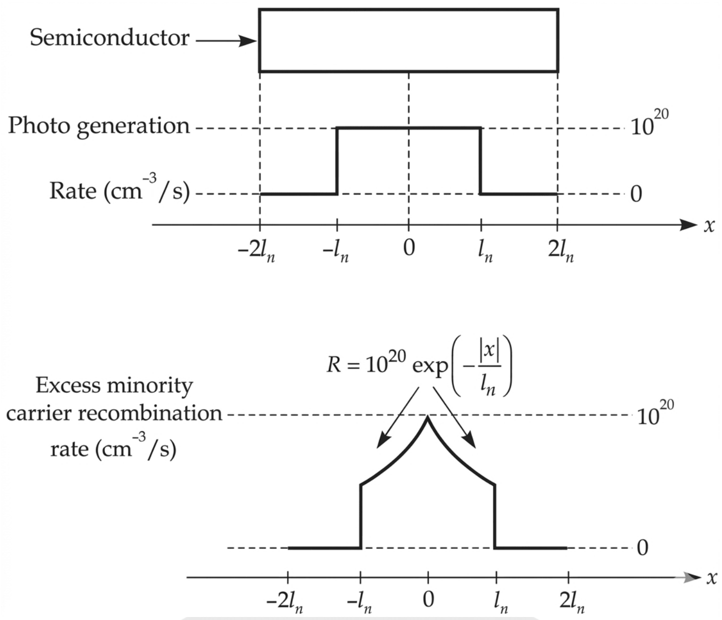

Fig. Solution Q1.22.4

Region 1: $0 < x < l_n$

From the continuity equation at steady state ($\frac{dn}{dt} = 0$):

$$0 = \frac{1}{q}\frac{dJ_n(x)}{dx} + G_n(x) - R_n(x)$$

$$D_n\frac{d^2\Delta n(x)}{dx^2} + G_n(x) - R_n(x) = 0$$

Given $G_n(x) = 10^{20}$ and $R_n(x) = 10^{20}e^{-x/l_n}$, we have:

$$\frac{d^2\Delta n(x)}{dx^2} = \frac{10^{20}}{D_n}\left[e^{-x/l_n} - 1\right]$$

Integrating once:

$$\frac{d\Delta n(x)}{dx} = \frac{10^{20}}{D_n}\left[-l_n e^{-x/l_n} - x + C\right]$$

Using boundary condition at $x = 0$: $\left.\frac{d\Delta n(x)}{dx}\right|_{x=0} = 0$

$$\frac{10^{20}}{D_n}\left[-l_n(1) - 0 + C\right] = 0 \implies C = l_n$$

$$\therefore \frac{d\Delta n(x)}{dx} = \frac{10^{20}}{D_n}\left[l_n(1 - e^{-x/l_n}) - x\right], \text{ for } 0 < x < l_n$$

Region 2: $l_n < x < 2l_n$

Here $R = G = 0$, so $\Delta n(x)$ is linear, and current density J_n is constant. Matching slopes at $x = l_n$:

$$\text{Slope} = \left.\frac{d\Delta n(x)}{dx}\right|_{x=l_n} = \frac{10^{20}}{D_n}\left[l_n(1 - e^{-1}) - l_n\right] = \frac{-10^{20}l_n e^{-1}}{D_n}$$

Current density at $x = \pm 2l_n$:

$$J_n(x = \pm 2l_n) = qD_n \left.\frac{d\Delta n(x)}{dx}\right|_{x=2l_n} = -q10^{20} l_n e^{-1}$$

Using given values $l_n = 10^{-4}$ cm:

$$|J_n| = 1.6 \times 10^{-19} \times 10^{20} \times 10^{-4} \times e^{-1} \text{ A/cm}^2$$

$$|J_n| = 1.6 \times 10^{-3} \times 0.3678 \text{ A/cm}^2 = 0.5886 \text{ mA/cm}^2$$

$$\boxed{0.59}$$

Key Points

- Continuity Equation : $D_n \frac{d^2\Delta n(x)}{dx^2} + G_n(x) - R_n(x) = \frac{dn}{dt}$.

Mistakes to be avoided

- Do not use the Recombination formula $R = \frac{\Delta n(x)}{\tau_n}$ since τ_n is not given.
- Don't forget to match the boundary conditions (slope continuity) at $x = l_n$.

Q1.21.1 MCQ 2021 1M Ans: C

Based on the given semiconductor parameters:
P-type dopant concentration $N_A = 10^{16} cm^{-3}$
Generation Rate $G = 10^{20} cm^{-3} s^{-1}$
Recombination lifetime $\tau = 100\mu s$
Intrinsic Concentration $n_i = 10^{10} cm^{-3}$

The steady-state excess carrier concentration (δ) is calculated by the product of the generation rate and the carrier lifetime:

$$\delta p = \delta n = \delta$$

$$\delta = G\tau = 10^{20} \times 100 \times 10^{-6} = 10^{16} \text{ cm}^{-3}$$

The equilibrium hole concentration for the p-type semiconductor is $p_{p0} \approx N_A = 10^{16}$ cm^{-3}. The total hole concentration (p_p) and electron concentration (n_p) under illumination are:

$$p_p = p_{p0} + \delta = 10^{16} + 10^{16} = 2 \times 10^{16} \text{ cm}^{-3}$$

$$n_p = n_{p0} + \delta \approx \delta = 10^{16} \text{ cm}^{-3}$$

(Since $n_{p0} = \frac{n_i^2}{p_{p0}} = \frac{10^{20}}{10^{16}} = 10^4$ cm^{-3}, which is much smaller than δ).

The product of the carrier concentrations is:

$$p_p n_p = (2 \times 10^{16}) \times (10^{16}) = 2 \times 10^{32} \text{ cm}^{-6}$$

$$\boxed{(C)\ 2 \times 10^{32} \text{ cm}^{-6}}$$

Key Points

- Steady-state excess carrier concentration is given by $\Delta n = G_L \tau_n$.
- Under non-equilibrium conditions (like illumination), the pn product exceeds n_i^2.

Mistakes to be avoided

- Do not assume $pn = n_i^2$ under illumination; this only holds true at thermal equilibrium.

Q1.21.2 MCQ 2021 1M Ans: A ● ○ ○

The built-in electric field in a semiconductor under equilibrium is induced by non-uniform doping, which causes a spatial variation in the energy band levels.

The relationship between the electric field E and the gradient of the valence band energy E_v is given by:

$$E = \frac{1}{q}\frac{dE_v}{dx}$$

From the provided figure, the change in energy over the distance L is Δ. Therefore, the gradient can be expressed as:

$$\frac{dE_v}{dx} = \frac{\Delta}{L}$$

Substituting this into the electric field equation:

$$E = \frac{1}{q}\frac{\Delta}{L}$$

$$\boxed{(A)\ \frac{\Delta}{qL}}$$

Key Points

- In equilibrium, the Fermi level E_f is constant, but E_c and E_v vary with position due to non-uniform doping.
- The electric field is defined by the slope of the band edges: $E = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$.

Q1.20.1 MCQ 2020 1M Ans: D ● ○ ○

To find the position of the intrinsic Fermi level (E_{Fi}) relative to the mid-gap energy, we use the standard semiconductor physics relation:

$$\frac{E_C + E_V}{2} - E_{Fi} = \frac{kT}{2}\ln\left(\frac{N_C}{N_V}\right)$$

Given the condition that the effective density of states of electrons is half that of holes:

$$N_C = \frac{N_V}{2} \implies \frac{N_C}{N_V} = 0.5$$

Assuming room temperature ($T = 300\,\text{K}$), the thermal voltage kT is approximately $0.026\,\text{eV}$. Substituting the values into the equation:

$$\frac{E_C + E_V}{2} - E_{Fi} = \frac{0.026}{2}\ln(0.5) = 0.013 \times (-0.693) \approx -0.00901\text{ eV} = -9.01\text{ meV}$$

Since magnitude of shift is asked for, answer is $9.01 meV$.

$$\boxed{(D)\ 9.01\text{ meV}}$$

Key Points

- The intrinsic Fermi level E_{Fi} is located exactly at the mid-gap only if $N_C = N_V$.
- Mid-gap energy is defined as $E_{midgap} = \frac{E_C+E_V}{2}$.
- If $N_V > N_C$, the intrinsic Fermi level shifts downward from the mid-gap toward the valence band.

Mistakes to be avoided

- Ensure the sign is correct; $\ln(0.5)$ is negative, indicating E_{Fi} is above the mid-gap in this specific mathematical arrangement (Midgap $- E_{Fi}$).
- Do not confuse kT (energy in eV) with kT/q (thermal voltage in V), although they share the numerical value 0.0259 at $300K$.

Q1.17.1 | NAT | 2017 | 2M | Ans: 1.5 to 1.7 | ● ● ○

The electric field E is calculated as the ratio of voltage V to the distance d:

$$E = \frac{V}{d} = \frac{5}{10^{-4}} = 5 \times 10^4 \text{ V/cm}$$

From the given curve, the slope m (representing mobility μ in the linear region) at $E = 5 \times 10^4$ V/cm is:

$$m = \frac{10^7 - 0}{5 \times 10^5} = 20 \quad (y = mx)$$

The drift velocity v_d is then:

$$v_d = \mu E = 20 \times E = 20 \times 5 \times 10^4 = 10^6 \text{ cm/s}$$

The current density J is given by:

$$J = n \times q \times v_d$$

$$J = 1 \times 10^{16} \times 1.6 \times 10^{-19} \times 10^6$$

$$J = 1.6 \times 10^3 \text{ A/cm}^2$$

$$J = 1.6\,\text{kA/cm}^2$$

$$\boxed{1.6}$$

Key Points

- Current density J is related to carrier concentration n and drift velocity v_d by $J = nev_d$.
- Drift velocity v_d is the product of mobility μ and electric field E in the linear velocity-field regime ($v_d = \mu E$).
- Ensure unit consistency, especially when dealing with cm and cm^2 in semiconductor physics.

Mistakes to be avoided

- Remember to to convert the final current density from A/cm^2 to kA/cm^2 as required by the magnitude of the answer.

Q1.17.2 NAT 2017 2M Ans: 15.9 to 16.1 ● ● ○

Net hole density varying in the direction of x is given by:

$$p_n(x) = p_{no} + \Delta p = p_{no} + G_L \tau_p$$

Since Generation Rate is said to be linearly decaying with x, the hole concentration is given as:

$$p_n(x) = p_{no} + G_{Lo} \tau_p \left(1 - \frac{x}{L}\right)$$

The diffusion current density $J_{p,\text{diff}}$ is calculated using the gradient of the hole concentration:

$$J_{p,\text{diff}} = -qD_p \frac{dp(x)}{dx} = -qD_p \left[\frac{-G_{Lo}\tau_p}{L}\right]$$

Substituting the given numerical values:

$$J_{p,\text{diff}} = \frac{1.6 \times 10^{-19} \times 100 \times 10^{17} \times 10^{-4}}{0.1 \times 10^{-4}} \text{ A/cm}^2$$

$$J_{p,\text{diff}} = 16 \text{ A/cm}^2$$

$$\boxed{16}$$

Key Points

- Under steady-state conditions with a linear generation profile, the excess carrier concentration often results in a constant diffusion current.

Mistakes to be avoided

- Be careful with the negative sign in the diffusion current formula; it usually cancels out with a negative concentration gradient to give a positive current in the direction of decreasing concentration.
- Ensure the diffusion constant D_p and length L are in consistent units (typically cm^2/s and cm) to match the A/cm^2 output.

Q1.17.3 MCQ 2017 1M Ans: A ● ○ ○

Silicon acts as an n-type dopant on Gallium sites (Group III) by providing an extra electron and a p-type dopant on Arsenic sites (Group V) by creating a hole. Therefore, statement **A** is correct.

$\boxed{(A) \text{ Silicon atoms act as } n\text{-type dopants in Arsenic sites and } p\text{-type dopants in Gallium sites}}$

Q1.16.1 MCQ 2016 2M Ans: A ● ● ○

Given the optical generation rate:

$$G_{opt} = 1.5 \times 10^{20} \text{ cm}^{-3}/\text{sec}$$

For t<0, steady state may be assumed due to continuous illumination before the radiation is switched OFF at $t = 0$. Hence, before turning OFF the light source, generation and recombination rates are equal.

For excess hole concentration Δp_{no}

$$G_{opt} = R = \frac{\Delta p_{no}}{\tau_p}$$

Substituting the given values (where $\tau_p = 0.1\ \mu$s):

$$1.5 \times 10^{20} = \frac{\Delta p_{no}}{0.1 \times 10^{-6}}$$

Excess hole concentration at $t = 0$ is

$$\Delta p_{no} = 1.5 \times 10^{13}\ \text{cm}^{-3}$$

After the light source is switched off at $t = 0$, Generation Rate $G = 0$, the excess hole concentration $\Delta p(t)$ decays with time only due to Recombination.

$$\frac{d\Delta p}{dt} = -\frac{\Delta p}{\tau_p}$$

which gives the solution as:

$$\Delta p(t) = \Delta p_{n_o} e^{-t/\tau_p}$$

Excess hole concentration at $t = 0.3\mu s$ is:

$$\Delta p(t) = 1.5 \times 10^{13} e^{-0.3/0.1}$$

$$\Delta p(t) = 1.5 \times 10^{13} e^{-3}$$

$$\Delta p(t) \approx 7.46 \times 10^{11}\ \text{cm}^{-3}$$

$$\boxed{(A)\ 1.5 \times 10^{13}\ \text{cm}^{-3} \text{ and } 7.46 \times 10^{11}\ \text{cm}^{-3}}$$

Mistakes to be avoided

- Distinguish between the steady-state concentration (when the light is on) and the instantaneous concentration during the decay phase.

Q1.16.2 | NAT | 2016 | 2M | Ans: 1.1 to 1.25 | ● ● ○

For a semiconductor in thermal equilibrium, the total current density J is zero. The net hole current density is the sum of the drift and diffusion components:

$$J = J_{p,\text{drift}} + J_{p,\text{diff}}$$

$$0 = q\mu_p p E - qD_p \frac{dp}{dx}$$

Rearranging to solve for the electric field E:

$$q\mu_p p E = qD_p \frac{dp}{dx}$$

$$E = \frac{D_p}{\mu_p} \frac{1}{p(x)} \frac{dp}{dx}$$

Using the Einstein relation $\frac{D_p}{\mu_p} = V_T$:

$$E = \frac{V_T}{p(x)} \frac{dp}{dx} = V_T \frac{d}{dx} \ln[p(x)]$$

Since the doping profile of semiconductor is given by $N_A(x)$, we replace $p(x)$ as $N_A(x)$:

$$E = V_T \frac{d}{dx} \ln[N_A(x)]$$

From the provided log-scale doping profile, the gradient of the natural log of doping concentration over the distance Δx (where 1 μm = 10^{-4} cm) is:

$$E = V_T\left[\frac{\ln(10^{16}) - \ln(10^{14})}{(2-1)\times 10^{-4}}\right]$$

$$E = 0.0259\left[\frac{36.84 - 32.24}{1\times 10^{-4}}\right]$$

$$E = 1196\,\text{V/cm} = 1.196\,\text{kV/cm}$$

$$\boxed{1.196}$$

Key Points

- A non-uniform doping profile in equilibrium creates a built-in electric field to counteract the diffusion of carriers.

Mistakes to be avoided

- Use ln (base e) as required by the derivative of $\ln(x)$, not $\log_{10}$.
- Use the standard room temperature value $V_T \approx 0.0259$ V unless otherwise specified.

Q1.16.3 MCQ 2016 1M Ans: A ● ○ ○

When a pentavalent impurity is introduced into an intrinsic semiconductor, it has five valence electrons. Four electrons form covalent bonds with neighboring semiconductor atoms, while the fifth electron is loosely bound to the impurity atom.

This creates a new discrete energy level, known as the **donor energy level** (E_d), which resides in the forbidden energy gap just below the conduction band (E_c). Because the energy difference $E_c - E_d$ is very small, these fifth electrons can be easily excited into the conduction band at room temperature, and become free electrons, thus making the semiconductor **n-type**.

$$\boxed{(A)\ \text{Intrinsic semiconductor doped with pentavalent atoms to form } n\text{-type semiconductor}}$$

Q1.15.1 MCQ 2015 2M Ans: C ● ● ○

From the problem description and the energy band diagram in the provided image:

$$n(x) = 10^{15}\exp\left(\frac{q\alpha x}{kT}\right)\ \text{cm}^{-3}$$

$$\frac{D}{\mu} = \frac{kT}{q} \implies \mu_n = \frac{D_n}{V_T} = \frac{36}{0.026} \approx 1384.6\ \text{cm}^2/\text{Vs}.$$

The electric field is related to the slope of the conduction band E_C:

$$E = \frac{1}{q}\frac{dE_C}{dx}$$

From the figure, the slope of the energy band is −0.1 eV/cm.

$$E = \frac{1}{q}(-0.1\ \text{eV/cm}) = -0.1\ \text{V/cm}$$

At $x = 0$: $n(0) = 10^{15}\exp(0) = 10^{15}\ \text{cm}^{-3}$.Gradient $\frac{dn}{dx} = n(x)\left(\frac{q\alpha}{kT}\right) = n(x)\frac{\alpha}{V_T}$ At $x = 0$:

$$\frac{dn}{dx} = 10^{15}\times\frac{0.1}{0.026} \approx 3.846\times 10^{15}\ \text{cm}^{-4}$$

The total current density is $J_n = J_{\text{drift}} + J_{\text{diffusion}}$

$$J_n = qn\mu_n E + qD_n \frac{dn}{dx}$$

Substitute the values at $x = 0$:

$$J_{drift} = (1.6 \times 10^{-19}) \times (10^{15}) \times (1384.6) \times (-0.1) \approx -0.02215 \text{ A/cm}^2$$

$$J_{diff} = (1.6 \times 10^{-19}) \times (36) \times (3.846 \times 10^{15}) \approx 0.02215 \text{ A/cm}^2$$

Thus total current density is

$$J_n = -0.02215 \text{ A/cm}^2 + 0.02215 \text{ A/cm}^2 = 0$$

$$\boxed{(C)\,0}$$

Key Points

- Non-uniform Doping leads to a built-in electric field and bending of energy bands.
- The slope of the energy bands (like E_c or E_v) is directly proportional to the internal electric field.

Q1.15.2 | NAT | 2015 | 2M | Ans: 95 to 105 | ● ○ ○

The electric field E inside the semiconductor bar is given as:

$$E = 10\,\text{V/cm}$$

The drift velocity v_d of the electron is calculated using the mobility μ and the electric field E:

$$v_d = \mu E = 1 \times 10^4 \text{ cm/sec}$$

The average time t taken by the electrons to move from one end of the bar of length L to the other end is:

$$t = \frac{L}{v_d} = 100\,\mu\text{s}$$

$$\boxed{100}$$

Q1.15.3 | NAT | 2015 | 1M | Ans: 1.8 to 2 | ● ○ ○

As per the given graph, the mobility of electrons at a concentration of $10^{16}\,\text{cm}^{-3}$ is $1200\,\text{cm}^2/\text{V s}$.
The conductivity σ_N for an n-type semiconductor is calculated using the donor concentration N_D, the electronic charge q, and the electron mobility μ_n:

$$\sigma_N = qN_D\mu_n$$

Substituting the numerical values:

$$\sigma_N = 1.6 \times 10^{-19} \times 10^{16} \times 1200 = 1.92\,\text{S cm}^{-1}$$

$$\boxed{1.92}$$

Q1.15.4 NAT 2015 1M Ans: 14 ● ○ ○

Given Data:

- Generation rate (G): $10^{20}\,\text{cm}^{-3}\text{s}^{-1}$
- Minority carrier lifetime (τ_p): $1\,\mu\text{s} = 10^{-6}\,\text{s}$

In the steady state, the rate of generation (G) is equal to the rate of recombination (R). For minority carriers (holes in n-type silicon), the excess hole concentration (Δp) is given by:

$$\Delta p = G\tau_p$$

Substituting the values:

$$\Delta p = 10^{20} \times 10^{-6} = 10^{14}\,\text{cm}^{-3}$$

Since the sample is n-type and the generation of carriers is significant, the steady-state hole concentration (p) is approximately equal to the excess hole concentration (Δp), assuming the equilibrium hole concentration (p_0) is negligible in comparison.

$$p \approx 10^{14}$$

Comparing this to the form 10^x:

$$x = 14$$

$$\boxed{14}$$

Key Points

- For extrinsic semiconductors under high injection or significant generation, the minority carrier concentration is dominated by the excess carriers.

Q1.15.5 NAT 2015 1M Ans: 0.50 to 0.54 ● ○ ○

The resistivity (ρ) of an n-type semiconductor is calculated using the donor concentration and electron mobility.

$$\rho = \frac{1}{\sigma_N} = \frac{1}{qN_D\mu_n}$$

Substituting the given values:

$$\rho = \frac{1}{10^{16} \times 1.6 \times 10^{-19} \times 1200} = 0.52\,\Omega\text{-cm}$$

$$\boxed{0.52}$$

Q1.14.1 MCQ 2014 2M Ans: D ● ○ ○

The bands tilt upward because electron energy is inversely related to electrostatic potential ($E = -qV$). Since the left side is connected to the negative terminal (lower potential), it has higher energy. This makes **Option D** the only diagram where E_C, E_F, and E_V correctly reflect this upward shift.

$$\boxed{D}$$

Q1.14.2 NAT 2014 2M Ans: 224.9 to 225.1 ● ○ ○

For a compensated p-type semiconductor, the minority carrier electron concentration is given by

$$n_o = \frac{n_i^2}{(N_A - N_D)}$$

Substituting the given values:

$$n_o = \frac{2.25 \times 10^{20}}{(10^{18} - 10^{15})} = 225\,\text{cm}^{-3}$$

$$\boxed{225}$$

Key Points

- In a compensated semiconductor, the net carrier concentration is determined by the difference between acceptor and donor concentrations.

Q1.14.3 NAT 2014 1M Ans: 1.12 to 1.14 ● ○ ○

The critical wavelength (λ_c) for a semiconductor, which corresponds to the minimum photon energy required to excite an electron across the bandgap, is calculated as follows:

$$\lambda_c = \frac{1.242}{E_g(\text{eV})}\,\mu\text{m} = \frac{1.242}{1.1}\,\mu\text{m} = 1.129\,\mu\text{m}$$

$$\boxed{1.129}$$

Key Points

- The relationship between photon wavelength and energy is derived from $E = \frac{hc}{\lambda}$.
- When energy (E_g) is expressed in electron-volts (eV) and wavelength in micrometers (μm), the constant hc is approximately 1.242.
- Photons with wavelengths longer than λ_c do not have sufficient energy to cause intrinsic excitation.

Mistakes to be avoided

- Do not confuse μm with nm; if using nanometers, the constant becomes 1242.

Q1.14.4 MCQ 2014 2M Ans: A ● ○ ○

Intrinsic resistivity ρ_i is inversely proportional to intrinsic carrier concentration n_i, which varies as $\exp(-E_g/2kT)$. Plotting $\ln(\rho_i)$ against $1/T$ yields a line with a slope proportional to $E_g/2k$, allowing for the estimation of the bandgap energy.

Thus **Option A** is correct.

$$\boxed{(A)\ \text{band gap energy of silicon } (E_g)}$$

Q1.14.5 MCQ 2014 1M Ans: A ● ● ○

The condition for a photon to be absorbed by a semiconductor and create an electron-hole pair is that its energy must be greater than or equal to the bandgap energy (E_g). This corresponds to a critical wavelength (λ_c) condition:

$$\lambda_c \leq \frac{1.242}{E_g(\text{eV})}\,\mu\text{m}$$

Given the bandgap energy $E_g = 1.42\,\text{eV}$, we calculate the maximum allowable wavelength:

$$\lambda_c \leq \frac{1.242}{1.42}$$

$$\lambda_c \leq 0.87\,\mu\text{m}$$

$$\boxed{(A)\,0.42\mu m < \lambda_c < 0.87\mu m}$$

Key Points

- Absorption occurs only when the incident photon wavelength is shorter than or equal to the threshold wavelength.

Q1.14.6 NAT 2014 1M Ans: 12.9 to 13.1 ● ○ ○

Using the Einstein relation for a semiconductor, the diffusion coefficient (D_p) and mobility (μ_p) of holes are related to the thermal voltage (V_T) as follows:

$$\frac{D_p}{\mu_p} = V_T$$

Given the hole mobility $\mu_p = 500\,\text{cm}^2/\text{Vs}$ and assuming a thermal voltage $V_T = 26\,\text{mV}$ at room temperature:

$$D_p = \mu_p V_T = 500 \times 26 \times 10^{-3} = 13\,\text{cm}^2/\text{s}$$

$$\boxed{13}$$

Q1.14.7 MCQ 2014 1M Ans: D ● ○ ○

The recombination rate in a semiconductor is defined as $R = \Delta n/\tau$, where Δn is the excess minority carrier concentration and τ is the lifetime. This formula shows that the rate is directly proportional to **D** the excess minority carrier concentration.

$$\boxed{(D)\text{ the excess minority carrier concentration}}$$

Q1.14.8 NAT 2014 2M Ans: 3990 to 4010 ● ○ ○

The electron diffusion current density is given by the relation:

$$J = qD_n\frac{dn}{dx}$$

where the electron diffusion coefficient D_n is determined using the Einstein relation:

$$D_n = \mu_n V_T$$

Given the values:

$$D_n = \frac{1000\,\text{cm}^2}{\text{V s}} \times 25 \times 10^{-3}\,\text{V} = 25\,\text{cm}^2/\text{s}$$

Substituting the parameters into the current density formula:

$$J = 1.6 \times 10^{-19} \times 25 \times 1 \times 10^{21} = 4000\,\text{A}/\text{cm}^2$$

$$\boxed{4000}$$

Q1.14.9 MCQ 2014 2M Ans: D ● ○ ○

Since $N_D \gg n_i$, the equilibrium electron concentration is approximately equal to the donor concentration:

$$n \simeq N_D = 2.25 \times 10^5 \text{ cm}^{-3}$$

The equilibrium hole concentration is determined using the mass action law:

$$p = \frac{n_i^2}{N_D} = \frac{(1.5 \times 10^{10})^2}{2.25 \times 10^{15}} = 1 \times 10^5 \text{ cm}^{-3}$$

$$\boxed{(D)\, n_0 = 2.25 \times 10^{15}\, cm^{-3}\,, p_0 = 1 \times 10^5\, cm^{-3}}$$

Q1.14.10 MCQ 2014 2M Ans: A ● ● ○

The electron concentration at the edge of the depletion region on the p-side under forward bias V_f is given by:

$$n_p = n_{p0} \exp\left[\frac{qV_f}{kT}\right]$$

where the equilibrium minority electron concentration n_{p0} is:

$$n_{p0} = \frac{n_i^2}{N_A}$$

Substituting the given values:

$$n_{p0} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4 \text{ cm}^{-3}$$

We then have the carrier concentration at the edge:

$$n_p = 2.25 \times 10^4 \exp\left[\frac{0.3}{0.026}\right] = 2.306 \times 10^9 \text{ cm}^{-3}$$

$$\boxed{(A)\, 2.3 \times 10^9 cm^{-3}}$$

Key Points

- Minority carrier concentration increases exponentially with forward bias voltage.

Q1.11.1 MCQ 2011 1M Ans: C ● ○ ○

Method 1

Drift current density is defined as $J_{drift} = \sigma E$, where σ is the conductivity and E is the electric field. Since conductivity depends on the carrier concentration ($\sigma = qn\mu_n + qp\mu_p$), the drift current depends on both the **electric field** and the **carrier concentration**.

Method 2

The drift current density can also be expressed as $J_{drift} = qnv_d$, where v_d is the drift velocity. Because drift velocity is directly proportional to the electric field ($v_d = \mu E$), the total current is a function of both the **carrier concentration** (n) and the **electric field** (E).

$$\boxed{(C) \text{ both the electric field and the carrier concentration}}$$

Mistakes to be avoided

- Do not confuse drift current with diffusion current; diffusion current depends on the concentration gradient, not the electric field directly.

Q1.10.1 MCQ 2010 2M Ans: C ● ○ ○

The sample is in thermal equilibrium. The electric field E is calculated as:

$$E = \frac{V}{d} = \frac{1}{1 \times 10^{-6}} = 10^6\,\text{V/m}$$

Converting the units to kV/cm:

$$E = 10\,\text{kV/cm}$$

$$\boxed{(C)\ 10\,\text{kV/cm}}$$

Q1.10.2 MCQ 2010 2M Ans: A ● ○ ○

The drift current density J is given by:

$$J = \sigma E$$

Since the conductivity σ for an n-type semiconductor is $\sigma = qn\mu_n \approx qN_D\mu_n$, we have:

$$J = N_D q \mu_n E$$

Substituting the given values:

$$J = 10^{16} \times (1.6 \times 10^{-19}) \times (1350) \times (10 \times 10^3) = 2.16 \times 10^4\,\text{A/cm}^2$$

$$\boxed{(A)\ 2.16 \times 10^4\,\text{A/cm}^2}$$

Key Points

- For extrinsic semiconductors, conductivity is dominated by the majority carrier concentration.

Q1.09.1 MCQ 2009 1M Ans: A ● ○ ○

The ratio of the diffusion constant D to the mobility μ is equal to the thermal voltage V_T:

$$\frac{D}{\mu} = V_T$$

Taking the reciprocal of the above expression, we get the ratio of mobility to the diffusion constant:

$$\frac{\mu}{D} = \frac{1}{V_T}$$

The unit of thermal voltage V_T is Volts (V). Therefore, the units for the ratio $\frac{\mu}{D}$ are:

$$\text{units : V}^{-1}$$

$$\boxed{(A)\ \text{V}^{-1}}$$

Q1.09.2 MCQ 2009 1M Ans: C ● ○ ○

- Silicon atom concentration in Si is about 5×10^{22}, cm^{-3}, so A is not possible.
- In n-type silicon, hole concentration is very small, far below 4×10^{19}, cm^{-3} , so B is impossible.
- An n-type Si crystal typically has donor dopant concentration around this order. Thus, C is correct.
- Each Silicon atom has 4 valence electrons, so the valence electrons concentration is about $2 \times 10^{23} cm^{-3}$, so D is incorrect.

$$\boxed{(C)\text{ Dopant atoms}}$$

Q1.08.1 MCQ 2008 1M Ans: C ● ○ ○

Since boron is a p-type impurity, the Fermi level shifts downward toward the valence band. The shift relative to the intrinsic Fermi level is given by:

$$E_i - E_f = kT \ln \frac{N_A}{n_i}$$

Substituting the given values:

$$E_i - E_f = 25 \times 10^{-3} \ln \frac{4 \times 10^{17}}{1.5 \times 10^{10}} = 0.427\,\mathrm{eV}$$

$$\boxed{(C)\text{ goes down by 0.31 eV}}$$

Mistakes to be avoided

- Do not confuse the p-type formula $(E_i - E_f)$ with the n-type formula $(E_f - E_i)$.

Q1.08.2 MCQ 2008 1M Ans: A ● ○ ○

Since boron is a p-type impurity, doping silicon with boron makes it a p^+ semiconductor.

$$\boxed{(A)\text{ A silicon wafer heavily doped with boron is a } p^+ \text{ substrate}}$$

Q1.07.1 MCQ 2007 1M Ans: D ● ○ ○

By the law of electrical neutrality, the total positive charge density must equal the total negative charge density:

$$p + N_D = n + N_A$$

Given, $N_D = 0$. Additionally, $N_A \gg n_i$, the intrinsic carrier concentration is negligible. Thus:

$$p \approx N_A$$

Using the mass action law, which states that the product of the electron and hole concentrations is constant at a given temperature:

$$np = n_i^2$$

So, the minority carrier electron concentration is:

$$n = \frac{n_i^2}{p} = \frac{n_i^2}{N_A}$$

$$\boxed{(D)\ \frac{n_i^2}{N_A}}$$

Mistakes to be avoided

- Do not confuse the mass action law with charge neutrality; they are independent physical principles used together to find carrier concentrations.

Q1.06.1 MCQ 2006 1M Ans: D ● ○ ○

For an n-type semiconductor, the conductivity σ_n is dominated by electrons:

$$\sigma_n = nq\mu_n$$

For an intrinsic semiconductor, the conductivity σ_i accounts for both electrons and holes:

$$\sigma_i = n_i q(\mu_n + \mu_p)$$

Taking the ratio of the two conductivities:

$$\frac{\sigma_n}{\sigma_i} = \frac{n\mu_n}{n_i(\mu_n + \mu_p)}$$

Substituting the given concentrations and simplifying the mobility term:

$$\frac{\sigma_n}{\sigma_i} = \frac{4.2 \times 10^8 \times \mu_n}{1.5 \times 10^4 \times \mu_n \left(1 + \frac{\mu_p}{\mu_n}\right)}$$

Given the mobility ratio $\frac{\mu_p}{\mu_n} = 1.4$:

$$\frac{\sigma_n}{\sigma_i} = \frac{4.2 \times 10^8}{1.5 \times 10^4 \times 1.4} = 2 \times 10^4$$

$$\boxed{(D)\ 20000}$$

Key Points

- Conductivity $\sigma = nq\mu_n + pq\mu_p$. In n-type, $n \gg p$, so $\sigma \approx nq\mu_n$.

Q1.06.2 MCQ 2006 1M Ans: B ● ● ○

In the Hall effect, when a magnetic field is applied perpendicular to the direction of current flow, a transverse electric field is established such that the net force on the charge carriers becomes zero. At equilibrium:

$$\text{Electric force} + \text{magnetic force} = 0$$

$$q\vec{E} + q(\vec{v} \times \vec{B}) = 0$$

$$\vec{E} = -(\vec{v} \times \vec{B})$$

Using the property of cross products ($\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$):

$$\vec{E} = \vec{B} \times \vec{v}$$

$$\boxed{(B)\ \vec{B} \times \vec{v}}$$

Key Points

- The total force on a charge q moving with velocity $\vec{v}$ in electric and magnetic fields is $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$.

Mistakes to be avoided

- Be careful with the order of the cross product; $\vec{v} \times \vec{B}$ is not equal to $\vec{B} \times \vec{v}$.
- Remember that the electric field acts in the opposite direction of the magnetic force to achieve balance.

Q1.06.3 MCQ 2006 1M Ans: A ● ○ ○

The injected minority carrier current in an extrinsic semiconductor under low-level injection is the **diffusion current**.

$$\boxed{(A)\ \text{Diffusion current}}$$

Q1.06.4 MCQ 2006 1M Ans: B ● ○ ○

According to the mass action law, the product of the equilibrium concentrations of electrons (n) and holes (p) in a semiconductor is a constant at a given temperature:

$$np = n_i^2$$

where n_i is the intrinsic carrier concentration. We can express the minority carrier concentration as:

$$p = \frac{n_i^2}{n}$$

Since n_i is constant, the relationship between the hole concentration and the electron concentration is:

$$p \propto \frac{1}{n}$$

$$\boxed{(B)\ \text{Inversely proportional to the doping concentration}}$$

Q1.05.1 MCQ 2005 1M Ans: B ● ○ ○

The conductivity of a semiconductor depends on the carrier concentration and the mobility of the charge carriers. For an n-type semiconductor, the conductivity σ_n is dominated by electrons:

$$\sigma_n = nq\mu_n$$

Similarly, for a p-type semiconductor with the same doping concentration ($n = p$), the conductivity σ_p is dominated by holes:

$$\sigma_p = pq\mu_p$$

Taking the ratio of the conductivities for equal doping concentrations:

$$\frac{\sigma_p}{\sigma_n} = \frac{\mu_p}{\mu_n}$$

Given

$$\frac{\mu_n}{\mu_p} = 3$$

Therefore we have

$$\frac{\sigma_p}{\sigma_n} = \frac{1}{3}$$

$$\boxed{(B)\ \frac{1}{3}}$$

Mistakes to be avoided

- Do not assume $\sigma_p = \sigma_n$ even if doping concentrations are equal, as mobilities differ.

Q1.05.2 MCQ 2005 1M Ans: A ● ○ ○

The abundance of Silicon is the primary technical reason for its dominance in the industry.

$$\boxed{(A) \text{ Abundance of Silicon on the surface of the Earth}}$$

Q1.05.3 MCQ 2005 1M Ans: C ● ○ ○

The bandgap of Silicon at room temperature is 1.1 eV.

$$\boxed{(C)\ 1.1eV}$$

Q1.04.1 MCQ 2004 1M Ans: A ● ○ ○

The bandgap energy (E_g) of a semiconductor material is related to the cutoff wavelength (λ) by the following relation:

$$E_g = \frac{hc}{\lambda(\mu\text{m})}\,\text{eV}$$

Given the cutoff wavelength $\lambda = 1.1\ \mu$m and the bandgap energy $1.12\ eV$. The constant hc is calculated as:

$$1.12\ eV = \frac{hc}{1.1\ \mu\text{m}}\,\text{eV}$$

$$hc = 1.232\,\text{eV}\ \mu m$$

For new cutoff wavelength $\lambda = 0.87\ \mu$m, bandgap energy is:

$$E_g = \frac{1.232}{0.87(\mu\text{m})}\,\text{eV} = 1.416\,\text{eV}$$

$$\boxed{(A)\ 1.416\,eV}$$

Mistakes to be avoided

- Do not confuse the units; the resulting energy E_g is specifically in electron-volts (eV).

Q1.04.2 MCQ 2004 1M Ans: B ● ○ ○

The resistivity (ρ) of an n-type semiconductor is given by:

$$\rho = \frac{1}{nq\mu_n}$$

For an n-type semiconductor with donor concentration N_D, assuming complete ionization at room temperature, the electron concentration is approximately equal to the donor concentration:

$$n \approx N_D$$

Rearranging the formula to solve for the donor concentration N_D:

$$N_D = \frac{1}{q\mu_n\rho}$$

Substituting the given values ($q = 1.6 \times 10^{-19}$ C, $\mu_n = 1250\,\text{cm}^2/\text{V s}$, and $\rho = 0.5\,\Omega$ cm):

$$N_D = \frac{1}{1.6 \times 10^{-19} \times 1250 \times 0.5} = 10^{16}/\text{cm}^3$$

$$\boxed{(B)\ 1 \times 10^{16}\ /cm^3}$$

Key Points

- The assumption $n \approx N_D$ holds true when N_D is much greater than the intrinsic carrier concentration n_i.

Q1.04.3 MCQ 2004 1M Ans: C ● ○ ○

The impurity commonly used for realizing the base region of a silicon n - p - n transistor is Boron since Boron is a trivalent dopant for Silicon.

$$\boxed{(C)\ \text{Boron}}$$

Q1.03.1 MCQ 2003 1M Ans: D ● ● ○

In the absence of an electric field, the current density J in the silicon sample is solely due to the diffusion of carriers. For electrons, the diffusion current density J_n is given by:

$$J_n = qD_n \frac{dn}{dx}$$

Variation in electron concentration

$$\Delta n = n(x_2) - n(x_1) = (6 \times 10^{16} - 10^{17})\,\text{cm}^{-3} = -4 \times 10^{16}\,\text{cm}^{-3}$$

Calculating the concentration gradient $\frac{dn}{dx}$:

$$\frac{dn}{dx} = \frac{\Delta n}{\Delta x} = \frac{-4 \times 10^{16}}{2 \times 10^{-4}} = -2 \times 10^{20}\,\text{cm}^{-4}$$

Substituting the values into the current density formula:

$$J_n = (1.6 \times 10^{-19}) \times (35) \times (-2 \times 10^{20}) = -1120\,\text{A/cm}^2$$

$$\boxed{(D)\ -1120\,A/cm^2}$$

Q1.03.2 MCQ 2003 1M Ans: A ● ○ ○

To find the resistance R of the n-type silicon bar, we first determine the conductivity σ of the material. Since it is an n-type semiconductor, the conductivity is dominated by the majority carriers (electrons):

$$\sigma \approx nq\mu_n$$

Substituting given values:

$$\sigma = (5 \times 10^{20}) \times (1.6 \times 10^{-19}) \times 0.13$$

$$\sigma = 80 \times 0.13 = 10.4\,\text{S/m}$$

The resistance R is given by the formula:

$$R = \frac{\rho L}{A} = \frac{L}{\sigma A}$$

Calculating the resistance with given values and obtained conductivity:

$$R = \frac{10^{-3}}{10.4 \times 10^{-10}} = \frac{10^7}{10.4} \approx 10^6\,\Omega$$

$$\boxed{(A)\ 10^6\ \text{Ohm}}$$

Key Points

- The formula for resistance of a bar of length L and cross-sectional area A, made of material having resistivity ρ is given as: $R = \frac{\rho L}{A} = \frac{L}{\sigma A}$

Q1.02.1 | MCQ | 2002 | 1M | Ans: A | ● ○ ○

According to the Law of Mass Action,

$$n_i^2 = np$$

Where n_i is the intrinsic concentration. To find the minority carrier concentration (holes, p) in an n-type semiconductor, the formula is rearranged as:

$$p = \frac{n_i^2}{n}$$

Given the values from the calculation:

$$p = \frac{1.5 \times 10^{16} \times 1.5 \times 10^{16}}{5 \times 10^{20}} = 4.5 \times 10^{11}/\text{cm}^3$$

$$\boxed{(A)\ 4.50 \times 10^{11}/m^3}$$

Mistakes to be avoided

- The Law of Mass Action does not apply under non-equilibrium conditions, such as under heavy illumination or carrier injection.

Q1.98.1 | MCQ | 1998 | 1M | Ans: B | ● ○ ○

Since kT/q represents the thermal voltage (V_T), its reciprocal carries the unit of $1/\text{V}$.

$$\boxed{(B)\ \text{V}^{-1}}$$

Q1.95.1 | MCQ | 1995 | 1M | Ans: C | ● ○ ○

The velocity of a charge carrier is defined as the distance traveled per unit time. Based on the given parameters:

$$\text{Velocity} = \frac{\text{Distance}}{\text{Time}}$$

Substituting the values from the problem:

$$v_d = \frac{1}{20 \times 10^{-6}} = 50,000\,\text{cm/sec}$$

The relationship between drift velocity (v_d), mobility (μ), and electric field (E) is given by:

$$v_d = \mu E$$

Rearranging the formula to solve for mobility:

$$\mu = \frac{v_d}{E} = \frac{50,000}{10} = 5,000\,\text{cm}^2/\text{V-sec}$$

$$\boxed{(C)\ 5000}$$

Key Points

- Drift velocity (v_d) is the average velocity attained by charged particles in a material due to an electric field.
- Mobility (μ) characterizes how quickly a charge carrier can move through a metal or semiconductor when pulled by an electric field.

Q1.95.2 MCQ 1995 1M Ans: C

Given the carrier concentrations in the semiconductor:

$$p = 2.25 \times 10^{15}/\text{cm}^3$$

$$n_i = 1.5 \times 10^{10}/\text{cm}^3$$

According to the Mass Action Law:

$$n = \frac{n_i^2}{p}$$

Substituting the given values into the equation:

$$n = \frac{(1.5 \times 10^{10})^2}{2.25 \times 10^{15}} = \frac{2.25 \times 10^{20}}{2.25 \times 10^{15}} = 10^5/\text{cm}^3$$

$$\boxed{(C)\ 10^5/\text{cm}^3}$$

Q1.95.3 MCQ 1995 1M Ans: C

At temperatures above 0 K, the Fermi level is defined as the energy level that has a 50% probability of being occupied by an electron.

$$\boxed{(C)\ 0.5}$$

Q1.95.4 MCQ 1995 1M Ans: D

The relationship between drift velocity v_d and the applied electric field E is governed by carrier mobility μ:

$$v_d = \mu E$$

The dependence of mobility on the electric field is illustrated in the figure below:

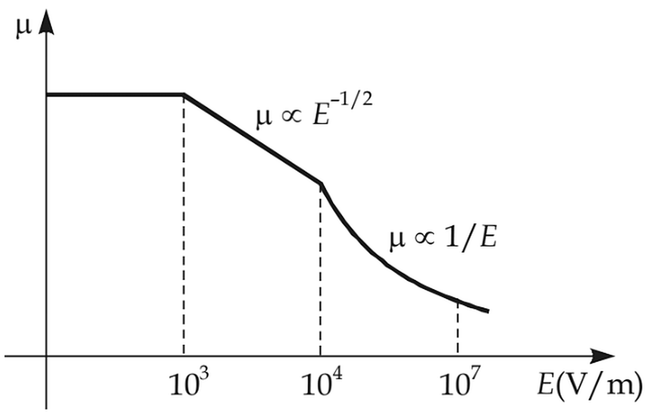

Fig. Solution Q1.95.4

Based on the variation of μ with E:

1. **Low Electric Field ($E < 10^3$ V/m):** The mobility of the charge carrier remains almost constant. Consequently, the drift velocity (v_d) increases linearly with the electric field ($\mu \propto$ constant $\implies v_d \propto E$).

2. **High Electric Field ($E > 10^4$ V/m):** For large applied electric fields, the mobility decreases significantly. Specifically, when $\mu \propto 1/E$, the drift velocity gradually saturates to a constant value (v_{sat}), meaning it becomes independent of further increases in the electric field.

$$\boxed{(D)}$$

$$\boxed{\text{increases linearly with electric field at low values of electric field and gradually saturates at higher values of electric field}}$$

Key Points

- At low fields, scattering is dominated by lattice vibrations (phonons) and impurities, keeping μ constant.
- At very high fields, carriers reach a scattering-limited velocity known as saturation velocity (v_{sat}).

Mistakes to be avoided

- Do not assume v_d increases infinitely with E; physical limitations lead to velocity saturation.

Q1.94.1 | MCQ | 1994 | 1M | Ans: False | ● ○ ○

The given statement is false. For any given semiconductor material, the electron mobility (μ_n) is always significantly higher than the hole mobility (μ_p):

$$\mu_n > \mu_p$$

The conductivity of a semiconductor depends on the carrier concentration, the charge of the carrier, and the mobility. For an n-type semiconductor, the conductivity (σ_n) is:

$$\sigma_n = nq\mu_n$$

For a p-type semiconductor, the conductivity (σ_p) is:

$$\sigma_p = pq\mu_p$$

Given that the dopant concentrations are the same ($n = p$) and the electronic charge $q = 1.6 \times 10^{-19}$ C is constant, it follows that:

$$\text{Since } \mu_n > \mu_p \implies \sigma_n > \sigma_p$$

Therefore, for equal doping concentrations, an n-type semiconductor will always have higher conductivity than a p-type semiconductor.

$$\boxed{\text{False}}$$

Q1.92.1 | MCQ | 1992 | 2M | Ans: A | ● ○ ○

Under steady-state conditions, the rate at which excess carriers are generated is balanced by the rate at which they recombine. The generation rate (G) can be expressed in terms of the excess carrier concentration and the carrier lifetime.

Given data:

- Excess electron concentration, $\Delta n = 10^{15}/\text{cm}^3$
- Minority (hole) carrier lifetime, $\tau_p = 10\,\mu\text{s} = 10 \times 10^{-6}$ s

Since electron-hole pairs are generated together, $\Delta n = \Delta p$. The generation rate is given by:

$$\text{Generation rate } G = \frac{\Delta p}{\tau_p}$$

Substituting the given values:

$$G = \frac{10^{15}}{10 \times 10^{-6}} = 10^{20} \text{ e-h pairs/cm}^3\text{/s}$$

$$\boxed{(A) \text{ is } 10^{20}\ e - h \text{ pairs/cm}^3\text{/s}}$$

Key Points

- In steady-state, the net rate of change of carriers is zero, meaning Generation Rate (G) = Recombination Rate (R).
- Carrier Lifetime is the average time a carrier exists before recombining. For minority carriers, the recombination rate is typically defined as $R = \Delta p/\tau_p$.
- In bulk semiconductors, excess electrons and holes are created in pairs ($\Delta n = \Delta p$) to maintain charge neutrality.

Mistakes to be avoided

- Ensure you are using the excess carrier concentration (Δp) and not the total concentration for recombination rate calculations.

Q1.91.1 | MCQ | 1991 | 2M | Ans: B | ● ○ ○

When a semiconductor is doped with both donor and acceptor impurities, the net carrier concentration depends on the difference between the concentrations of the two types of dopants. This process is known as compensation.

Since $N_A > N_D$, the acceptor atoms neutralize the donor atoms, and the remaining concentration determines the type of the semiconductor:

$$\text{Net carrier concentration} = N_A - N_D$$

Substituting the values:

$$\text{Net concentration} = (2 \times 10^{16} - 10^{16})/\text{cm}^3 = 10^{16}/\text{cm}^3$$

Because the resulting majority carriers are holes, the material is a p-type semiconductor with a concentration of $10^{16}/\text{cm}^3$.

$$\boxed{(B)\ p - \text{type with carrier concentration of } 10^{16}/cm^3}$$

Q1.90.1 | MSQ | 1990 | 2M | Ans: A,C | ● ● ○

The relationship between drift velocity (v_d), mobility (μ), and the applied electric field (E) is given by:

$$v_d = \mu E$$

In a semiconductor, the behavior of charge carriers changes significantly under high electric field conditions:

1. **Mobility Degradation:** At high electric fields, the mobility of charge carriers is no longer constant. It begins to decrease as the electric field increases due to increased scattering rates (primarily optical phonon scattering). In this regime, the mobility is inversely proportional to the electric field:

$$\mu \propto \frac{1}{E}$$

2. **Velocity Saturation:** Substituting the high-field mobility relation into the drift velocity equation:

$$v_d = \mu E \propto \left(\frac{1}{E}\right) E \approx \text{constant}$$

This implies that the drift velocity of the charge carriers reaches a maximum limit, known as the saturation velocity (v_{sat}), and does not increase further even if the electric field is increased.

Based on the analysis, both the decrease in mobility and the saturation of drift velocity are correct characteristics of high-field transport.

$$\boxed{(A)\text{the mobility of charge carriers decreases}}$$

$$\boxed{(C)\text{the velocity of the charge carriers saturates.}}$$

Mistakes to be avoided

- Do not assume Ohm's law ($v_d \propto E$) remains valid at very high electric fields.
- Ensure proper distinction between mobility (μ) and velocity (v_d) when describing their behavior versus the electric field.

Q1.89.1 MCQ 1989 2M Ans: A ● ○ ○

The Fermi level position is given by

$$E_i - E_F = kT \ln\left(\frac{N_A - N_D}{n_i}\right)$$

, which directly depends on the net doping concentration $N_A - N_D$. Since $N_A > N_D$, the semiconductor is p-type and the Fermi level shifts toward the valence band based on this difference.

$$\boxed{(A)\ N_A - N_D}$$

Q1.89.2 MCQ 1989 2M Ans: C ● ○ ○

When a semiconductor is illuminated by light with photon energy greater than the bandgap energy ($E_{ph} > E_g$), valence electrons absorb the energy and move to the conduction band. This process generates electron-hole pairs (EHPs).

Due to the nature of this generation process, for every electron that is excited to the conduction band, a corresponding hole is created in the valence band. Therefore, the increase in carrier concentrations must be equal:

$$\Delta n = \Delta p$$

where,

- Δn = increase in electron concentration due to illumination
- Δp = increase in hole concentration due to illumination

$$\boxed{(C)\ \Delta n = \Delta p}$$

Q1.87.1 MCQ 1987 2M Ans: B ● ○ ○

The probability P_1 of an electron occupying E_1 and P_2 of E_2 being empty are equal ($P_1 = P_2$) due to the symmetry of the Fermi-Dirac distribution around the Fermi level. Mathematically, $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$, meaning the probability of finding an electron at a level above E_F equals the probability of finding a hole at the same distance below it.

$$\boxed{(B)\ P_1 = P_2}$$

Q1.87.2 MCQ 1987 2M Ans: C ● ○ ○

According to the mass action law, the product of the free electron concentration (n) and the free hole concentration (p) is constant for a given semiconductor at a fixed temperature:

$$np = n_i^2$$

The intrinsic carrier concentration (n_i) is strongly dependent on temperature (T). The relationship is given by:

$$n_i^2 \propto T^3 e^{-E_g/kT} \implies n_i \propto T^{3/2}$$

For an intrinsic semiconductor, since $n = p = n_i$, the carrier concentration n follows the same temperature dependence:

$$n \propto T^{3/2}$$

$$\boxed{(C)\ \text{temperature of the semiconductor}}$$

Key Points

- Intrinsic carrier concentration increases exponentially with temperature.

Q1.87.3 MCQ 1987 2M Ans: A ● ○ ○

The Einstein relation provides a fundamental link between the diffusion constant (D) and the mobility (μ) of charge carriers in a semiconductor. The ratio is proportional to the absolute temperature:

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

The thermal voltage (V_T) is defined as:

$$V_T = \frac{kT}{q} \approx \frac{T\ (\text{in Kelvin})}{11600}$$

At room temperature ($T = 300\,\text{K}$), $V_T \approx 25.86\,\text{mV}$.

$$\boxed{(A)\ \text{depends upon the temperature of the semiconductor.}}$$

Key Points

- This relation implies that carriers with higher mobility will also diffuse more rapidly.

Q1.87.4 MCQ | 1987 | 2M | Ans: C | ● ○ ○

Semiconductors are classified into Direct Band Gap (DBG) and Indirect Band Gap (IBG) materials based on the alignment of the conduction band minimum and valence band maximum in momentum space.

- **Direct Band Gap (e.g., GaAs):** The maximum of the valence band and the minimum of the conduction band occur at the same value of the crystal momentum (k-vector). During recombination, an electron can transition directly, releasing energy primarily in the form of **light** (photons).
- **Indirect Band Gap (e.g., Si, Ge):** The extrema occur at different k-values. Recombination requires a change in momentum (involving phonons), and energy is primarily released as **heat**.

Because DBG semiconductors exhibit high radiative recombination efficiency and short carrier lifetimes, they are used for fabricating Lasers and LEDs.

$\boxed{(C) \text{ exhibit short carrier life time and they are used for fabricating lasers.}}$

DEBUG INFO

Chapter I

Basic Semiconductor Physics

Total Questions : 75

MCQ : 55

MSQ : 4

NAT : 16

SOLUTIONS

II

PN Junctions

Topics

1. Basic Semiconductor Junctions
2. PN Junction Under Equilibrium
3. PN Junction Under Biasing

Q2.26.1 MCQ 2026 1M Ans: C ● ○ ○

In a heavily doped p-n junction, the depletion region width is extremely narrow. When a reverse bias is applied, the high electric field generated across this thin layer allows electrons to move from the valence band of the p-side to the conduction band of the n-side through a process known as quantum mechanical tunneling. This phenomenon is characteristic of Zener breakdown.

$$\boxed{(C)\text{Tunneling}}$$

Mistakes to be avoided

- Do not confuse Zener breakdown with Avalanche breakdown, which is caused by impact ionization in lightly doped junctions.

Q2.26.2 MSQ 2026 1M Ans: A ● ● ○

In a forward-biased p-n junction, the carrier concentrations depart from their equilibrium values, necessitating the use of quasi-Fermi levels E_{FN} and E_{FP} to describe the electron and hole distributions respectively. The separation between these two levels is directly proportional to the applied forward bias voltage V.

The relationship between the quasi-Fermi level separation and the applied voltage is given by:

$$E_{FN} - E_{FP} = qV$$

Given an applied forward bias voltage $V = 2\,\text{V}$, the magnitude of the difference between the quasi-Fermi levels is:

$$|E_{FN} - E_{FP}| = q(2\,\text{V}) = 2\,\text{eV}$$

$$\boxed{(A)\,2\text{ eV}}$$

Key Points

- **Quasi-Fermi Levels:** Used to describe the population of electrons and holes when a semiconductor is not in thermal equilibrium (e.g., under bias or illumination).
- **Bias Relationship:** The separation $E_{FN} - E_{FP}$ at any point in the depletion region is equal to the energy equivalent of the applied potential qV.

Mistakes to be avoided

- Do not confuse the separation of quasi-Fermi levels with the built-in potential V_{bi}.

Q2.25.1 NAT 2025 2M Ans: 0.23 to 0.24 ● ○ ○

The forward bias current I_f is given by the diode equation:

$$I_f = I_o\left[e^{V/V_T} - 1\right]$$

First, calculate the thermal voltage V_T at $T = 300\,\text{K}$:

$$V_T = \frac{kT}{q} \approx 25.86\,\text{mV}$$

Substituting the given values into the diode equation:

$$100 \times 10^{-3} = 10 \times 10^{-6}\left[e^{V/V_T} - 1\right]$$

$$10001 = e^{V/V_T}$$

Taking the natural logarithm on both sides:

$$V = 25.86 \times 10^{-3} \ln(10001) \approx 0.238\,\text{V}$$

$$\boxed{0.238}$$

Key Points

- **Diode Current Equation:** $I = I_o(e^{V/\eta V_T} - 1)$, where η is the ideality factor (assumed 1 here).

Mistakes to be avoided

- **Approximation:** For large forward currents where $I_f \gg I_o$, the "-1" in the formula can sometimes be neglected, but for precision in NAT questions, keep it until the final steps.

Q2.21.1 NAT 2021 2M Ans: 8.1 to 8.4 ● ○ ○

Although the N-side has slightly higher doping concentration, its width is much smaller

$$(0.2\mu m)$$

compared to the P-side

$$(1.2\mu m)$$

. Since the doping levels are of comparable order, the depletion widths on both sides are also comparable. Hence, the narrower N-region reaches complete depletion earlier than the wider P-region.

The depletion width on the n-side is given by the formula:

$$x_n = \sqrt{\frac{2\epsilon_{si}\epsilon_o}{q}\left(\frac{N_A}{N_D}\right)\left(\frac{1}{N_A + N_D}\right)(\phi_{bi} + V_R)}$$

Where V_R is the magnitude of the reverse bias potential. Substituting the given values, we equate it to the total n-side width:

$$0.2 \times 10^{-4} = \sqrt{\frac{2 \times 12 \times 8.85 \times 10^{-14}}{1.6 \times 10^{-19}} \frac{5 \times 10^{16}}{10 \times 10^{16}} \frac{1}{(5 \times 10^{16} + 10 \times 10^{16})}(0.8 + V_R)}$$

Squaring both sides and simplifying:

$$4 \times 10^{-10} = \frac{21.24 \times 10^{-13}}{1.6 \times 10^{-19}} \frac{1}{2} \frac{1}{15 \times 10^{16}}(0.8 + V_R)$$

$$4 \times 10^{-10} = \frac{21.24 \times 10^{-13}}{48 \times 10^{-3}}(0.8 + V_R)$$

$$0.8 + V_R = 9.039$$

$$V_R = 9.039 - 0.8 = 8.239\,\text{V}$$

$$\boxed{8.239}$$

Key Points

- **Depletion Width Distribution:** The depletion region extends further into the lightly doped side. The relationship is $N_A x_p = N_D x_n$. However, since here n-side is much narrower than p-side, and the doping concentrations on both sides are of the same order, the n-region gets depleted first.

Mistakes to be avoided

- **Bias Sign:** For reverse bias, the potential term is $(\phi_{bi} + V_R)$. For forward bias, it would be $(\phi_{bi} - V_F)$.

Q2.20.1 MCQ 2020 1M Ans: B ● ● ○

In a forward-biased p-n junction, recombination occurs within the depletion region to return carrier concentrations to their equilibrium values. The recombination rate U_{max} is influenced by the carrier concentrations which depend exponentially on the applied forward bias V_a.

At equilibrium, the product of carrier concentrations is n_i^2. Under forward bias, this becomes:

$$np = n_i^2 e^{V_a/V_T}$$

Since $V_a > 0$, the carrier product exceeds n_i^2, leading to a recombination current I_{recomb}. Recombination occurs to bring carrier concentration to equilibrium state.

Regarding the statements provided in the analysis:

- As n_i increases, np increases, thereby increasing the recombination current I_{recomb}. Thus if n_i increases, U_{max} decreases. $\therefore$ A is TRUE.
- Statement B suggests that maximum recombination occurs at the edge of the depletion region; however, the highest electric field and the highest recombination rate U_{max} actually occur at the metallurgical junction ($x = 0$). $\therefore$ B is FALSE.
- The recombination current I_{recomb} depends exponentially on the applied voltage V_a. $\therefore$ C is TRUE.
- If temperature T increases, the thermal velocity and n_i both increase, which leads to an increase in the recombination current I_{recomb} and thus recombination rate U_{max}. $\therefore$ D is TRUE.

$$\boxed{(B)\ U_{max} \text{ occurs at the edges of the depletion region in the device.}}$$

Key Points

- Bulk traps or trap energy levels within the bandgap facilitate the recombination or generation of carriers.
- Recombination current in the depletion region typically follows an $e^{V_a/\eta V_T}$ dependency, where η is the ideality factor (often ≈ 2 for recombination).
- The recombination rate R is generally highest where the carrier concentrations and trap distributions are most favorable, typically near the junction center.

Mistakes to be avoided

- Do not assume the recombination rate is uniform throughout the depletion region; it varies with the local electric field and carrier profiles.

Q2.20.2 MCQ 2020 2M Ans: MTA ● ○ ○

The depletion (transition) capacitance C_D for a p-n junction is given by:

$$C_D = \frac{A\epsilon}{W}$$

For a one-sided p^+-n junction, the depletion width W is expressed as:

$$W = \sqrt{\frac{2\epsilon(V_{bi} - V)}{eN_D}}$$

where V is the applied potential from anode to cathode. Substituting W into the capacitance formula:

$$C_D = \frac{A\epsilon}{\sqrt{\frac{2\epsilon(V_{bi}-V)}{eN_D}}} \implies \frac{1}{C_D^2} = \frac{2}{A^2\epsilon e N_D}(V_{bi} - V)$$

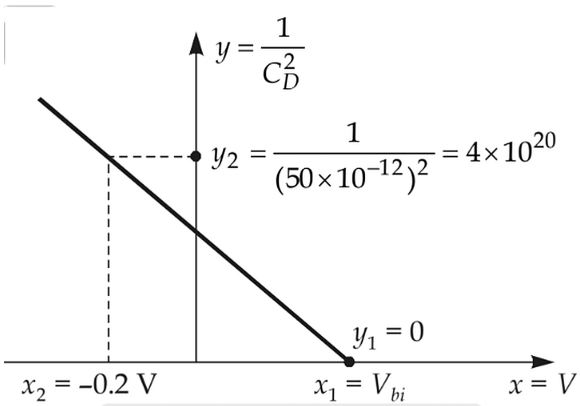

Fig. Solution Q2.20.2

From the provided graph of $y = \frac{1}{C_D^2}$ versus $x = V$:

- The value of $\frac{1}{C_D^2}$ becomes zero at the x-intercept, which corresponds to $V = V_{bi}$.
- At $x_1 = V_{bi}$, $y_1 = 0$.
- At $x_2 = -0.2$ V, the value is $y_2 = \frac{1}{(50\times10^{-12})^2} = 4 \times 10^{20}\,\text{F}^{-2}$.

The slope of the line is calculated as:

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 \times 10^{20} - 0}{-0.2 - V_{bi}}$$

Since the built-in potential V_{bi} is not explicitly provided in the problem statement, the slope and subsequent parameters cannot be uniquely determined. As per the GATE official answer key, this question was awarded **Marks to All (MTA)**.

Key Points

- **Depletion Capacitance:** C_D is inversely proportional to the square root of the reverse bias voltage.
- **C-V Characterization:** A plot of $1/C_D^2$ vs. V yields a straight line where the x-intercept represents the built-in potential V_{bi}.
- **Doping Determination:** The slope of the $1/C_D^2$ plot is inversely proportional to the doping concentration N_D.

Mistakes to be avoided

- Ensure the applied voltage V is treated correctly; for reverse bias, V is negative, making $(V_{bi} - V)$ positive.
- Do not confuse depletion capacitance (dominant in reverse bias) with diffusion capacitance (dominant in forward bias).

Q2.19.1 NAT | 2019 | 2M | Ans: 34 to 38 | ● ○ ○

The thermal voltage V_T at a temperature $T = 300\,\text{K}$ is calculated using the Boltzmann constant k and the electron charge q:

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.602 \times 10^{-19}}\,\text{V} = 25.843\,\text{mV}$$

The diode current I is given by the standard diode equation. For a reverse bias condition where the current is -75% of the reverse saturation current I_0:

$$I = I_0\left(e^{V/V_T} - 1\right) = -\frac{3}{4}I_0$$

Simplifying the equation to solve for the voltage V:

$$e^{V/V_T} - 1 = -0.75$$

$$e^{V/V_T} = 0.25$$

$$V = V_T \ln(0.25) = 25.843\,\text{mV} \times (-1.386) = -35.83\,\text{mV}$$

The magnitude of the reverse bias potential V_R is:

$$V_R = |V| = 35.83\,\text{mV}$$

$$\boxed{35.83}$$

Key Points

- In reverse bias, V is negative, resulting in a current I that approaches $-I_0$.

Mistakes to be avoided

- The question asks for the magnitude of the reverse bias; the final answer should be positive.

Q2.19.2 MCQ | 2019 | 1M | Ans: A | ● ○ ○

At equilibrium, the Fermi level must remain flat throughout the entire pnn^+p^{++} structure. Option A is the only diagram where E_F is constant while the conduction and valence bands bend according to the doping changes.

Also, the bands move closer to E_F in the heavily doped n^+ and p^{++} regions, which correctly reflects stronger donor/acceptor doping and the built-in potential variations.

$$\boxed{A}$$

Q2.18.1 NAT | 2018 | 2M | Ans: 0.11 to 0.13 | ● ○ ○

The built-in potential V_{bi} for a semiconductor junction with varying dopant concentrations is determined by the ratio of the acceptor concentrations N_{A2} and N_{A1}.

The formula for the built-in potential is:

$$V_{bi} = \frac{kT}{q}\ln\left(\frac{N_{A2}}{N_{A1}}\right)$$

Given the numerical values from the calculation:

$$V_{bi} = \frac{1.38 \times 3}{1.6 \times 100}\ln(100)\,\text{V} = 0.1192\,\text{V} \simeq 0.12\,\text{V}$$

$$\boxed{0.12}$$

Key Points

- The built-in potential arises due to the diffusion of carriers caused by the concentration gradient between N_{A2} and N_{A1}.

Q2.18.2 NAT 2018 1M Ans: 0.4 to 0.43 ● ○ ○

The depletion width W_{dep} of a p-n junction is given by the following expression:

$$W_{dep} = \sqrt{\frac{2\epsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_{bi} - V_{FB})}$$

Given that the built-in potential $V_{bi} = 0.65\,\text{V}$, and the depletion width changes from $1\,\mu\text{m}$ to $0.6\,\mu\text{m}$ upon applying a forward bias V_{FB}, we can establish a ratio:

$$\frac{\sqrt{0.65 - V_{FB}}}{\sqrt{0.65}} = \frac{0.6\,\mu\text{m}}{1\,\mu\text{m}} = 0.6$$

Squaring both sides to solve for V_{FB}:

$$\frac{0.65 - V_{FB}}{0.65} = 0.36$$

$$V_{FB} = 0.65(1 - 0.36) = 0.65 \times 0.64 = 0.416\,\text{V} \simeq 0.42\,\text{V}$$

$$\boxed{0.42}$$

Key Points

- **Depletion Width Relation:** $W_{dep} \propto \sqrt{V_{bi} - V_{applied}}$.

Q2.18.3 MCQ 2018 1M Ans: D ● ○ ○

In a p-n junction as shown in the diagram:

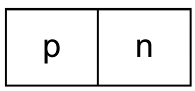

Fig. Solution Q2.18.3

- **Diffusion:** Holes move from the p-side (high concentration) to the n-side ($\longrightarrow$), and electrons move from the n-side to the p-side ($\longleftarrow$).
- **Drift:** The built-in electric field ($\longleftarrow$) causes minority holes to drift from the n-side to the p-side ($\longleftarrow$) and minority electrons to drift from the p-side to the n-side ($\longrightarrow$).
- **Current Directions:**
  - Hole diffusion current flows in the direction of hole movement ($\longrightarrow$).
  - Electron diffusion current flows opposite to electron movement ($\longrightarrow$).
  - Hole drift current flows in the direction of hole movement ($\longleftarrow$).
  - Electron drift current flows opposite to electron movement ($\longleftarrow$).

$$\boxed{(D)\ \text{On an average, electrons drift and diffuse in the same direction.}}$$

Mistakes to be avoided

- Do not confuse particle flow direction with current flow direction for electrons; current always flows opposite to the direction of electron motion.
- Note that the question asks for the FALSE option, not the TRUE one.

Q2.17.1

MCQ | 2017 | 2M | Ans: A | ● ○ ○

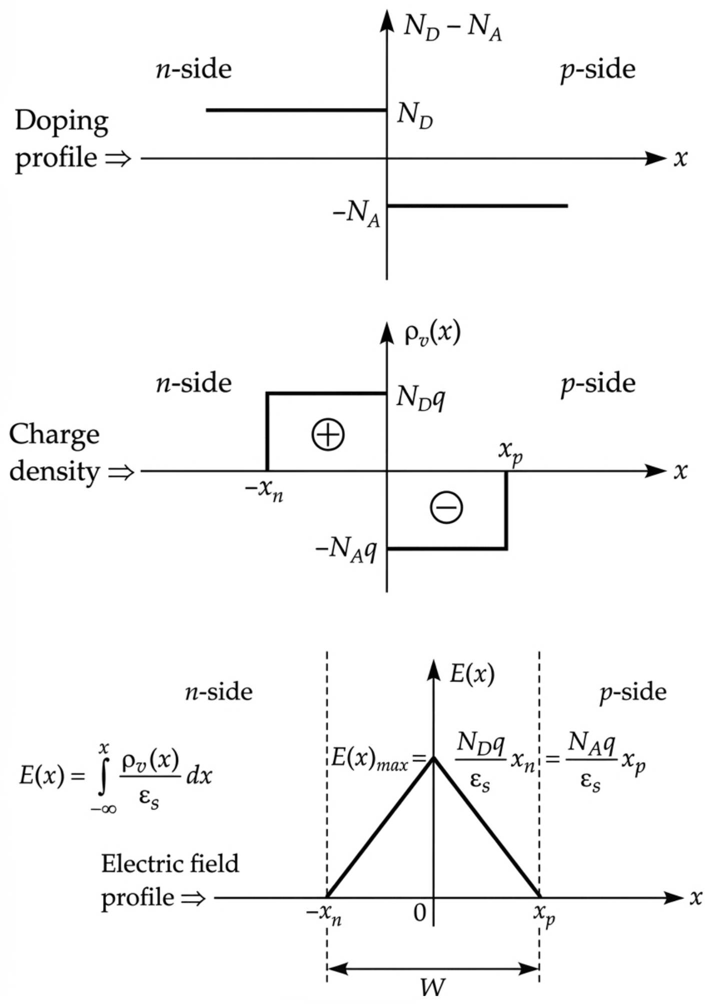

Fig. Solution Q2.17.1

Q2.17.2

NAT | 2017 | 2M | Ans: 10 | ● ○ ○

The depletion width W for a p-n junction under reverse bias is given by:

$$W = \sqrt{\frac{2\epsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_{bi} + V_R)}$$

where V_R is the magnitude of the applied reverse bias potential.
Since $V_R \gg V_{bi}$ is given, the formula may be written as:

$$W = \sqrt{\frac{2\epsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_R}$$

Under reverse biased conditions, the junction capacitance C is related to the depletion width W as:

$$C = \frac{\epsilon A}{W} \implies C \propto \frac{1}{W}$$

The ratio of two capacitances C_2 and C_1 can be expressed in terms of the corresponding depletion widths W_1 and W_2:

$$\frac{C_2}{C_1} = \frac{W_1}{W_2}$$

Substituting the expression for W with different doping concentrations (assuming $N_A = N_D$ for simplicity in the ratio setup shown in the figure):

$$\frac{C_2}{C_1} = \sqrt{\frac{\frac{2\epsilon V_R}{e}\left(\frac{2}{10^{14}}\right)}{\frac{2\epsilon V_R}{e}\left(\frac{2}{10^{16}}\right)}} = \sqrt{\frac{10^{16}}{10^{14}}} = \sqrt{100} = 10$$

$$\boxed{10}$$

Mistakes to be avoided

- The formula used assumes $V_R \gg V_{bi}$; if the applied voltage is comparable to the built-in potential, V_{bi} must be included in the square root.

Q2.17.3 MCQ 2017 1M Ans: D ● ○ ○

The built-in potential V_0 (or contact potential) for a semiconductor junction with varying doping concentrations is given by:

$$V_0 = V_T \ln \frac{n_1}{n_2} = V_T \ln \frac{N_{D1}}{N_{D2}}$$

Substituting the given values:

$$V_0 = 0.025 \ln \frac{10^{18}}{10^{15}} = 0.025 \times 6.907 \approx 0.173\,\text{V}$$

$$\boxed{(D)\, 0.173\, V}$$

Q2.16.1 MCQ 2016 1M Ans: C ● ○ ○

A larger bandgap semiconductor has a lower intrinsic carrier concentration, so the diode needs a higher forward voltage to conduct significantly.

Since the turn-on voltage shifts right from $X \rightarrow Y \rightarrow Z$, the bandgaps increase in the same order:

$$E_{gX} < E_{gY} < E_{gZ}$$

$$\boxed{(C)\, E_{gX} < E_{gY} < E_{gZ}}$$

Q2.16.2 NAT 2016 2M Ans: 30 to 34 ● ○ ○

The magnitude of the electric field $|\mathcal{E}|$ in the depletion region of a p-n junction is related to the doping concentration and the depletion width. For the n-side, the relationship is given by:

$$|\mathcal{E}| = \frac{q}{\varepsilon} N_D X_N$$

The calculation is as follows:

$$50 \times 10^3 = \frac{1.6 \times 10^{-19}}{8.85 \times 10^{-14} \times 11.7} \times 10^{17} \times X_N$$

On solving for X_N:

$$X_N = 3.2356 \times 10^{-6}\,\text{cm} = 32.36\,\text{nm}$$

$$\boxed{32.36}$$

Key Points

- **Electric Field in Depletion Region**: The electric field is maximum at the metallurgical junction and varies linearly toward the edges of the depletion region.
- **Charge Neutrality**: In the depletion region, the total negative charge on the p-side must equal the total positive charge on the n-side ($qN_A x_P = qN_D x_N$).

Q2.16.3 MCQ 2016 2M Ans: C ● ○ ○

In any type of P-N junction, the breakdown voltage V_{BR} is inversely proportional to the sum of inverse of doping concentrations.

$$V_{BR} \propto \frac{1}{N_D} + \frac{1}{N_A}$$

For a junction where the n-side is the lightly doped region with concentration N_D, the relationship can be approximated as:

$$V_{BR} \propto \frac{1}{N_D}$$

More specifically, the breakdown voltage can be expressed in terms of the critical electric field E_{crit}:

$$V_{BR} = \frac{\epsilon E^2}{2qN_D}$$

From this expression, it is evident that for a given semiconductor material and critical field:

$$V_{BR} \times N_D = \text{constant}$$

Thus, if the doping concentration is increased, the breakdown voltage decreases proportionally to maintain this constant relationship.

$$\boxed{(C)\; N_D \times V_{BR} = constant}$$

Key Points

- **Breakdown Voltage Dependency:** V_{BR} is determined by the doping level of the more lightly doped side of the junction.
- **Impact of Doping:** Increasing the doping concentration N_D reduces the depletion width, which increases the electric field for a given voltage, leading to a lower breakdown voltage.

Q2.16.4 NAT 2016 2M Ans: -5 to -4.6

To find the charge per unit junction area in the depletion layer on the p-side, we first determine the built-in potential V_{bi} and the total depletion width W.

The built-in potential V_{bi} is calculated as:

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$V_{bi} = 26 \times 10^{-3} \ln\left[\frac{10^{16} \times 10^{17}}{(1.5 \times 10^{10})^2}\right] = 0.757\,\text{V}$$

The total depletion width W is:

$$W = \sqrt{\frac{2\varepsilon}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right) V_{bi}} = 3.3255\,\mu\text{m}$$

The depletion width on the p-side W_p is:

$$W_p = \frac{W N_D}{N_A + N_D} = \frac{3.3255 \times 10^{-6} \times 10^{17}}{10^{16} + 10^{17}} = 0.3023\,\mu\text{m}$$

The charge per unit junction area on the p-side is:

$$Q_p = -qN_A W_p$$

$$Q_p = -(1.6 \times 10^{-19}) \times 10^{17} \times (0.3023 \times 10^{-6}) = -4.8368\,\text{nC/cm}^2$$

Rounding to two decimal places:

$$\boxed{-4.84}$$

Q2.15.1 MCQ 2015 2M Ans: C

- On the left side, the slope is positive, meaning positive space charge → ionized donor ions → n-type region. On the right side, the slope is negative, meaning negative space charge → ionized acceptor ions → p-type region. A is TRUE, and thus incorrect.

- Slope of Electric field graph is constant. This implies uniform doping on both n and p regions. B is TRUE, thus incorrect.

- The built-in potential (V_{bi}) for a p-n junction can be calculated from the area under the electric field profile. In a linear approximation, the relationship is given by:

  $$V_{bi} = \frac{1}{2} \times E_{max} \times W$$

  Substituting the given values:

  $$V_{bi} = \frac{1}{2} \times (10^6\,\text{V/m}) \times (1.1 \times 10^{-6}\,\text{m}) = 550\,\text{mV}$$

  .

  The value of V_{bi} mentioned in C is 700 mV. Thus C is FALSE and correct.

- Charge neutrality gives $N_A x_p = N_D x_n$

  Given that $\frac{x_p}{x_n} = 10$, thus $\frac{N_D}{N_A}$ should also be 10 , which is satisfied by values in option D. So option D is TRUE, thus incorrect.

$$\boxed{(C)\ \text{The potential difference across the depletion region is 700 mV}}$$

Key Points

- The built-in potential V_{bi} is the integral of the electric field across the depletion region.

Q2.15.2 NAT 2015 2M Ans: 28 to 30

The hole current density (J_p) injected from the P-region to the N-region in a p-n junction diode under forward bias is given by:

$$J_p = \frac{qn_i^2 D_p}{N_d L_p}\left[\exp\left(\frac{V_{fb}}{\eta V_T}\right) - 1\right]$$

The diffusion length L_p is calculated as:

$$L_p = \sqrt{\tau_p D_p} = \sqrt{100 \times 10^{-6} \times 36} = 0.06\,\text{cm}$$

Substituting the known values into the current density equation :

$$J_p = \frac{1.6 \times 10^{-19} \times (10^{10})^2 \times 36}{10^{15} \times 0.06}\left[\exp\left(\frac{0.208}{0.026}\right) - 1\right] = 28.6076\ \text{nA/cm}^2$$

$$\boxed{28.6076}$$

Key Points

- The diffusion length L_p represents the average distance a hole travels before recombining in the N-region.

Q2.15.3 NAT 2015 2M Ans: 2.4 to 2.6

The junction capacitance (C_j) of a p-n junction diode is inversely proportional to the square root of the total potential across the junction (for an abrupt junction). The relationship is given by:

$$C_j \propto \frac{1}{\sqrt{V_{bi} + V_R}}$$

where V_{bi} is the built-in potential, and V_R is the applied reverse bias voltage. Taking the ratio of capacitance at two different reverse bias voltages V_{R1} and V_{R2}:

$$\frac{C_{2j}}{C_{1j}} = \sqrt{\frac{V_{bi} + V_{R1}}{V_{bi} + V_{R2}}}$$

Given the values from the problem context, where the ratio of the total potentials simplifies to $\frac{2}{8}$:

$$C_{2j} = C_{1j}\sqrt{\frac{2}{8}} = \frac{C_{1j}}{2}$$

If the initial capacitance C_{1j} is 5 pF:

$$C_{2j} = \frac{5\,\text{pF}}{2} = 2.5\,\text{pF}$$

$$\boxed{2.5}$$

Key Points

- Junction capacitance (also called depletion capacitance) dominates in reverse bias.

Mistakes to be avoided

- Do not forget to include the built-in potential (V_{bi}) in the denominator; the capacitance is not simply proportional to $1/\sqrt{V_R}$.

Q2.15.4 MCQ 2015 1M Ans: A ● ○ ○

Negative differential resistance occurs in a tunnel diode, where both P and N regions are very heavily doped, making the depletion region extremely thin. This allows quantum tunneling of carriers, causing current to decrease with increasing voltage over a certain region.

$$\boxed{(A)\text{ both the P-region and the N-region are heavily doped}}$$

Q2.14.1 MCQ 2014 2M Ans: D ● ○ ○

The built-in potential (V_{bi}) of a p-n junction is determined by the doping concentrations of the p and n sides and the intrinsic carrier concentration of the material.

$$V_{bi} = V_T \ln\left[\frac{N_a N_d}{n_i^2}\right]$$

Substituting the given values:

$$V_{bi} = 0.026 \ln\left[\frac{10^{16} \times 5 \times 10^{18}}{2.25 \times 10^{20}}\right] = 0.86\text{ V}$$

The depletion width (W) at equilibrium is given by:

$$W = \sqrt{\frac{2\epsilon}{q}\left[\frac{1}{N_a} + \frac{1}{N_d}\right] V_{bi}}$$

$$W = \sqrt{\frac{2 \times 1.04 \times 10^{-12}}{1.6 \times 10^{-19}}\left[\frac{1}{5 \times 10^{18}} + \frac{1}{10^{16}}\right] \times 0.86}$$

$$W = 3.3 \times 10^{-5}\text{ cm}$$

$$\boxed{(D)\, 0.86\, V \text{ and } 3.3 \times 10^{-5} cm}$$

Key Points

- In a one-sided junction ($N_d \gg N_a$), the formula simplifies to $W \approx \sqrt{\frac{2\epsilon V_{bi}}{q N_a}}$.

Q2.14.2 NAT 2014 2M Ans: -32 to -30 ● ○ ○

For an abrupt P^+N junction, the doping concentration on the p-side (N_a) is much higher than the doping concentration on the n-side (N_d). Consequently, the depletion region extends almost entirely into the n-side.

$$N_a \gg N_d \implies W \simeq x_n$$

The maximum electric field (E_{max}) occurs at the junction ($x = 0$) and is given by:

$$E_{max} = \frac{-qN_d x_n}{\epsilon_s}$$

Substituting the given values into the equation:

$$E_{max} = \frac{-1.6 \times 10^{-19} \times 10^{16} \times 0.2 \times 10^{-4}}{1.044 \times 10^{-12}} = -30.65 \times 10^{3} \text{ V/cm}$$

The peak value of the electric field is approximately -30.65 kV/cm.

$$\boxed{-30.65}$$

Key Points

- The electric field is maximum at the metallurgical junction and decreases linearly to zero at the edges of the depletion region.
- The negative sign indicates that the electric field vector points from the n-side (positive space charge) to the p-side (negative space charge).

Q2.14.3 MCQ 2014 2M Ans: B ● ● ○

We know that the charge neutrality condition in a p-n junction relates the doping concentrations to the depletion widths:

$$\frac{N_A}{N_D} = \frac{x_n}{x_p}$$

Adding 1 to both sides to relate the total depletion width ($W = x_n + x_p$):

$$\frac{N_A}{N_D} + 1 = \frac{x_n}{x_p} + 1 \implies \frac{N_A + N_D}{N_D} = \frac{x_n + x_p}{x_p}$$

Substituting the given values ($N_A + N_D \approx 10^{17}$ cm^{-3}, $N_D = 10^{16}$ cm^{-3}, and $W = 3\,\mu$m):

$$\frac{10^{17}}{10^{16}} = \frac{3\,\mu\text{m}}{x_p} \implies x_p = 0.3\,\mu\text{m}$$

The maximum electric field ($E_{\max}$) at the junction is given by:

$$E_{\max} = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}$$

Substituting the numerical values:

$$E_{\max} = \frac{1.6 \times 10^{-19} \times 9 \times 10^{16} \times 0.3 \times 10^{-4}}{1.04 \times 10^{-12}}$$

$$E_{\max} = 4.15 \times 10^{5} \text{ V/cm}$$

$$\boxed{(B)\, 0.3\mu m \text{ and } 2.3 * 10^5 V/cm}$$

Key Points

- Charge neutrality requires $N_A x_p = N_D x_n$.
- The electric field is maximum at the metallurgical junction ($x = 0$).

Q2.13.1 MCQ 2013 1M Ans: A ● ○ ○

In a forward-biased p-n junction diode, the applied external voltage opposes the built-in potential, reducing the height of the potential barrier. This allows majority carriers (electrons from the n-side and holes from the p-side) to diffuse across the junction.

Once these majority carriers cross the junction, they become minority carriers in the opposite region. For example, electrons entering the p-region are minority carriers that eventually recombine with the abundant holes there. The continuous process of diffusion of majority carriers and subsequent recombination maintains the forward current flow.

$\boxed{(A) \text{ injection, and subsequent diffusion and recombination of minority carriers}}$

Key Points

- Forward bias narrows the depletion region and lowers the contact potential.
- Current in a p-n junction is composed of both hole and electron components.
- The diffusion current dominates over the drift current under forward bias conditions.

Mistakes to be avoided

- Do not confuse this with reverse bias, where the current is primarily due to the drift of minority carriers (leakage current).
- Remember that majority carriers "diffuse" because they move from a high concentration area to a low concentration area across the junction.

Q2.11.1 MCQ 2011 1M Ans: B ● ○ ○

In the forward bias region, the diode conducts once the barrier potential is overcome. In the reverse bias region, the diode remains non-conductive (behaving like an open circuit) until the reverse voltage reaches the breakdown threshold.

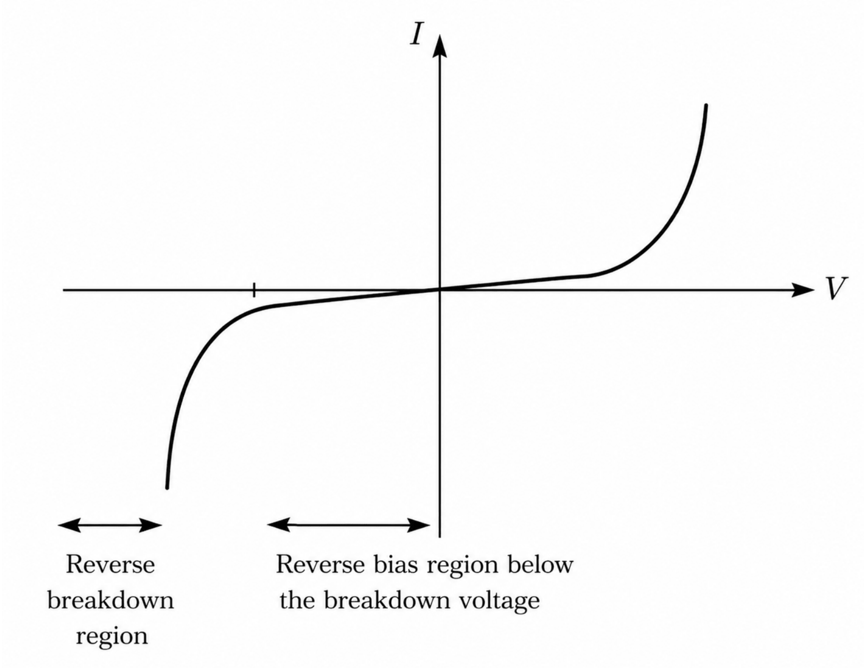

Fig. Solution Q2.11.1

- **Reverse Bias Region (Below Breakdown):** Here, the current is negligible (leakage current), effectively behaving as an **open circuit**.

- **Breakdown Region:** Beyond the breakdown voltage , the current increases sharply. This is the region where Zener diodes are specifically designed to operate for voltage regulation.

$$\boxed{(B) \text{ reverse breakdown region}}$$

Mistakes to be avoided

- Ensure you distinguish between the "Reverse Bias Region" (low current) and the "Breakdown Region" (high current).

Q2.11.2 MCQ 2011 1M Ans: D ● ● ○

The temperature dependence of the forward voltage (V) across a p-n junction diode under a constant forward current is characterized by a negative temperature coefficient.

The rate of change of voltage with respect to temperature is given by:

$$\frac{dV}{dT} = -2.5\,\text{mV/°C}$$

For a temperature increase of $\Delta T = 10\,°\text{C}$, the change in forward voltage (ΔV) can be calculated as:

$$\Delta V = \frac{dV}{dT} \times \Delta T$$

$$\Delta V = -2.5\,\text{mV/°C} \times 10\,°\text{C} = -25\,\text{mV}$$

The negative sign indicates that the voltage decreases. Therefore, the voltage across the p-n junction will decrease by 25 mV.

$$\boxed{(D) \text{ decreases by 25 mV}}$$

Key Points

- The forward voltage of a silicon diode decreases by approximately 2 mV to 2.5 mV for every 1 °C increase in temperature.
- This negative temperature coefficient occurs because the intrinsic carrier concentration n_i increases exponentially with temperature, which significantly increases the reverse saturation current I_s and consequently lowers the required forward bias voltage for a given current.

Q2.10.1 MCQ 2010 1M Ans: C ● ○ ○

The physical and electrical characteristics of a p-n junction are heavily influenced by the doping concentration.

- **Depletion Width (d):** The width of the depletion region is inversely proportional to the square root of the doping concentration ($d \propto 1/\sqrt{N}$). Therefore, as doping increases, the depletion width d **decreases**.
- **Depletion Capacitance (C):** The junction behaves like a parallel plate capacitor where $C = \epsilon A/d$. Since d decreases with higher doping, the capacitance C **increases**.
- **Breakdown Voltage:** Increased doping leads to a narrower depletion region, which increases the electric field intensity for the same applied voltage. This results in a **decrease** in the reverse breakdown voltage (often transitioning from Avalanche to Zener breakdown).

$$\boxed{(C) \text{ Reverse breakdown voltage is lower and depletion capacitance is higher}}$$

Key Points

- Highly doped junctions (Zener diodes) have very thin depletion layers, facilitating tunneling.
- Junction capacitance is also known as transition or space-charge capacitance.

Q2.09.1

MCQ | 2009 | 1M | Ans: B | ● ○ ○

The electric field $E(x)$ in the depletion region of a p-n junction is derived from Poisson's equation. For a step junction, the electric field varies linearly with distance.

The expression for the electric field on the n-side is given by:

$$E = \frac{qN_d}{\epsilon}(x - W_n)$$

The maximum electric field (E_{max}) occurs at the metallurgical junction ($x = 0$):

$$E_{max} = \frac{qN_d}{\epsilon}(-W_n)$$

Substituting the given values:

$$E_{max} = \frac{-1.6 \times 10^{-19} \times 10^{17} \times 10^{-5}}{12 \times 8.85 \times 10^{-14}} \approx -0.15\,\text{MV/cm}$$

Also, direction of electric field is from n-region to p-region is due to the positive donor ions in the n-side of the depletion region and the negative acceptor ions in the p-side of the acceptor region.

$$\boxed{(B)\ 0.15\ \text{MVcm}^{-1}\text{, directed from n-region to p-region}}$$

Q2.09.2

MCQ | 2009 | 1M | Ans: B | ● ○ ○

The depletion width on the n-side (W_n) of a p-n junction is related to the junction potential (V_j) and the donor doping concentration (N_d). In cases where the n-side is heavily doped, the depletion region is very narrow.

The relationship is given by the formula:

$$W_n = \left(\frac{2\epsilon V_j}{eN_d}\right)^{1/2}$$

Rearranging this expression to solve for the junction potential (V_j):

$$V_j = \frac{eN_d W_n^2}{2\epsilon_0 \epsilon_r}$$

Based on the provided parameters for the calculation:

$$V_j \approx 0.76\,\text{V}$$

$$\boxed{(B)\ \text{is } 0.76\ \text{V}}$$

Q2.08.1 MCQ | 2008 | 1M | Ans: A | ● ○ ○

Zener breakdown occurs due to quantum tunneling across a very thin depletion region, which requires heavy doping and hence an abrupt junction. But S2 is incorrectly worded: tunneling requires a narrow potential barrier/depletion region, not a "narrow energy barrier," so only S1 is true.

$$\boxed{(A)\text{ Only S1 is true}}$$

Q2.08.2 MCQ | 2008 | 1M | Ans: D | ● ○ ○

Channel Length Modulation occurs in MOSFETs, not in P-N junction diodes.

$$\boxed{(D)\text{ Channel Length Modulation}}$$

Q2.07.1 MCQ | 2007 | 1M | Ans: A | ● ○ ○

The depletion width (W) of an abrupt p-n junction depends on the junction potential (V_j), which is the sum of the built-in potential (V_0) and the applied reverse bias voltage (V_R). For an abrupt p-n junction, the depletion width is proportional to the square root of the total junction potential:

$$W \propto V_j^{1/2} \implies W = KV_j^{1/2}$$

Where K is a proportionality constant.

We can establish two equations based on the given operating conditions:

- **Condition 1:** At $V_{R1} = 1.2\,\text{V}$ with $V_0 = 0.8\,\text{V}$, the depletion width is $W_1 = 2\,\mu\text{m}$.

$$2\,\mu\text{m} = K(0.8 + 1.2)^{1/2} = K(2.0)^{1/2} \quad \text{— (i)}$$

- **Condition 2:** At $V_{R2} = 7.2\,\text{V}$ with $V_0 = 0.8\,\text{V}$, the new depletion width is $W_2 = x$.

$$x = K(0.8 + 7.2)^{1/2} = K(8.0)^{1/2} \quad \text{— (ii)}$$

Dividing equation (ii) by equation (i):

$$\frac{x}{2\,\mu\text{m}} = \frac{K(8.0)^{1/2}}{K(2.0)^{1/2}}$$

$$\frac{x}{2\,\mu\text{m}} = \sqrt{\frac{8.0}{2.0}} = \sqrt{4} = 2$$

$$x = 2 \times 2\,\mu\text{m} = 4\,\mu\text{m}$$

$$\boxed{(A)\,4\,\mu m}$$

Mistakes to be avoided

- Ensure that you apply the correct power law exponent (1/2 for abrupt junctions vs. 1/3 for linearly graded junctions).

Q2.07.2 MCQ | 2007 | 1M | Ans: C | ● ○ ○

Electrical field is always maximum at the junction.

$$\boxed{(C)\ \text{the } p^{+}n \text{ junction}}$$

Q2.05.1 MCQ | 2005 | 1M | Ans: B | ● ○ ○

The capacitance per unit area ($\frac{C}{A}$) of a parallel plate capacitor is given by the formula:

$$\frac{C}{A} = \frac{\epsilon_0 \epsilon_r}{d}$$

Substituting the given values into the formula:

$$\frac{C}{A} = \frac{8.85 \times 10^{-12} \times 11.7}{10 \times 10^{-6}}$$

$$\frac{C}{A} = 1.03545 \times 10^{-5}\ \text{F/m}^2 \approx 10\mu\text{F/m}^2$$

$$\boxed{(B)\ 10\,\mu F}$$

Q2.05.2 MCQ | 2005 | 1M | Ans: B | ● ○ ○

The reverse saturation current (I_Q) of a semiconductor diode or transistor depends strongly on temperature. It approximately doubles for every 10 °C rise in temperature.

The relation between the reverse saturation currents at two different temperatures T_1 and T_2 is given by:

$$I_{QT2} = I_{QT_1} \times 2^{\frac{T_2 - T_1}{10}}$$

Substituting the temperature values into the exponent:

$$\frac{T_2 - T_1}{10} = \frac{40 - 20}{10} = 2$$

Therefore, the equation becomes:

$$I_{q40\,°\text{C}} = I_{q20\,°\text{C}} \times 2^2$$

Given that the calculated or final current at 40 °C is:

$$I_{q40\,°\text{C}} = 40\,\text{pA}$$

$$\boxed{(B)\ 40\,pA}$$

Key Points

- A well-known rule of thumb in semiconductor physics is that I_0 doubles for every 10 °C increase in temperature:

$$I_0(T_2) = I_0(T_1)2^{\Delta T/10}$$

Q2.04.1 MCQ 2004 1M Ans: D ● ○ ○

The junction capacitance (C_j) of a linearly graded or uniformly doped p-n junction under reverse bias is inversely proportional to the reverse voltage. For a step (abrupt) junction, the junction capacitance is given by:

$$C_j \propto V_i^{-1/2}$$

Where V_i is the total reverse bias potential (including the built-in potential).

Taking the ratio of the junction capacitance at two different voltage levels V_1 and V_2:

$$\frac{C_{j2}}{C_{j1}} = \sqrt{\frac{V_1}{V_2}}$$

Substituting the voltage ratio into the equation:

$$\frac{C_{j2}}{C_{j1}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

Solving for the final junction capacitance C_{j2}:

$$C_{j2} = \frac{C_{j1}}{2} = \frac{1\,\text{pF}}{2} = 0.5\,\text{pF}$$

$$\boxed{(D)\,0.5\,\text{pF}}$$

Key Points

- The depletion or junction capacitance C_j arises due to the charge storage variation in the depletion region under reverse bias conditions.
- Generally, the relationship is expressed as $C_j = \frac{C_{j0}}{\left(1+\frac{V_R}{V_{bi}}\right)^m}$, where $m = 1/2$ for an abrupt junction and $m = 1/3$ for a linearly graded junction.

Q2.04.2 MCQ 2004 1M Ans: B ● ○ ○

In a p-n junction, charge neutrality must be maintained across the depletion region. Therefore, the total uncompensated positive charge on the n-side must equal the total uncompensated negative charge on the p-side:

$$qN_D w_n = qN_A w_p$$

Rearranging the charge neutrality equation gives the ratio of the depletion widths:

$$\frac{w_n}{w_p} = \frac{N_A}{N_D}$$

Given that the total depletion width $w \simeq w_n$ because $w_n \gg w_p$, we can solve for w_p:

$$w_p = \frac{w_n \times N_D}{N_A}$$

Substituting the given values into the equation:

$$w_p = \frac{3\,\mu\text{m} \times 10^{16}}{9 \times 10^{16}} = \frac{3}{9}\,\mu\text{m} = 0.33\,\mu\text{m} \simeq 0.3\,\mu\text{m}$$

Thus, the correct option is B.

$$\boxed{(B)\,0.3\,\mu\text{m}}$$

Key Points

- **Charge Neutrality Condition:** The depletion layer widths are inversely proportional to the doping concentrations on respective sides ($N_A w_p = N_D w_n$).

Mistakes to be avoided

- Avoid confusing the inverse relationship; the lightly doped side always gets the larger depletion width.

Q2.03.1 MCQ 2003 1M Ans: C ● ○ ○

The diode current equation is expressed as:

$$I = I_0 \left[exp\left(\frac{V_D}{\eta V_T}\right) - 1 \right]$$

η is the ideality factor (where $\eta = 1$ for Germanium and $\eta = 2$ for Silicon).
For the Silicon diode:

$$I = I_{O\,\text{Si}} \left[exp\left(\frac{V_{D1}}{2V_T}\right) - 1 \right] \text{— (i)}$$

For the Germanium diode:

$$I = I_{O\,\text{Ge}} \left[exp\left(\frac{V_{D2}}{V_T}\right) - 1 \right] \text{— (ii)}$$

Equating equation (i) and equation (ii):

$$I_{O\,\text{Si}} \left[exp\left(\frac{V_{D1}}{2V_T}\right) - 1 \right] = I_{O\,\text{Ge}} \left[exp\left(\frac{V_{D2}}{V_T}\right) - 1 \right]$$

Rearranging the terms to find the ratio of the reverse saturation currents $\frac{I_{O\,\text{Ge}}}{I_{O\,\text{Si}}}$:

$$\frac{I_{O\,\text{Ge}}}{I_{O\,\text{Si}}} = \frac{exp\left(\frac{V_{D1}}{2V_T}\right) - 1}{exp\left(\frac{V_{D2}}{V_T}\right) - 1}$$

Substituting the given values into the ratio expression and evaluating:

$$\frac{I_{O\,\text{Ge}}}{I_{O\,\text{Si}}} = \frac{992257}{248.44} \approx 3993.95 \approx 4 \times 10^3$$

Hence, the final simplified ratio is:

$$\boxed{(C)\, 4 \times 10^3}$$

Key Points

- **Ideality Factor (η):** Accounts for carrier recombination in the depletion region. It typically equals 1 for Ge and varies from 1 to 2 for Si depending on the current density levels.
- **Reverse Saturation Current Relationship:** Due to differences in bandgap energy (E_g), the reverse saturation current of Germanium is several orders of magnitude higher than that of Silicon ($I_{O\,\text{Ge}} \gg I_{O\,\text{Si}}$).

Mistakes to be avoided

- Do not neglect the -1 term in the diode equation unless the exponents are confirmed to be extremely large ($e^{V_D/\eta V_T} \gg 1$). In this case, keeping it ensures absolute calculation precision.

Q2.02.1 MCQ 2002 1M Ans: B ● ○ ○

The forward voltage drop (V_D) across a p-n junction diode decreases linearly with an increase in temperature if the forward current I is maintained constant.

For either Silicon (Si) or Germanium (Ge), the temperature coefficient of the forward voltage drop is approximately:

$$\frac{dV_D}{dT} \simeq -2\,\text{mV/°C}$$

Calculating the change in temperature (ΔT):

$$\Delta T = T_2 - T_1 = 40\,°\text{C} - 20\,°\text{C} = 20\,°\text{C}$$

The change in forward voltage (ΔV_D) due to this temperature rise is:

$$\Delta V_D = \frac{dV_D}{dT} \times \Delta T = -2\,\text{mV/°C} \times 20\,°\text{C} = -40\,\text{mV}$$

Therefore, the new forward voltage drop V_{D2} at 40 °C is:

$$V_{D2} = V_{D1} + \Delta V_D = 700\,\text{mV} - 40\,\text{mV} = 660\,\text{mV}$$

$$\boxed{(B)\ 660\ \text{mV}}$$

Mistakes to be avoided

- Avoid adding the voltage drop change instead of subtracting it; remember that an increase in temperature reduces the required forward voltage drop for a constant current.

Q2.98.1 MCQ 1998 2M Ans: B ● ● ○

Initially, the state of the circuit dictates that the diode behaves under a specific biasing condition.

The diode is forward biased in the starting period.

However, precisely at the switching instant $t = 0$, the transient conditions or the specific input voltage level ensures that the diode is reverse biased.

Consequently, evaluating the circuit loop equation under this state yields the transient current:

$$I = -100\,\text{mA}$$

Therefore, the correct option is B.

$$\boxed{(B)\ \text{-100 mA}}$$

Key Points

- **Diode Switching Transients:** When a diode transitions from a forward-biased state to a reverse-biased state, it does not turn off instantly due to stored minority carrier charges in the depletion and neutral regions.
- **Reverse Recovery Phase:** At $t = 0$, during the reverse recovery time (t_{rr}), a transient reverse current flows through the device. The magnitude of this current is typically limited by the external circuit loop resistance.

Mistakes to be avoided

- Do not assume the diode current instantly drops to zero at $t = 0$. The stored spatial charges must be swept out first, resulting in a negative transient current.

Q2.98.2 MCQ | 1998 | 1M | Ans: B | ● ○ ○

The ideal diode current equation under forward bias conditions is given by:

$$I = I_0 \left(e^{\frac{V_d}{\eta V_T}} - 1 \right)$$

where
* I is the forward diode current.
* I_0 is the reverse saturation current.
* V_d is the forward voltage drop across the diode.
* V_T is the thermal voltage.
* η is the ideality factor.

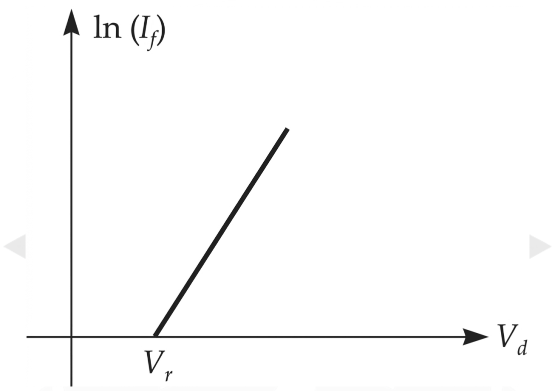

Fig. Solution Q2.98.2

For a typical forward bias operating region, the exponential term is significantly greater than unity ($e^{\frac{V_d}{\eta V_T}} \gg 1$). Thus, the equation simplifies to:

$$I \approx I_0 e^{\frac{V_d}{\eta V_T}}$$

Taking the natural logarithm (ln) on both sides to study the relationship between the log-current and voltage:

$$\ln(I) = \ln\left(I_0 e^{\frac{V_d}{\eta V_T}} \right)$$

$$\ln(I) = \ln(I_0) + \frac{V_d}{\eta V_T}$$

Rearranging the terms into the slope-intercept form of a straight line equation ($y = mx + c$):

$$\frac{V_d}{\eta V_T} = \ln(I) - \ln(I_0)$$

$$V_d = \eta V_T \ln(I) - \eta V_T \ln(I_0)$$

Comparing this with $y = mx + c$, where $y = V_d$ and $x = \ln(I)$:
* Slope $(m) = \eta V_T$
* Intercept $(c) = -\eta V_T \ln(I_0)$

Alternatively, if plotting $\ln(I_f)$ versus V_d as standard practice:

$$\ln(I_f) = \left(\frac{1}{\eta V_T} \right) V_d + \ln(I_0)$$

This confirms that the relationship between the log value of forward current ($\ln I$) and the forward voltage (V_d) is strictly linear.

Therefore, the correct option is B.

$$\boxed{(B)\ \log I \text{ vs } V}$$

Q2.95.1 MCQ 1995 1M Ans: C ● ○ ○

The depletion layer capacitance (also known as transition capacitance, C_T or C_j) of a p-n junction diode under reverse bias varies according to the voltage profile across the junction.

The general relationship between the depletion capacitance and the reverse bias voltage V_R is expressed as:

$$C_T \propto V_R^{-n}$$

Where n represents the junction grading coefficient.

The value of n depends directly on the nature of the semiconductor doping profile. For a step-graded or abrupt p-n junction, the grading factor is:

$$n = \frac{1}{2}$$

Substituting $n = \frac{1}{2}$ for the abrupt junction case given in the problem:

$$C_T \propto V_R^{-1/2}$$

Thus, the grading coefficient value corresponds to option C.

$$\boxed{(C)\, C_J \propto V_R^{-1/2}}$$

Q2.95.2 MCQ 1995 1M Ans: A ● ○ ○

A Zener diode exhibits a breakdown mechanism that depends directly on the doping concentration of the p-n junction.

When both the p-side and n-side are heavily doped, the depletion region becomes extremely narrow. Under a high reverse-bias voltage, a strong electric field is established across this narrow gap. This intense field is sufficient to force valence electrons to breach their covalent bonds and cross the barrier.

This physical phenomenon is defined as the tunneling of charge carriers across the junction, which leads the junction directly into zener breakdown.

Therefore, the correct option is A.

$$\boxed{(A) \text{ tunneling of charge carriers across the junction.}}$$

Q2.95.3 MCQ 1995 1M Ans: D ● ○ ○

The built-in diffusion potential, V_{bi} across an unbiased p-n junction forms due to the diffusion of majority carriers across the metallurgical interface, and is given by:

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

At a constant operating temperature, the thermal voltage term $V_T = \frac{kT}{q}$ and the intrinsic carrier concentration n_i remain fixed. Under these conditions, the contact potential relationship varies logarithmically with respect to the product of the doping concentrations:

$$V_{bi} \propto \ln(N_A N_D)$$

Therefore, an increase in either the acceptor doping concentration (N_A) or the donor doping concentration (N_D) results in a corresponding increase in the contact potential across the junction.

Also, intrinsic concentration n_i depends on bandgap E_g as

$$n_i \propto e^{-\frac{E_g}{kT}}$$

At constant temperature, with decrease in bandgap, n_i increases. This causes a decrease in diffusion potential since the relation can be observed as:

$$V_{bi} \propto ln\left(\frac{1}{n_i^2}\right)$$

Thus, the only true statement is option D.

$$\boxed{(D) \text{ increases with increase in doping concentration}}$$

Q2.93.1 MSQ | 1993 | 2M | Ans: A, B | ● ○ ○

The built-in potential of a PN junction is

$$V_{bi} = \frac{kT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right)$$

- **(A) True:** At equilibrium, the Fermi level becomes flat across the junction, and the built-in voltage equals the initial difference in Fermi levels of the P and N sides expressed in volts.
- **(B) True:** From the formula, increasing doping concentrations N_A or N_D increases V_{bi}.
- **(C) False:** Although $\frac{kT}{q}$ increases with temperature, the intrinsic carrier concentration n_i rises much faster, making V_{bi} decrease overall with temperature.
- **(D) False:** Built-in potential is related to the difference, not the average, of the Fermi levels

$$\boxed{(A) \text{ is equal to the difference in the Fermi-level of the two sides, expressed in volts}}$$

$$\boxed{(B) \text{ increases with the increase in the doping levels of the two sides.}}$$

Q2.91.1 NAT | 1991 | 2M | Ans: Sol | ● ● ○

When a diode is instantaneously switched from a conduction state, it needs some time to return to a non-conduction state. Thus, the diode behaves as a short circuit for a little time, even in the reverse direction. This is due to the accumulation of stored excess minority carrier charge when the diode is forward biased.

The time required to return back to the state of non-conduction is called the 'Reverse recovery time' (t_{rr}), which is the sum of 'storage time' (t_s) and 'transition time' (t_t).

- Storage time (t_s): It is the period for which the diode remains in the conduction state even in the reverse direction.
- Transition time (t_t): It is the time elapsed in returning back to the state of non-conduction.

For $0 < t < t_o$, $V_m = 20\,\text{V}$, and the diode 'D' is forward biased:

$$i = \frac{V_m}{R} = \frac{20\,\text{V}}{10\,\text{k}\Omega} = 2\,\text{mA}$$

For $t > t_o$, the input voltage switches to $V_m = -20\,\text{V}$.
During the storage time ($t_o < t < t_o + t_s$):

$$i = \frac{V_m}{R} = \frac{-20\,\text{V}}{10\,\text{k}\Omega} = -2\,\text{mA}$$

During the transition time ($t_o + t_s < t < t_o + t_s + t_t$): The current '$i$' decreases exponentially to ($-I_o$).

For $t > t_o + t_s + t_t$:

$$i = -I_o$$

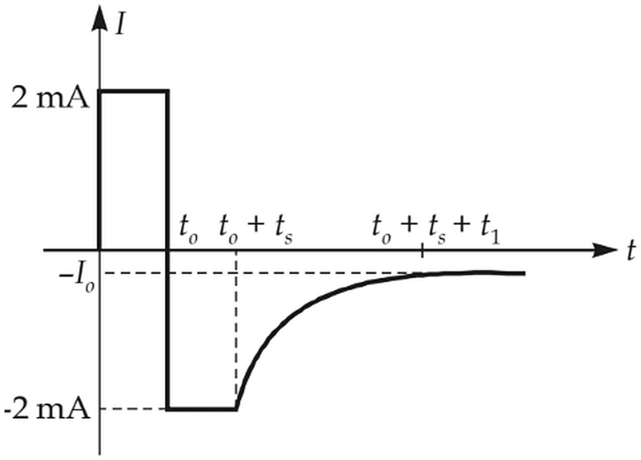

Fig. Solution Q2.91.1

Key Points

- During the storage time interval, the diode acts as a virtual short circuit in the reverse direction due to stored minority carriers.
- The peak reverse current depends on the external circuit resistance and the reverse voltage applied.

Mistakes to be avoided

- Do not assume the diode stops conducting immediately when the applied voltage goes negative; minority carrier storage delays the turn-off process.

Q2.91.2 MCQ 1991 2M Ans: B ● ○ ○

For an abrupt p-n junction, the junction capacitance (C_j) is inversely proportional to the square root of the total reverse bias potential (V).

$$C_j \propto \frac{1}{\sqrt{V}}$$

Taking the ratio of the capacitances for two different biasing states:

$$\frac{C_2}{C_1} = \sqrt{\frac{V_1}{V_2}} = \sqrt{\frac{1+0}{1+99}} = \sqrt{\frac{1}{100}} = \frac{1}{10}$$

Given that the initial capacitance $C_1 = 1\text{ nF/cm}^2$, we can calculate the final capacitance C_2:

$$C_2 = \frac{C_1}{10} = \frac{1}{10} = 0.1\text{ nF/cm}^2$$

$$\boxed{(B)\, 0.1}$$

Q2.90.1 MCQ | 1990 | 2M | Ans: A | ● ○ ○

In a step-graded diode (abrupt p-n junction), the total negative charge per unit area on the p-side must exactly equal the total positive charge per unit area on the n-side to maintain macro-charge neutrality.

By using the charge density condition or charge neutrality condition:

$$qN_AW_P = qN_DW_N$$

Rearranging the terms gives the relationship between the depletion widths and the doping concentrations:

$$\frac{W_N}{W_P} = \frac{N_A}{N_D}$$

Given that the donor concentration is four times the acceptor concentration ($N_D = 4N_A$):

$$\frac{W_N}{W_P} = \frac{N_A}{4N_A} = \frac{1}{4} = 0.25$$

$$\boxed{(A)\,0.25}$$

Mistakes to be avoided

- Do not invert the ratio; ensure that the depletion layer width of the n-side (W_N) corresponds inversely to the donor concentration (N_D) in the denominator.

Q2.90.2 MCQ | 1990 | 1M | Ans: B | ● ○ ○

The depletion capacitance is related to reverse bias voltage as:

$$C \propto \frac{1}{\sqrt{\text{Reverse bias voltage}}}$$

Therefore, as the reverse bias voltage increases, the depletion width W widens, which causes the depletion capacitance C to decrease.

$$\boxed{(B)\text{ the depletion capacitance decreases with increase in the reverse bias.}}$$

Q2.89.1 MCQ | 1989 | 2M | Ans: MTA | ● ○ ○

In a P^+N junction, the heavily doped P^+ side injects very few electrons into itself, while most excess charge storage occurs due to holes injected into the lightly doped N region. Hence, switching speed mainly depends on the **minority carrier lifetime in the N region**, which is not among the given options.

$$\boxed{MTA}$$

Q2.89.2 MCQ | 1989 | 2M | Ans: C | ● ○ ○

Both P and N regions are heavily doped in a Zener diode to facilitate tunneling.

$$\boxed{(C)\text{ both P and N regions are heavily doped}}$$

Q2.88.1 MCQ 1988 2M Ans: B ● ○ ○

Diode D_1 is in forward bias, and diode D_2 is in reverse bias. Therefore, the current through diode D_1 is the forward bias current I_f, and the current through diode D_2 is the reverse saturation current I_o.

The total current I flowing through the configuration is given by:

$$I = I_f + I_o$$

Using the ideal diode equation for the forward-biased diode:

$$I_f = I_o\left(e^{V_d/\eta V_T} - 1\right) = I_o e^{V_d/\eta V_T} - I_o$$

Substituting I_f back into the total current expression:

$$I = \left(I_o e^{V_d/\eta V_T} - I_o\right) + I_o$$

$$I = I_o e^{V_d/\eta V_T}$$

Rearranging the terms to solve for the exponential factor:

$$e^{V_d/\eta V_T} = \frac{I}{I_o}$$

Taking the natural logarithm on both sides:

$$\frac{V_d}{\eta V_T} = \ln\left(\frac{I}{I_o}\right)$$

$$V_d = \eta V_T \ln\left(\frac{I}{I_o}\right)$$

The thermal voltage is defined as $V_T = \frac{KT}{q}$. For a Germanium (Ge) diode, the ideality factor is $\eta = 1$:

$$V_d = \frac{KT}{q}\ln\left(\frac{I}{I_o}\right)$$

Given that the total voltage drop across the combination is $V = V_d$:

$$V = \frac{KT}{q}\ln\left(\frac{I}{I_o}\right)$$

$$\boxed{(B)\ V = \frac{KT}{q}\ln\left(\frac{I}{I_o}\right)}$$

Mistakes to be avoided

- Ensure you do not drop the reverse saturation current I_o prematurely before combining the parallel branch expressions.
- Take note of the semiconductor material specified (Ge vs Si) as it directly changes the value of η.

Q2.88.2 MCQ 1988 2M Ans: D ● ○ ○

Based on the operational physics of semiconductor devices, the matching pairs are determined as follows:

- **Avalanche breakdown → Collision of carriers with crystal ions:** High reverse bias accelerates minority carriers to high velocities, causing them to collide with crystal atoms, breaking covalent bonds and generating electron-hole pairs via impact ionization.
- **Zener breakdown → Rupture of covalent bonds due to strong electric field:** In heavily doped junctions with narrow depletion regions, a high electric field directly breaks or ruptures covalent bonds, causing field emission of carriers.
- **Punch through → Early effect:** In a Bipolar Junction Transistor (BJT), as the reverse bias across the collector-base junction increases, the depletion region widens into the base. At a sufficiently high voltage, the base depletion region reaches all the way to the emitter junction, an extreme consequence of the Early effect known as punch-through.

Thus, the correct matching sequence is: 1-A, 2-C, 3-B.

$$\boxed{(D)\ \text{1-A, 2-C, 3-B}}$$

Q2.87.1 MCQ 1987 2M Ans: B ● ○ ○

The diffusion capacitance (C_D) decreases with decreasing forward current and increasing temperature.

The formula for diffusion capacitance is given by:

$$C_D = \frac{\tau}{r}$$

where the dynamic internal forward resistance (r) of the diode is defined as:

$$r = \frac{\eta V_T}{I_f}$$

Substituting the expression for r into the capacitance formula yields:

$$C_D = \frac{\tau I_f}{\eta V_T}$$

Since the thermal voltage is defined as $V_T = \frac{kT}{q}$, where k is the Boltzmann constant and T is the absolute temperature:

$$C_D = \frac{\tau I_f q}{\eta k T}$$

From this relation, it is clear that:

$$C_D \propto I_f$$

$$C_D \propto \frac{1}{T}$$

Therefore, the diffusion capacitance is directly proportional to the forward current (I_f) and inversely proportional to the absolute temperature (T).

$$\boxed{(B)\ \text{decreases with decreasing current and increasing temperature}}$$

Key Points

- Diffusion capacitance (C_D) is dominant under forward-biased conditions due to minority carrier storage.
- The parameter τ represents the mean carrier lifetime of the minority charge carriers.

Mistakes to be avoided

- Do not confuse the temperature dependence of diffusion capacitance ($C_D \propto T^{-1}$) with depletion capacitance, which has a different mathematical relationship governed by the built-in potential.

DEBUG INFO

Chapter II

PN Junctions

Total Questions : 56

MCQ : 42

MSQ : 2

NAT : 12

SOLUTIONS

III

BJTs

Topics

1. Introduction to BJTs and Mode of Operations
2. Ideal and Non-Ideal Characteristics of BJTs

Q3.26.1 MSQ 2026 2M Ans: C,D ● ○ ○

The given plot is extended backwards to determine the Early voltage values V_{A1} and V_{A2}.

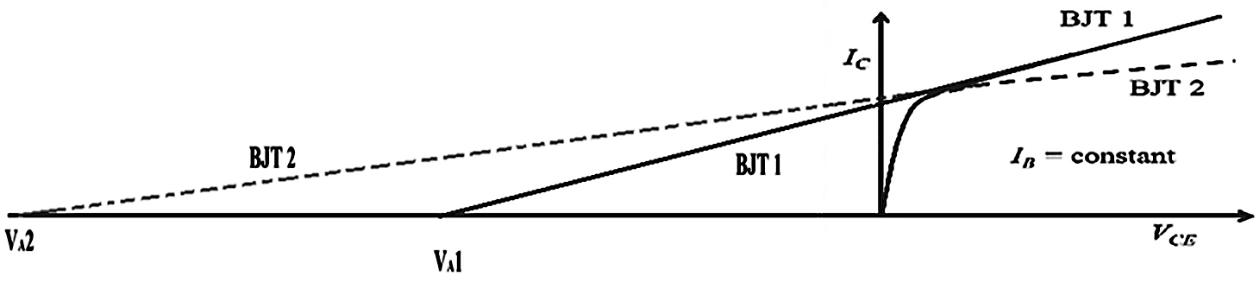

Fig. Solution Q3.26.1

- **(A) - False** As shown in the extended plot, $|V_{A1}| < |V_{A2}|$.
- **(B) - False** As shown in the extended plot, both V_{A1} and V_{A2} are finite.
- **(C) - True** As shown in the extended plot, $|V_{A1}| < |V_{A2}|$.
- **(D) - True** As shown in the extended plot, both V_{A1} and V_{A2} are finite.

$\boxed{(C)\ |V_{A1}| < |V_{A2}|}$

$\boxed{(D)\ |V_{A2}| \text{ is finite}}$

Key Points

- The Early voltage for a BJT can be obtained by extending the plot of I_C vs V_{CE} backwards, at the point where this extended plot crosses $I_C = 0$.

Q3.24.1 MSQ 2024 2M Ans: A,C,D ● ○ ○

- **(A) - True** — In high-level injection, excess carriers flood the base region, increasing recombination and reducing emitter efficiency. Hence base current rises faster than collector current, so $\beta = \frac{I_C}{I_B}$ decreases at high I_C.
- **(B) - False** — When emitter-base current is dominated by recombination-generation, base current varies more slowly than collector current as current increases. Therefore β actually *increases* with collector current in this low-current region.
- **(C) - True** — In saturation, both junctions are forward biased, causing excess charge storage and a large increase in base current. Since collector current no longer increases proportionally, the effective β becomes smaller than in active mode.
- **(D) - True** — A transistor with larger β has stronger transistor action, so avalanche-generated collector-base leakage gets amplified more strongly. This causes breakdown to occur at a lower collector-emitter voltage (BV_{CEO} decreases as β increases).

$\boxed{\text{(A) Under high-level injection condition in forward active mode, } \beta \text{ will decrease with increase in the magnitude of collector current.}}$

$\boxed{\text{(C) } \beta \text{ will be lower when the BJT is in saturation region compared to when it is in active region.}}$

$\boxed{\text{(D) A higher value of } \beta \text{ will lead To a lower value of the collector-to-emitter breakdown voltage.}}$

Key Points

- The avalanche breakdown voltage BV_{CEO} is related to BV_{CBO} by $BV_{CEO} \approx \dfrac{BV_{CBO}}{\beta^{1/n}}$ so a higher β yields a lower BV_{CEO}.

Mistakes to be avoided

- Do not confuse high-level injection (statement 1) with low-current recombination-generation (statement 2) — β behaves *oppositely* in these two regimes.

Q3.17.1 MCQ 2017 1M Ans: C ● ○ ○

When a BJT operates in the active region, increasing the reverse bias voltage across the collector-base junction causes the depletion region at that junction to widen.

This widening encroaches into the base, reducing the effective base width W_B. A narrower base means fewer minority carriers recombine before reaching the collector, so the base recombination current I_B decreases relative to the collector current I_C.

Since the common-emitter current gain is:

$$\beta = \frac{I_C}{I_B}$$

a reduction in base recombination increases β. This phenomenon is known as the **Early effect** (base-width modulation).

$$\boxed{(C) \text{ the effective base width decreases and common - emitter current gain increases}}$$

Key Points

- The Early effect describes the modulation of effective base width W_B by the collector-base reverse bias; as $|V_{CB}|$ increases, W_B decreases.
- Recombination in the base is proportional to W_B; a smaller W_B reduces I_B and thus raises $\beta = I_C/I_B$.
- The Early voltage V_A characterises this effect; a larger V_A implies less base-width modulation and a more ideal transistor.
- Base-width modulation also causes the output characteristics of a BJT to have a finite slope in the active region rather than being perfectly flat.

Mistakes to be avoided

- Do not conclude that increasing $|V_{CB}|$ increases I_C proportionally — the dominant effect here is the reduction of I_B through less recombination, not a direct increase in I_C.
- Do not confuse base-width modulation (Early effect) with punch-through, which occurs when $W_B \to 0$ and the device breaks down.

Q3.17.2 MCQ 2017 1M Ans: C ● ○ ○

Identifying the bias conditions of each junction:

$$\text{Emitter-base junction } (J_E) \to \text{Reverse Biased (RB)}$$

$$\text{Collector-base junction } (J_C) \to \text{Forward Biased (FB)}$$

Comparing with the BJT operating mode table:

Mode	J_E	J_C
Active	FB	RB
Saturation	FB	FB
Cut-off	RB	RB
Inverse Active	RB	FB

Since J_E is reverse biased and J_C is forward biased, the BJT operates in **Inverse Active Mode**.

$$\boxed{(C) \quad \text{Inverse Active}}$$

Q3.16.1 NAT | 2016 | 2M | Ans: 6.55 to 6.75 | ● ○ ○

The collector current is given by the diffusion current relation:

$$I_c = AqD_n \frac{dn}{dx}, \quad \text{where } D_n = \mu_n V_T$$

Substituting $D_n = \mu_n V_T = 800 \times 26 \times 10^{-3}\ \text{cm}^2/\text{s}$ and the given values , $dn/dx = 10^{14}/(0.5 \times 10^{-4})\ \text{cm}^{-4}$:

$$I_c = 0.001 \times (1.6 \times 10^{-19}) \times 800 \times 26 \times 10^{-3} \times \left[\frac{10^{14}}{0.5 \times 10^{-4}}\right] = 66.56 \times 10^{-4}\ \text{A} = 6.656\ \text{mA}$$

$$\boxed{6.656}$$

Key Points

- The collector current in a BJT is dominated by minority carrier diffusion across the base.
- The carrier gradient dn/dx is approximated as $\Delta n/W_B$, where W_B is the effective base width.

Q3.15.1 NAT | 2015 | 2M | Ans: 1465 to 1485 | ● ○ ○

The collector current in a BJT is given by:

$$I_C = I_S\, e^{V_{BE}/V_T}$$

The emitter current is related to the collector current by:

$$I_E = \frac{\beta + 1}{\beta} I_C = \frac{\beta + 1}{\beta} I_S (e^{V_{BE}/V_T})$$

I_E is maximum when $(\beta + 1)/\beta$ is maximum, which occurs at the smallest value of $\beta = 50$:

$$\frac{\beta + 1}{\beta} = \frac{51}{50} = 1.02$$

Substituting $I_S = 10^{-9} \times 10^{-6}$ A, $V_{BE} = 700$ mV, $V_T = 25$ mV:

$$I_E = 1.02 \times (10^{-9} \times 10^{-6}) \times e^{700\times10^{-3}/25\times10^{-3}} = 1475 \times 10^{-6}\ A$$

$$\boxed{1475}$$

Q3.15.2 MCQ | 2015 | 1M | Ans: D | ● ○ ○

When base width W_B is doubled, Early effect is still present but becomes less severe relative to the previous W_B.

The Early voltage V_A is inversely related to the sensitivity of the effective base width to changes in V_{CE}:

$$V_A \propto W_B$$

A larger W_B means that the same incremental change in V_{CE} causes a *proportionally smaller* fractional reduction in base width. This reduces base-width modulation, so the slope of I_C versus V_{CE} in the active region decreases. The mathematical dependency of Early voltage V_A on I_C-V_{CE} characteristic can be determined as:

$$I_C = I_{C0}\left(1 + \frac{V_{CE}}{V_A}\right)$$

The slope of the output characteristic is obtained by differentiating I_C with respect to V_{CE}:

$$\frac{dI_C}{dV_{CE}} = \frac{I_{C0}}{V_A}$$

At a given operating point, this is approximately written as:

$$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_A}$$

So Early voltage dependency is:

$$\text{Slope} = \frac{I_C}{V_A} \downarrow \quad \Longrightarrow \quad V_A \uparrow$$

Doubling W_B therefore causes the Early voltage to **increase**.

$$\boxed{(D) \quad V_A \text{ increases}}$$

Key Points

- The Early effect arises from base-width modulation: as V_{CE} increases, the CB depletion region widens, reducing W_B and increasing I_C.
- Early voltage $V_A \propto W_B$; a wider base is less susceptible to modulation by V_{CE}, giving a larger V_A and flatter I_C–V_{CE} curves.

Mistakes to be avoided

- Do not conclude that a wider base makes the Early effect more severe — a larger W_B means the depletion region encroachment is a *smaller fraction* of W_B, reducing the effect.
- The quantity related to Early voltage is the differential slope of the I_C vs V_{CE} characteristics, not the ordinary ratio $\dfrac{I_C}{V_{CE}}$.

Q3.14.1 NAT | 2014 | 2M | Ans: 378 to 381 | ● ● ○

The collector current of a BJT is given by:

$$I_C = I_S \exp\left[\frac{V_{BE}}{\eta V_T}\right]$$

For the two transistors:

$$I_{C_1} = I_{S_1} \exp\left[\frac{V_{BE_1}}{\eta V_T}\right] \quad \cdots (i)$$

$$I_{C_2} = I_{S_2} \exp\left[\frac{V_{BE_2}}{\eta V_T}\right] \quad \cdots (ii)$$

Applying the given condition $I_{C_1} = I_{C_2}$, equating (i) and (ii):

$$I_{S_1} \exp\left[\frac{V_{BE_1}}{\eta V_T}\right] = I_{S_2} \exp\left[\frac{V_{BE_2}}{\eta V_T}\right]$$

Rearranging:

$$V_{BE_1} - V_{BE_2} = \eta V_T \ln\left[\frac{I_{S_2}}{I_{S_1}}\right]$$

The question does not explicitly mention that the device is a Silicon BJT. The Silicon assumption may be inferred from the given intrinsic carrier concentration:

$$n_i = 1 \times 10^{10} cm^{-3}$$

which corresponds approximately to silicon at room temperature.

For a Silicon BJT, the ideality factor $\eta = 2$. Substituting $I_{S_2}/I_{S_1} = (300 \times 300)/(0.2 \times 0.2)$ and $V_T = 26\,\text{mV}$:

$$V_{BE_1} - V_{BE_2} = 2 \times 26 \times \ln\left[\frac{300 \times 300}{0.2 \times 0.2}\right] = 52 \times 7.717 = 380.28\,\text{mV}$$

$$\boxed{380.28}$$

Key Points

- The saturation current $I_S \propto A$ (junction area), so $\frac{I_{S_2}}{I_{S_1}} = \frac{A_2}{A_1}$.

Q3.14.2 NAT 2014 2M Ans: 5.7 to 5.9 ● ○ ○

The transconductance of a BJT is given by:

$$g_m = \frac{I_C}{V_T}$$

First, computing the collector current using $I_S = 10^{-13}$ A and $V_{BE} = 0.7\,\text{V}$, $V_T = 25\,\text{mV}$:

$$I_C = I_S\, e^{V_{BE}/V_T} = 10^{-13} \times e^{0.7/0.025} = 0.144\,\text{mA}$$

Substituting into the transconductance expression:

$$g_m = \frac{I_C}{V_T} = \frac{0.144\,\text{mA}}{25\,\text{mV}} = 5.76\,\text{mA/V}$$

$$\boxed{5.76}$$

Q3.11.1 MCQ 2011 1M Ans: D ● ○ ○

The collector current of a BJT including leakage is:

$$I_C = \beta I_B + (1+\beta) I_{CO}$$

Converting the given common-base current gain to common-emitter current gain:

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = \frac{0.98}{0.02} = 49$$

Substituting values:

$$I_C = 49 \times 20 + (1+49) \times 0.6 = 1.01\,\text{mA}$$

$$\boxed{(D)\ 1.01\,\text{mA}}$$

Q3.10.1 MCQ 2010 1M Ans: B ● ○ ○

The emitter injection efficiency for NPN transistor is given by:

$$\gamma = \frac{\text{Electron current injected from emitter to base}}{\text{Electron current injected from emitter to base + Hole current injected from base to emitter}}$$

$$\gamma = \frac{I_{NE}}{I_{NE} + I_{PE}} = \frac{1}{1 + \frac{I_{PE}}{I_{NE}}}$$

Now we know that, $\frac{I_{PE}}{I_{NE}} \propto \frac{N_B}{N_E}$. Thus the equation becomes:

$$\gamma = \frac{1}{1 + \frac{N_B}{N_E}}$$

To achieve maximum emitter efficiency $\gamma \to 1$, the denominator must approach unity, which requires:

$$\frac{N_B}{N_E} \to 0 \implies N_E \gg N_B$$

Therefore, the emitter doping must be much greater than the base doping to maximise γ. Also, collector doping is kept lesser than base doping to avoid widening of B-C depletion width on base side, which depletes base carriers.

$$\boxed{(B)\ N_E \gg N_B \text{ and } N_B > N_C}$$

Q3.06.1 MCQ 2006 1M Ans: B ● ○ ○

A large collector base reverse bias is the reason behind the early effect manifested by BJTs. As reverse biasing of the collector to base junction increases, the depletion region penetrates more into the base, as the base is lightly doped. This reduces the effective base width and hence the concentration gradient in the base increases. This reduction in the effective base width causes less recombination of carriers in the base region which results in an increase in collector current. This is known as the Early effect.

$$\boxed{(B) \text{ The reverse biasing of the base - collector junction}}$$

Q3.04.1 MCQ 2004 1M Ans: B ● ○ ○

The diffusion current density J_C is given by:

$$J_C = qD_n \frac{dn}{dx}$$

Substituting the given values for charge q, diffusion coefficient D_n, and the concentration gradient $\frac{dn}{dx}$:

$$J_C = 1.6 \times 10^{-19} \times 25 \times \frac{10^{14}}{0.5 \times 10^{-4}} = 8\ \text{A/cm}^2$$

$$\boxed{(B)\ 8\ \text{A/cm}^2}$$

Q3.04.2

MCQ | 2004 | 1M | Ans: D | ● ○ ○

The common-emitter current gain is related to the common-base current gain by:

$$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}$$

Analysing the effect of each parameter on β:

Increasing base width W_B: A wider base increases the probability of recombination of minority carriers before they reach the collector. This raises I_B and reduces α, thereby *decreasing* β.

Increasing base doping N_B: Higher base doping increases the majority carrier concentration in the base, which enhances recombination with the injected minority carriers from the emitter. This again reduces α and hence *decreases* β.

Both effects reduce β; therefore the correct statement is that β **decreases** when either base width or base doping is increased.

$$\boxed{(D)\text{ S1 is TRUE and S2 is FALSE}}$$

Key Points

- Base transport factor $b \approx 1 - \dfrac{W_B^2}{(2L_n^2)}$; a wider base reduces b and hence $\alpha = \gamma\, b$.
- Higher base doping N_B reduces minority carrier lifetime via increased recombination, shortening the diffusion length $L_n = \sqrt{D_n \tau_n}$ and further degrading α_T.
- For high β, BJT design requires: thin base ($W_B \ll L_n$), lightly doped base ($N_B \ll N_E$), and heavily doped emitter.

Mistakes to be avoided

- Do not confuse the effect of emitter doping with base doping; increasing *emitter* doping N_E increases γ and raises β, while increasing *base* doping N_B does the opposite.
- Do not assume β is independent of geometry; base width W_B directly controls the base transport factor and is a critical design parameter.

Q3.04.3

MCQ | 2004 | 1M | Ans: A | ● ○ ○

For $V_{BE} = 0.7$ V, Base-Emitter Junction is Forward-Biased , and for $V_{CB} = 0.2$ V, Base-Collector Junction is Reverse-Biased. For an $n-p-n$ transistor, this is the criteria for forward active mode.

$$\boxed{(A)\text{ normal active mode}}$$

Q3.99.1

MCQ | 1999 | 1M | Ans: C | ● ○ ○

In a transistor by changing the base-collector junction voltage, it is seen that the base width of the transistor is altered and this property is called as early effect.

$$\boxed{(C)\text{ large collector-base reverse bias}}$$

Q3.97.1

MCQ | 1997 | 2M | Ans: C | ● ○ ○

The emitter current of a BJT is given by:

$$I_e = I_o\left[e^{V_{be}/\eta V_T} - 1\right]$$

Setting up the ratio for two bias conditions where $I_2 = 2I_1$:

$$\frac{2I_1}{I_1} = \frac{I_o\left[e^{V_{be_2}/\eta V_T} - 1\right]}{I_o\left[e^{V_{be_1}/\eta V_T} - 1\right]}$$

Since $e^{V_{be}/\eta V_T} \gg 1$ for both bias points, the -1 terms are negligible:

$$2 = \frac{e^{V_{be_2}/\eta V_T}}{e^{V_{be_1}/\eta V_T}} = e^{(V_{be_2}-V_{be_1})/\eta V_T}$$

Taking the natural logarithm of both sides:

$$\frac{V_{be_2} - V_{be_1}}{\eta V_T} = \ln 2$$

$$V_{be_2} - V_{be_1} = \eta V_T \ln 2 = \eta V_T \times 0.693$$

For a Silicon BJT, $\eta = 1$ and $V_T = 26\,\text{mV}$:

$$V_{be_2} - V_{be_1} = 1 \times 0.026 \times 0.693 = 18\,\text{mV} \approx 20\,\text{mV}$$

$$\boxed{(C)\text{ increases by about 20 mV}}$$

Key Points

- The voltage required to double the current is $\Delta V_{be} = \eta V_T \ln 2 \approx 20\,\text{mV}$ for $\eta = 1$ at room temperature — a fundamental BJT design rule.

Q3.96.1 | MCQ | 1996 | 2M | Ans: B | ● ○ ○

In a transistor when both the junction (J_C and J_E) are forward bias and if collector junction voltage is greater than emitter junction voltage then this transistor is in reverse saturation region.

$$\boxed{(B)\text{ reverse saturation mode.}}$$

Q3.95.1 | MCQ | 1995 | 1M | Ans: D | ● ○ ○

Both the junctions are forward biased. When emitter base (E-B) junction and collector base (C-B) junction both are forward biased then the BJT is said to be operating in saturation region.

$$\boxed{(D)\text{ both the junctions are forward biased.}}$$

Q3.95.2 | MCQ | 1995 | 1M | Ans: C | ● ○ ○

The relationship between open base breakdown voltage (BV_{CEO}) of BJT with open emitter voltage (BV_{CBO}) is given by

$$BV_{CEO} = BV_{CBO}\sqrt{\frac{1}{\beta}}$$

Since $\beta > 1$, thus $BV_{CEO} < BV_{CBO}$.

$$\boxed{(C)\ BV_{CEO} < BV_{CBO}}$$

Q3.94.1 NAT 1994 2M Ans: A-2, B-3, C-1 ● ○ ○

- **A - 2:** As the base width of the BJT is reduced, the recombination current (base current I_B) decreases. As a result collector current (I_C) increases and consequently, the current gain $\alpha = \frac{I_C}{I_E} = \frac{I_C}{I_C + I_B}$ of the BJT increases.
- **B - 3:** If the emitter doping concentration to base doping concentration ratio is reduced then the emitter injection efficiency γ decreases, so the current gain α of BJT reduces.
- **C - 1:** If the collector doping concentration is increased then the breakdown voltage (V_{BR}) of a BJT will be reduced.

$$\boxed{A - 2\,,\ B - 3\,,\ C - 1}$$

Key Points

- The emitter injection efficiency for a PNP transistor is given as:

$$\gamma = \frac{I_{pE}}{I_{pE} + I_{nE}}$$

- In terms of device parameters , emitter injection efficiency is given by:

$$\gamma = \frac{1}{1 + \frac{D_E N_B x_B}{D_B N_E x_E}}$$

  Where N_E and N_B are the emitter and base doping concentrations, D_E and D_B are diffusion coefficients, and x_E and x_B are the space charge region widths.
- The relation between γ and α is given as $\alpha = \gamma b$, where b is the base transport factor.

Q3.92.1 NAT 1992 2M Ans: decrease ● ○ ○

As the reverse bias increase at CB (collector base) junction then the collector current (I_C) increases and the effective base width decreases, so the recombination in base decreases.

$$\boxed{\text{decrease}}$$

DEBUG INFO

Chapter III

BJTs

Total Questions : 22

MCQ : 14

MSQ : 2

NAT : 6

SOLUTIONS

IV

MOSCap and MOSFETs

Topics

1. Different Modes of Operations of MOSCAPs
2. MOS Physics

Q4.24.1 NAT | 2024 | 2M | Ans: 4.10 to 4.50 | ● ● ○

First, compute the oxide capacitance per unit area:

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{4 \times 8.85 \times 10^{-14}}{100 \times 10^{-9} \times 100} = 35.4 \times 10^{-9}\ \text{F/cm}^2$$

The flat-band voltage condition gives:

$$V_{FB} = \frac{Q_{ox}}{C_{ox}} + \phi_{ms} = 0$$

Since Q_{ox} is positive, corresponding negative voltage must be applied at gate to achieve flatband condition. Thus Q_{ox}/C_{ox} is negative:

$$\frac{Q_{ox}}{C_{ox}} = \frac{10^{-8}}{35.4 \times 10^{-9}} = \frac{10}{35.4}\ \text{V} \quad \rightarrow \quad -\frac{10}{35.4}\ \text{V (negative contribution)}$$

Substituting into the flat-band equation:

$$0 = -\frac{10}{35.4} + \phi_{ms} \implies \phi_{ms} = \frac{10}{35.4}$$

Using $\phi_{ms} = \phi_m - \phi_s$:

$$\phi_s = \phi_m - \frac{10}{35.4} = 4.6 - \frac{10}{35.4} = 4.317\ \text{V}$$

Therefore, the work function energy of the semiconductor:

$$\phi_s = 4.317 \approx 4.32\ \text{eV}$$

$$\boxed{4.32}$$

Key Points

- The metal–semiconductor work function difference is $\phi_{ms} = \phi_m - \phi_s$; knowing ϕ_m and ϕ_{ms} uniquely determines ϕ_s.

Mistakes to be avoided

- Do not confuse the sign of Q_{ox}/C_{ox}: positive oxide charge produces a *negative* contribution to V_{FB}.
- Ensure t_{ox} is converted consistently — mixing nm and cm without conversion is a frequent unit error.

Q4.23.1 MCQ | 2023 | 1M | Ans: B | ● ○ ○

Since $V_G < V_{FB}$, the device is in **accumulation mode**.
In accumulation mode, there is no depletion charge, therefore $W_d = 0$
For $V_{FB} < V_G < V_T$, the device enters **depletion and inversion mode**, so a depletion width exists.
The depletion width is given by:

$$W_d = \sqrt{\frac{2\varepsilon|\phi_s|}{qN_s}}, \quad |\phi_s| \propto V_G$$

When $V_G > V_T$, **strong inversion** is reached and the depletion width W_d remains constant regardless of further increase in V_G.

$$\boxed{(B)}$$

Mistakes to be avoided

- Do not confuse flat-band voltage V_{FB} with threshold voltage V_T; they define the boundaries of three distinct operating regimes.

Q4.22.1 NAT 2022 2M Ans: 5.5 to 5.7 ● ○ ○

The inversion charge density is given by:

$$Q_{in} = C_{ox}(V_G - V_T)$$

Rearranging to find V_T using the known condition $Q_{in} = -2.2\,\mu\text{C/cm}^2$ at $V_G = 2\,\text{V}$:

$$\frac{Q_{in}}{C_{ox}} = V_G - V_T$$

$$V_T = V_G - \frac{Q_{in}}{C_{ox}} = 2 - \frac{2.2}{1.7} = 2 - 1.294$$

$$V_T = 0.706\,\text{V}$$

Now, computing Q_{in} at $V_G = 4\,\text{V}$:

$$Q_{in} = C_{ox}(V_G - V_T) = 1.7 \times 10^{-6} \times (4 - 0.706)$$

$$Q_{in} = 1.7 \times 10^{-6} \times 3.294 = 5.5998\,\mu\text{C/cm}^2$$

$$\boxed{5.6}$$

Mistakes to be avoided

- Confirm that $V_G > V_T$ before applying the inversion charge formula; if $V_G \leq V_T$ the device is not in inversion.

Q4.21.1 MCQ 2021 2M Ans: A ● ● ○

Given for an N-channel MOSFET:

$$t_{ox} = 10\,\text{nm} = 10 \times 10^{-7}\,\text{cm}, \quad \frac{\partial V_T}{\partial |V_{BS}|} = 50\,\text{mV/V}, \quad |V_{BS}| = 2\,\text{V}$$

$$\varepsilon_o = 8.85 \times 10^{-14}\,\text{F/cm}, \quad \varepsilon_{r_{Si}} = 12, \quad \varepsilon_{r_{ox}} = 4$$

The threshold voltage including the body effect is:

$$V_T = \phi_{ms} + \frac{\sqrt{2\varepsilon_{si} q N_A (2\phi_B - V_{BS})}}{C_{ox}} + 2\phi_B$$

Since $|V_{BS}| = |V_{SB}|$, rewriting with $+|V_{SB}|$:

$$V_T = \phi_{ms} + \frac{\sqrt{2\varepsilon_{si} q N_A (2\phi_B + |V_{SB}|)}}{C_{ox}} + 2\phi_B$$

Differentiating with respect to $|V_{BS}|$ and applying $|V_{SB}| \gg 2\phi_B$:

$$\frac{\partial V_T}{\partial |V_{BS}|} = \frac{\sqrt{2\varepsilon_{si} q N_A}}{C_{ox}} \cdot \frac{1}{2\sqrt{2\phi_B + |V_{SB}|}} = \frac{\sqrt{2\varepsilon_{si} q N_A}}{C_{ox}} \cdot \frac{1}{2\sqrt{|V_{SB}|}}$$

Substituting $C_{ox} = \varepsilon_{ox}/t_{ox}$ and $|V_{SB}| = |V_{BS}|$:

$$50 \times 10^{-3} = \frac{\sqrt{2 \times 8.85 \times 10^{-14} \times 12 \times 1.6 \times 10^{-19} \times N_A}}{\varepsilon_{ox}/t_{ox}} \cdot \frac{1}{2\sqrt{|V_{BS}|}}$$

Squaring both sides:

$$\left(\frac{50 \times 10^{-3} \times 4 \times 8.85 \times 10^{-14}}{10 \times 10^{-7}}\right)^2 = \frac{2 \times 8.85 \times 10^{-14} \times 12 \times 1.6 \times 10^{-19} \times N_A}{4 \times 2}$$

Solving for N_A:

$$N_A = 7.375 \times 10^{15}\,\text{cm}^{-3}$$

$$\boxed{(A)\ 7.37 \times 10^{15}\,\text{cm}^{-3}}$$

Q4.20.1 MCQ 2020 2M Ans: A ● ○ ○

For a MOS capacitor, the threshold voltage is given by:

$$V_T = \phi_{ms} - \frac{Q_{ox}}{C_{ox}} - \frac{Q'_d}{C_{ox}} + 2\phi_{F_p}$$

The given parameters are:

$$\phi_m = 3.87, \quad \phi_s = 4.8$$

$$\text{Thus } \phi_{ms} = \phi_m - \phi_s - 0.93$$

Computing ϕ_{F_p}:

$$\phi_{F_p} = E_i - E_F = 0.5 - 0.2 = 0.3\,\text{V}$$

Substituting into the threshold voltage equation with $V_T = -0.16$ V and $Q_{ox} = 0$:

$$-0.16 = -0.93 - 0 - \frac{Q'_d}{C_{ox}} + 2 \times 0.3$$

Solving for $\frac{Q'_d}{C_{ox}}$:

$$\frac{Q'_d}{C_{ox}} = 0.6 + 0.16 - 0.93 = -0.17\,\text{V}$$

Therefore, the depletion charge:

$$Q_d = -0.17 \times C_{ox} = -0.17 \times 100 \times 10^{-9} = -1.7 \times 10^{-8}\,\text{C/cm}^2$$

Since magnitude is asked, final answer is:

$$\boxed{(A)\, 1.7 \times 10^{-8}}$$

Key Points

- ϕ_s is the semiconductor work function, and is given by the difference in energy between Vacuum Level and semiconductor Fermi level.
- The depletion charge Q_d is inherently negative for a p-type MOS structure (ionized acceptors carry negative charge).

Q4.19.1 MCQ 2019 1M Ans: B ● ● ○

From the graph, the capacitance is maximum at large negative gate voltage (point P). This indicates accumulation occurs for negative gate bias, which is possible only for a p-type substrate because holes (majority carriers) accumulate at the oxide-semiconductor interface. As the gate voltage becomes positive, holes are repelled from the interface, leading to depletion (point Q), and for sufficiently large positive gate voltage, inversion occurs (point R). In a high-frequency C-V measurement, the inversion carriers cannot follow the AC signal, so the capacitance remains at its minimum value. Therefore, the shape of the graph itself indicates that the MOS capacitor is an NMOS capacitor fabricated on a p-type substrate.

Analysing each labelled point on the high-frequency C–V curve of the MOS capacitor:

At P (large negative V_G) — Accumulation: Holes accumulate at the oxide–semiconductor interface, so capacitance is maximum ($\approx C_{ox}$).

At Q ($V_G = 0$) — Flat-band: Since $\Phi_{ms} = 0$ and there is no oxide charge, flat-band occurs exactly at zero gate voltage.

Between Q and R (positive V_G) — Depletion: Holes are repelled from the interface, creating a depletion region; capacitance decreases as the depletion width widens.

At R (large positive V_G) — Inversion: Minority electrons form an inversion layer. In the high-frequency C–V measurement, inversion charge cannot follow the AC signal, so capacitance stays at its minimum value C_{min}.

The correct identification of the sequence flat-band → inversion → accumulation on the curve is:

$$\boxed{(B)\, Q,\ R,\ P}$$

Key Points

- In a p-type MOS capacitor: accumulation ($V_G \ll 0$) gives $C \approx C_{ox}$; depletion ($0 < V_G < V_T$) reduces C; inversion ($V_G > V_T$) gives C_{min}.
- At high frequency, the inversion layer charge cannot respond to the AC signal, so C does not recover and remains at C_{min} — unlike low-frequency C–V where C rises back to C_{ox}.
- The total MOS capacitance in depletion is $C = C_{ox}C_{dep}/(C_{ox} + C_{dep})$, which decreases as depletion width W_d increases.

Mistakes to be avoided

- Do not apply low-frequency C–V behaviour (where C recovers at inversion) to a high-frequency measurement — at high frequency C stays at C_{min} in inversion.
- Do not assume flat-band always occurs at $V_G = 0$; it only does so when both $\phi_{ms} = 0$ and $Q_{ox} = 0$.
- For a p-type substrate, inversion occurs at *positive* V_G, not negative — confusing substrate type reverses the entire C–V curve.

Q4.17.1 MCQ 2017 2M Ans: B ● ○ ○

Given that $W_2 = 2W_1$ and the overdrive voltages are equal:

$$V_{GS2} - V_{T2} = V_{GS1} - V_{T1}$$

Using the MOSFET drain current relation in saturation, $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2$, the ratio of drain currents is:

$$\frac{I_{D2}}{I_{D1}} = \frac{W_2}{W_1} \times \left(\frac{V_{GS2} - V_{T2}}{V_{GS1} - V_{T1}}\right)^2 = 2 \times 4 = 8$$

$$I_{D2} = 8\, I_{D1}$$

Using the transconductance relation $g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)$, the ratio of transconductances is:

$$\frac{g_{m2}}{g_{m1}} = \frac{W_2}{W_1} \times \frac{V_{GS2} - V_{T2}}{V_{GS1} - V_{T1}} = 2 \times 2 = 4$$

$$g_{m2} = 4\, g_{m1}$$

$$\boxed{(B)\ 8\, I_{D1} \text{ and } 4\, g_{m1}}$$

Mistakes to be avoided

- Do not confuse W/L scaling with W scaling alone – here only W changes while L is assumed equal for both devices.

Q4.17.2 NAT 2017 2M Ans: 6.85 to 6.95 ● ● ○

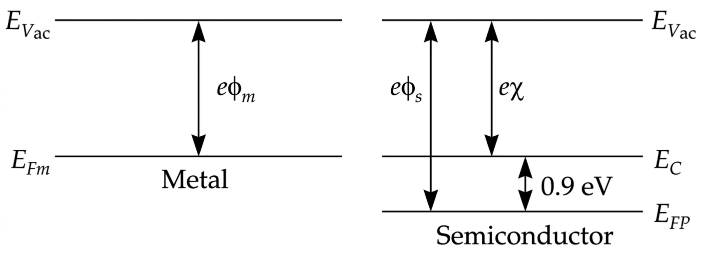

Fig. Solution Q4.17.2

From the energy band diagram , the given parameters are:

$$e\phi_m = 4.1\,\text{eV}, \quad e\chi = 4\,\text{eV}, \quad E_C - E_{FP} = 0.9\,\text{eV}$$

Computing the semiconductor work function:

$$e\phi_s = e\chi + E_C - E_{FP} = 4 + 0.9 = 4.9\,\text{eV}$$

$$\phi_m = 4.1\,\text{V}, \quad \phi_s = 4.9\,\text{V}$$

The metal–semiconductor work function difference:

$$\phi_{ms} = \phi_m - \phi_s = 4.1 - 4.9 = -0.8\,\text{V}$$

Computing the oxide capacitance per unit area:

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{0.1 \times 10^{-4}} = 34.515 \times 10^{-9}\,\text{F/cm}^2$$

The flat-band voltage condition:

$$V_{FB} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}$$

Setting $V_{FB} = 0$ and solving for ϕ'_{ox}/C_{ox}:

$$\frac{Q_{ox}}{C_{ox}} = -0.8 + 1 = 0.2\,\text{V}$$

Therefore, the oxide charge density:

$$Q_{ox} = 0.2 \times C_{ox} = 0.2 \times 34.515 \times 10^{-9}\,\text{C/cm}^2 = 6.903\,\text{nC/cm}^2$$

$$\boxed{6.903}$$

Key Points

- The semiconductor work function is $e\phi_s = e\chi + (E_C - E_{FP})$, combining electron affinity and the Fermi level offset from the conduction band.

Mistakes to be avoided

- Do not set $Q_{ox}/C_{ox} = \phi_{ms}$ directly; the sign convention in $V_{FB} = \phi_{ms} - Q_{ox}/C_{ox}$ must be respected when solving for the oxide charge.

Q4.16.1 NAT 2016 2M Ans: 1.55 to 1.65 ● ● ○

In Fig. I, Capacitors C_1 and C_2 are in series. Thus equivalent capacitance of Fig. I is given by:

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

Now C_1 and C_2 are to be calculated.

$$C_1 = \frac{\epsilon_1 A}{t_1} = \frac{4\epsilon_0}{10^{-7}\,\text{cm}} = 4 \times 10^7 \epsilon_0\, F$$

$$C_2 = \frac{\epsilon_2 A}{t_2} = \frac{20\epsilon_0}{3 \times 10^{-7}\,\text{cm}} = \frac{20}{3} \times 10^7 \epsilon_0\, F$$

Thus Fig. I capacitance can be calculated as:

$$C = \frac{4 \times 10^7 \epsilon_0 \times \frac{20}{3} \times 10^7 \epsilon_0}{\frac{32}{3} \times 10^7 \epsilon_0} = 2.5 \times 10^7 \epsilon_0$$

Fig. II capacitance is given as:

$$C_{Eq} = \frac{\epsilon_1 A}{t_{Eq}} = \frac{4\epsilon_0}{t_{Eq}} F$$

Equating the capacitances of both figures, we get:

$$2.5 \times 10^7 \epsilon_0 = \frac{4\epsilon_0}{t_{Eq}}$$

$$t_{Eq} = \frac{4}{2.5 \times 10^7} = 1.6 \text{ nm}$$

$$\boxed{1.6}$$

Q4.16.2 MCQ | 2016 | 1M | Ans: A | ● ○ ○

The given MOS capacitor is in Inversion Mode of Operation since it is observable that at Oxide-Semiconductor interface, the polarity of the semiconductor has inverted, i.e., in the bulk, the Fermi Level of the semiconductor is close to Conduction Band, implying that it is an n-type semiconductor. On the other hand, near the interface, the Fermi Level crosses the intrinsic Fermi Level of the semiconductor and is now near the Valence Band, implying that it is p-type at the interface.

$$\boxed{(A) \text{ Inversion}}$$

Q4.16.3 MCQ | 2016 | 2M | Ans: B | ● ● ○

The relation between inversion charge density Q_{inv} and inversion carrier density N_{inv} is

$$Q_{inv} = -q \times N_{inv}$$

In a MOS capacitor, the inversion charge density is proportional to the overdrive voltage:

$$(V_G - V_{Th}) \propto Q_{inv}$$

which leads to

$$(V_G - V_{Th}) \propto N_{inv}$$

Taking the ratio for two bias conditions:

$$\frac{V_{G1} - V_{Th}}{V_{G2} - V_{Th}} = \frac{N_{inv1}}{N_{inv2}}$$

Substituting $V_{G1} = 0.8\,\text{V}$, $V_{G2} = 1.3\,\text{V}$, $N_{inv1} = 2 \times 10^{11}\,\text{cm}^{-2}$, $N_{inv2} = 4 \times 10^{11}\,\text{cm}^{-2}$:

$$\frac{0.8 - V_{Th}}{1.3 - V_{Th}} = \frac{2 \times 10^{11}}{4 \times 10^{11}} = \frac{1}{2}$$

Solving for V_{Th}:

$$2(0.8 - V_{Th}) = 1.3 - V_{Th} \implies 1.6 - 2V_{Th} = 1.3 - V_{Th} \implies V_{Th} = 0.3\,\text{V}$$

Now applying the same proportionality for $V_{G3} = 1.8\,\text{V}$:

$$\frac{V_{G2} - V_{Th}}{V_{G3} - V_{Th}} = \frac{N_{inv2}}{N_{inv3}}$$

$$\frac{1.3 - 0.3}{1.8 - 0.3} = \frac{4 \times 10^{11}}{N_{inv3}}$$

Solving for Q_3:

$$N_{inv3} = 4 \times 10^{11} \times \frac{1.5}{1.0} = 6 \times 10^{11}\,\text{cm}^{-2}$$

$$\boxed{(B)\, 6 \times 10^{11}\,\text{cm}^{-2}}$$

Key Points

- In a MOS capacitor operating in inversion, $Q_{inv} = C_{ox}(V_G - V_{Th})$, so inversion charge scales linearly with overdrive voltage $(V_G - V_{Th})$.

Mistakes to be avoided

- Note that Q here has units of cm^{-2} (sheet charge density), not C/cm^2; do not confuse the two representations.

Q4.16.4 MCQ 2016 1M Ans: D ● ○ ○

The channel resistance of a MOSFET operating in the linear (triode) region is:

$$r_{ds} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)}$$

Analysing how r_{ds} changes with each parameter:

Option A — Increasing W: $r_{ds} \propto 1/W$, so $W \uparrow \Rightarrow r_{ds} \downarrow$. **Correct.**
Option B — Decreasing V_T: $(V_{GS} - V_T) \uparrow \Rightarrow r_{ds} \downarrow$. **Correct.**
Option C — Increasing L: $r_{ds} \propto L$, so $L \uparrow \Rightarrow r_{ds} \uparrow$. **Correct.**
Option D — Increasing V_{GS}: $(V_{GS} - V_T) \uparrow \Rightarrow r_{ds} \downarrow$, so resistance *decreases*, not increases. **Wrong statement.**

$$\boxed{(D) \text{ If } V_{GS} \text{ is increased, the resistance increases.}}$$

Mistakes to be avoided

- Do not mix up W and L dependencies: wider channels ($W \uparrow$) decrease resistance, while longer channels ($L \uparrow$) increase it.

Q4.16.5 NAT 2016 2M Ans: 28 to 29 ● ○ ○

In the saturation region, the drain conductance is given by:

$$g_d = \lambda I_{OS}$$

Expanding using the saturation drain current expression:

$$g_d = \lambda \left[\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2\right]$$

Substituting $\lambda = 0.09$, $\mu_n C_{ox} = 70 \times 10^{-6}$ A/V^2, $W/L = 4$, $V_{GS} = 1.8$ V, $V_T = 0.3$ V:

$$g_d = 0.09 \times \left[\frac{1}{2} \times 70 \times 10^{-6} \times 4 \times (1.8 - 0.3)^2\right] = 28.35 \times 10^{-6}$$

$$\boxed{28.35}$$

Key Points

- In the saturation region, the ideal MOSFET has $g_d = 0$; channel-length modulation introduces a finite output conductance $g_d = \lambda I_{DS}$.
- The channel-length modulation parameter λ (in V^{-1}) characterises the slope of the I_{DS}–V_{DS} curve in saturation; a larger λ implies a less ideal current source.

- The saturation current $I_{DS} = \frac{1}{2}\mu_n C_{ox}(W/L)(V_{GS} - V_T)^2$ must be computed first before multiplying by λ to obtain g_d.

Mistakes to be avoided

- Do not confuse g_d (output/drain conductance = $\partial I_{DS}/\partial V_{DS}$) with g_m (transconductance = $\partial I_{DS}/\partial V_{GS}$); they are distinct small-signal parameters.

Q4.16.6 | MCQ | 2016 | 1M | Ans: C | ● ○ ○

P: True — As channel length decreases, short-channel effects increase leakage, so OFF-state current increases.
Q: False — Smaller channel length causes output resistance r_o to decrease.
R: False — Threshold voltage does not remain constant; it decreases due to threshold voltage roll-off/DIBL.
S: True — Reducing channel length lowers channel resistance, so ON current increases.
Hence, the incorrect statements are Q and R.

$$\boxed{(C)\text{ Q and R}}$$

Q4.15.1 | NAT | 2015 | 2M | Ans: 0.018 to 0.026 | ● ○ ○

The channel-length modulation parameter λ is defined as:

$$\lambda = \frac{\Delta I_D}{I_D\, \Delta V_{DS}}$$

Substituting $\Delta I_D = 0.02\,\text{mA}$, $I_D = 1\,\text{mA}$, and $\Delta V_{DS} = 1\,\text{V}$:

$$\lambda = \frac{0.02\,\text{mA}}{1\,\text{mA} \times 1\,\text{V}} = 0.02\,V^{-1}$$

$$\boxed{0.02}$$

Key Points

- λ is the channel-length modulation parameter (units: V^{-1}) and represents the fractional change in I_D per unit change in V_{DS} in the saturation region.

Mistakes to be avoided

- Do not confuse $\Delta I_D/\Delta V_{DS}$ (which gives g_d in siemens) with $\lambda = \Delta I_D/(I_D \Delta V_{DS})$ (which gives λ in V^{-1}); the denominator must include I_D.

Q4.15.2 | MCQ | 2015 | 2M | Ans: A | ● ○ ○

Since $V_{GS} = V_{DS}$, the MOSFET operates in the **saturation region**.
In saturation, the drain current is:

$$I_D = k(V_{GS} - V_T)^2$$

With $V_{GS} = V_{DD} - V_{SS}$, substituting:

$$I_D = k(V_{DD} - V_{SS} - V_T)^2$$

As V_{SS} increases, the overdrive voltage $(V_{DD} - V_{SS} - V_T)$ decreases, so I_D decreases. Since I_D depends on the *square* of the overdrive voltage, the decrease is **non-linear**.

$$\boxed{(A)}$$

Q4.15.3 NAT 2015 2M Ans: 4.33 ● ○ ○

The ratio of maximum to minimum MOS capacitance is derived as:

$$\frac{C_{\max}}{C_{\min}} = \frac{\frac{\varepsilon_{ox}}{t_{ox}}}{\frac{\varepsilon_{ox}}{t_{ox}} \times \frac{\varepsilon_S}{X_{d\max}} \div \left(\frac{\varepsilon_{ox}}{t_{ox}} + \frac{\varepsilon_S}{X_{d\max}}\right)}$$

Simplifying:

$$\frac{C_{\max}}{C_{\min}} = \left[1 + \frac{X_{d\max}}{t_{ox}} \times \frac{\varepsilon_{ox}}{\varepsilon_S}\right]$$

Substituting $X_{d\max}/t_{ox} = 100/10 = 10$ and $\varepsilon_{ox}/\varepsilon_S = 1/3$:

$$\frac{C_{\max}}{C_{\min}} = \left[1 + \frac{100}{10} \times \frac{1}{3}\right] = \left[1 + \frac{10}{3}\right] = 1 + 3.33 = 4.33$$

$$\boxed{4.33}$$

Key Points

- $C_{\max} = C_{ox} = \varepsilon_{ox}/t_{ox}$ occurs in accumulation, where the semiconductor contributes no series capacitance.
- $C_{\min}$ occurs at strong inversion (high frequency) where the total capacitance is C_{ox} in series with $C_{dep} = \varepsilon_S/X_{d\max}$:

$$C_{\min} = \frac{C_{ox}C_{dep}}{C_{ox} + C_{dep}}$$

Q4.15.4 NAT 2015 2M Ans: 19 to 21 ● ○ ○

Under channel-length modulation, the drain current in saturation is:

$$I_D = I_{D\text{sat}}(1 + \lambda V_{DS})$$

Differentiating with respect to V_{DS}:

$$\frac{dI_D}{dV_{DS}} = \frac{1}{r_o} = \lambda I_{D\text{sat}}$$

Therefore, the output resistance is:

$$r_o = \frac{1}{\lambda I_{D\text{sat}}}$$

Substituting $\lambda I_{D\text{sat}} = 0.05 \times 10^{-3}$ A/V:

$$r_o = \frac{1}{0.05 \times 10^{-3}} = 20\, k\Omega$$

$$\boxed{20}$$

Q4.14.1 NAT 2014 2M Ans: 2.3 to 2.5 ● ○ ○

The peak electric field in the oxide region is related to the semiconductor electric field by:

$$E_{ox} = \frac{\varepsilon_S}{\varepsilon_{ox}} E_S$$

First, computing the semiconductor electric field E_S:

$$E_S = \frac{0.2 \times 2}{0.5} = 0.8\,\text{V}/\mu\text{m}$$

Substituting $\varepsilon_S/\varepsilon_{ox} = 12/4 = 3$ and $E_S = 0.8\,\text{V}/\mu\text{m}$:

$$E_{ox} = \frac{12}{4} \times 0.8 = 2.4$$

$$\boxed{2.4}$$

Key Points

- The boundary condition at the oxide–semiconductor interface requires continuity of the normal component of electric displacement: $\varepsilon_{ox} E_{ox} = \varepsilon_S E_S$.

Q4.14.2 NAT 2014 2M Ans: 0.06 to 0.08 ● ○ ○

In the linear (triode) region, the drain current is:

$$I_D = K_n \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Differentiating with respect to V_{GS}:

$$\frac{\partial I_D}{\partial V_{GS}} = K_n V_{DS}$$

Using the given values $\partial I_D / \partial V_{GS} = 10^{-3}$ and $V_{DS} = 0.1$ V:

$$\frac{10^{-3}}{0.1} = K_n \implies K_n = 10^{-2}\,\text{A/V}^2$$

In the saturation region, the drain current is:

$$I_D = \frac{K_n}{2} (V_{GS} - V_{th})^2$$

Taking the square root:

$$\sqrt{I_D} = \sqrt{\frac{K_n}{2}} (V_{GS} - V_{th})$$

Differentiating with respect to V_{GS}:

$$\frac{\partial \sqrt{I_D}}{\partial V_{GS}} = \sqrt{\frac{K_n}{2}}$$

Substituting $K_n = 10^{-2}$:

$$\frac{\partial \sqrt{I_D}}{\partial V_{GS}} = \sqrt{\frac{10^{-2}}{2}} = \sqrt{5 \times 10^{-3}} = 0.0707 \sqrt{\text{A}}/\text{V}$$

$$\boxed{0.0707}$$

Key Points

- $K_n = \mu_n C_{ox}(W/L)$ is extracted from the linear region by exploiting $\partial I_D/\partial V_{GS} = K_n V_{DS}$, which eliminates V_{th} from the calculation.
- In saturation, $\sqrt{I_D}$ is linear in $(V_{GS} - V_{th})$; the slope $\partial\sqrt{I_D}/\partial V_{GS} = \sqrt{K_n/2}$ is a constant independent of bias point — useful for graphical extraction of K_n and V_{th}.

Q4.14.3 NAT 2014 2M Ans: 499 to 501 ● ○ ○

The voltage-controlled resistance r_{DS} in the linear region is given by:

$$r_{DS} = \frac{1}{\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)}$$

Substituting $\mu_n C_{ox} = 800 \times 10^{-8}$ A/V^2, $W/L = 100$, and $V_{GS} - V_t = 2 + 0.5 = 2.5$ V:

$$r_{DS} = \frac{1}{800 \times 10^{-8} \times 100 \times (2 + 0.5)} = \frac{1}{2 \times 10^{-3}} = 500\Omega$$

$$\boxed{500}$$

Key Points

- In the deep linear (triode) region where $V_{DS} \ll V_{GS} - V_t$, the MOSFET behaves as a voltage-controlled resistor: $r_{DS} = \dfrac{1}{\mu_n C_{ox}(W/L)(V_{GS} - V_t)}$.
- r_{DS} is inversely proportional to $(V_{GS} - V_t)$; increasing V_{GS} reduces r_{DS}, making the channel more conductive.
- A larger W/L ratio directly reduces r_{DS}, which is why wide short-channel devices are used as low-resistance switches.

Q4.14.4 MCQ 2014 2M Ans: A ● ○ ○

The threshold voltage of an n-channel enhancement MOSFET is given by:

$$V_T = \phi_{ms} - \frac{Q_{ox}}{C_{ox}} - \frac{Q'_d}{C_{ox}} + 2\phi_F$$

Fixed positive charges $Q_{ox} > 0$ present in the gate oxide contribute a negative term $-Q_{ox}/C_{ox}$ to V_T.

Therefore, the presence of fixed positive oxide charges **decreases** the threshold voltage. For an n-channel enhancement MOSFET, a lower V_T means the device turns on at a smaller gate voltage — in extreme cases, fixed oxide charges can even convert an enhancement-mode device into a depletion-mode device.

Options B, C, and D are incorrect:

- Channel-length modulation is caused by V_{DS} variation, not oxide charge.
- Substrate leakage current is unrelated to fixed oxide charges.
- Accumulation capacitance $C_{ox} = \varepsilon_{ox}/t_{ox}$ depends only on oxide thickness and permittivity, not on fixed charge.

$$\boxed{(A)\ \text{A decrease in the threshold voltage}}$$

Q4.12.1 MCQ 2012 1M Ans: A ● ○ ○

The source-body junction capacitance is given by:

$$C_j = \frac{\varepsilon A}{d} = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

where $\varepsilon_r = 11.7$, the junction area is:

$$A = 1\,\mu\text{m} \times 0.2\,\mu\text{m} = 0.2 \times 10^{-12}\,\text{m}^2$$

and d is the depletion width of the p-n junction:

$$d = 10\,\text{nm} = 10^{-8}\,\text{m}$$

Substituting $\varepsilon_0 = 8.9 \times 10^{-12}$ F/m:

$$C_j = \frac{8.9 \times 10^{-12} \times 11.7 \times 0.2 \times 10^{-12}}{10^{-8}} = 2.082 \times 10^{-15}\,\text{F} \approx 2\,\text{fF}$$

$$\boxed{(A)\,2\,\text{fF}}$$

Q4.12.2 MCQ 2012 1M Ans: A ● ○ ○

The gate-source capacitance C_g is modelled as a parallel-plate capacitor:

$$C_g = \frac{\varepsilon_1 A_1}{d_1} = \frac{\varepsilon_0 \varepsilon_{r1} A_1}{d_1}$$

Computing the overlap area A_1 with gate width $1\,\mu$m and overlap $\delta = 20$ nm:

$$A_1 = 1\,\mu\text{m} \times \delta = 1 \times 10^{-6} \times 20 \times 10^{-9} = 2 \times 10^{-14}\,\text{m}^2$$

The oxide thickness:

$$d_1 = 1\,\text{nm} = 10^{-9}\,\text{m}$$

Substituting $\varepsilon_0 = 8.9 \times 10^{-12}$ F/m:

$$C_g = \frac{8.9 \times 10^{-12} \times 3.9 \times 2 \times 10^{-14}}{10^{-9}} = 0.69 \times 10^{-15}\,\text{F} \approx 0.7\,\text{fF}$$

$$\boxed{(A)\,0.7\ \text{fF}}$$

Key Points

- The gate-source overlap capacitance arises from the physical overlap of the gate electrode over the source diffusion region; it is an unwanted parasitic that limits high-frequency performance.

Mistakes to be avoided

- Do not use the Silicon permittivity ($\varepsilon_r = 11.7$) here; the gate-source capacitance is across the oxide (SiO_2), so $\varepsilon_r = 3.9$.
- Do not confuse the gate-channel capacitance $C_{ox} \times W \times L$ with the gate-source overlap capacitance $C_{ox} \times W \times \delta$; they involve different area definitions.

Q4.12.3 MCQ 2012 1M Ans: D ● ○ ○

The given parameters are:

$$n_i = 10^{10}\,\text{cm}^{-3}, \quad A = 1 \times 10^{-12}\,\text{m}^2, \quad d = 10^{-6}\,\text{m}$$

Computing the volume of the silicon block:

$$V = A \times d = 10^{-12} \times 10^{-6} = 10^{-18}\,\text{m}^3 = 10^{-12}\,\text{cm}^3$$

With donor doping $N_D = n = 10^{19}\,\text{cm}^{-3}$, the minority hole concentration is found using the mass action law:

$$p = \frac{n_i^2}{N_D} = \frac{(10^{10})^2}{10^{19}} = \frac{10^{20}}{10^{19}} = 10\,\text{cm}^{-3}$$

The total number of holes in volume V is:

$$H = p \times V = 10\,\text{cm}^{-3} \times 10^{-12}\,\text{cm}^3 = 10^{-11}$$

Since the number of holes cannot be a decimal (fractional) number, there are effectively zero holes in this volume:

$$\boxed{(D)\,0}$$

Key Points

- Carrier concentration is a statistical average over a large volume; in extremely small volumes, the expected number of minority carriers can be less than one, meaning there is a high probability of finding *zero* carriers at any instant.

Mistakes to be avoided

- Do not confuse N_D (donor concentration) with n_i (intrinsic concentration); for heavy n-type doping, $n \approx N_D \gg n_i$.

Q4.10.1 MCQ 2010 1M Ans: A ● ○ ○

The bulk electron mobility in silicon at room temperature is about $\mu_n \approx 1350; \text{cm}^2/\text{V}{\cdot}\text{s}$.

But in a MOSFET inversion layer, electrons move very close to the Si–SiO(2) interface, where surface scattering and vertical electric field reduce mobility significantly. So inversion-layer mobility is much lower, typically around 100–300$\text{cm}^2/\text{V}{\cdot}\text{s}$

Hence, 150 $\text{cm}^2/\text{V}{\cdot}\text{s}$ is the realistic value.

$$\boxed{(A)\;150\,\text{cm}^2/\text{V} - \text{s}}$$

Q4.09.1 MCQ 2009 1M Ans: D ● ○ ○

S2 is true because channel potential gradually rises from source (0) to drain (V_{DS}). Due to this, the effective gate-to-channel voltage $V_{GC} = V_G - V(x)$ decreases toward the drain, so inversion charge decreases from source to drain (S1 true).

$$\boxed{(D)\text{ Both S1 and S2 are true, and S2 is a reason for S1}}$$

Q4.08.1 MCQ 2008 1M Ans: B

The transconductance is defined as:

$$g_m = \left.\frac{\partial i_d}{\partial V_{GS}}\right|_{V_{DS}}$$

In the saturation region, drain current is given as $I_D = K(V_{GS} - V_T)^2$. Differentiating with respect to V_{GS} to obtain transconductance:

$$g_m = \frac{\partial K(V_{GS} - V_T)^2}{\partial V_{GS}} = 2K(V_{GS} - V_T)$$

$$\boxed{(B)\, 2K(V_{GS} - V_T)}$$

Key Points

- In saturation, $g_m = 2K(V_{GS} - V_T) = \sqrt{4KI_D}$; these three equivalent forms are all useful depending on what is known.
- Unlike the linear region (where $g_m \propto V_{DS}$ as well), in saturation g_m depends only on the overdrive voltage and device parameter K.

Q4.08.2 MCQ 2008 1M Ans: A

In the linear (triode) region, the drain current is:

$$I_D = K_n\left[(V_{gs} - V_t)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

The transconductance g_m is defined as $\partial I_D/\partial V_{gs}$ at constant V_{DS}:

$$g_m = \frac{\partial I_D}{\partial V_{gs}} = K_n V_{DS}$$

Since K_n and V_{DS} are constants at a fixed operating point, for the ideal long-channel MOSFET model, g_m is independent of V_{GS} at a fixed V_{DS}.

However, the question specifically asks for the **measured transconductance**. As discussed in standard device texts such as Razavi's Fundamentals of Microelectronics, practical MOSFETs exhibit mobility degradation. At low gate voltages, g_m increases as inversion charge builds up. As V_{GS} increases further, the increasing vertical electric field reduces carrier mobility, causing g_m to saturate and eventually decrease. Therefore, the expected measured g_m-V_{GS} characteristic:

- starts from a small value,
- increases with V_{GS},
- reaches a maximum,
- then gradually decreases due to mobility degradation.

This behavior corresponds to Option A.

$$\boxed{(A)}$$

Key Points

- The distinction between linear and saturation region expressions for g_m is important; the two regions yield the same linear V_{GS} dependence but with different proportionality constants.

Q4.07.1 MCQ 2007 1M Ans: C ● ○ ○

S1 is false because the C–V curve shows high capacitance at negative (V) and low capacitance at positive (V), which corresponds to a p-type substrate, not n-type.

S2 is true because positive oxide charge reduces the required gate voltage for a given surface condition, causing the entire C–V curve to shift left.

$$\boxed{(C)\ \text{S1 is false and S2 is true}}$$

Q4.07.2 MCQ 2007 1M Ans: B ● ○ ○

The minimum capacitance in a MOS capacitor (series combination of $C_{\max}$ and C_d) is:

$$C_{\min} = \frac{C_{\max} C_d}{C_{\max} + C_d}$$

Rearranging to solve for the depletion capacitance C_d:

$$C_d = \frac{C_{\max} C_{\min}}{C_{\max} - C_{\min}}$$

Substituting $C_{\max} = 7\,\text{pF}$ and $C_{\min} = 1\,\text{pF}$:

$$C_d = \frac{7 \times 1}{7 - 1} = \frac{7}{6} = 1.166\,\text{pF}$$

Using $C_d = \varepsilon_{si} A / w$ to find the maximum depletion width w, with area $A = 10^{-4} \times 10^{-4} = 10^{-8}\,\text{m}^2$:

$$w = \frac{\varepsilon_{si} A}{C_d} = \frac{10^{-10} \times 10^{-8}}{1.166 \times 10^{-12}} = \frac{10^{-18}}{1.166 \times 10^{-12}} = 0.857\,\mu\text{m}$$

$$\boxed{(B)\ 0.857\,\mu\text{m}}$$

Key Points

- $C_{\max} = C_{ox}$ occurs in accumulation (no depletion capacitance in series); $C_{\min}$ occurs at strong inversion where C_{ox} and C_d are in series.
- The maximum depletion width $w = \varepsilon_{si} A / C_d$ follows from the parallel-plate model for the depletion region.

Mistakes to be avoided

- Do not use $C_d = C_{\max} - C_{\min}$ directly; the series combination formula must be rearranged properly to extract C_d.
- Ensure ε_{si} is converted consistently: $10^{-12}\,\text{F/cm} = 10^{-10}\,\text{F/m}$; mixing cm and m leads to a 10^4 error in w.
- Do not confuse $C_{\min}$ with C_d; $C_{\min}$ is the *total* measured capacitance at inversion, while C_d is just the depletion component.

Q4.07.3 MCQ 2007 1M Ans: A ● ○ ○

The oxide thickness is found using the parallel-plate capacitance formula:

$$C = \frac{\varepsilon A}{d}$$

Rearranging for d:

$$d = \frac{\varepsilon A}{C} = \frac{3.5 \times 10^{-13} \times 1 \times 10^{-4} \times 10^{-4}}{7 \times 10^{-12} \times 10^{-2}}$$

$$d = \frac{3.5 \times 10^{-21}}{7 \times 10^{-14}} = 5 \times 10^{-8}\,\text{m} = 50\,\text{nm}$$

$$\boxed{(A)\ 50\,\text{nm}}$$

Q4.05.1 MCQ | 2005 | 1M | Ans: A | ● ○ ○

In accumulation mode for NMOS having p-substrateV_G is negative. When negative V_G is applied to the gate electrode, the holes in the p-type substrate are attracted to the semiconductor oxide interface. This condition is called carrier accumulation on the surface.

$\boxed{(A)\text{ holes}}$

Q4.04.1 MCQ | 2004 | 1M | Ans: D | ● ● ●

The threshold voltage of an N-MOS capacitor is given by:

$$V_T = V_{FB} + 2\phi_F + \frac{Q_B}{C_{ox}}$$

where $C_{ox} = \varepsilon_{ox}/t_{ox}$ and V_{FB} is the flatband voltage.

Statement $S1$: As the oxide thickness increases, C_{ox} decreases, causing the term $\frac{Q_B}{C_{ox}}$ to increase. Hence, the magnitude of the threshold voltage increases.

Statement $S2$: Similarly, increasing the substrate doping concentration increases both the Fermi potential (ϕ_F) and the depletion charge (Q_B), which again increases the magnitude of the threshold voltage.

For a PMOS capacitor, the threshold voltage is negative, but the same physical arguments apply. Increasing the oxide thickness or the substrate doping concentration increases the magnitude of the threshold voltage (that is, it becomes more negative).

$\boxed{(D)\text{ Both S1 and S2 are false}}$

Q4.03.1 MCQ | 2003 | 1M | Ans: D | ● ○ ○

The threshold voltage including the body effect is given by:

$$V_T = V_{T_0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$$

where γ is the substrate bias coefficient and V_{SB} is the source-to-body voltage.

Given $V_{SB} > 0$, thus V_T **increases**.

$\boxed{(D)\text{ increases}}$

Q4.01.1 MCQ | 2001 | 1M | Ans: B | ● ○ ○

At the edge of saturation i.e. when drain to source voltage reaches $V_{DS_{sat}}$ the inversion layer charge at the drain end becomes zero (ideally). The channel is said to be pinched off at the drain end. If the drain to source voltage V_{DS} is increased even further beyond the saturation edge so that $V_{DS} > V_{DS_{sat}}$, an even larger portion of the channel becomes pinched off and effective channel length is reduced.

$\boxed{(B)\text{ drain voltage}}$

Q4.01.2 MCQ | 2001 | 1M | Ans: B | ● ○ ○

MOSFET can be used as a **Voltage-Controlled Capacitor** since by altering the voltage at the Gate terinal, the charge accumulated at Oxide-Semiconductor interface can be altered, which is a property of capacitors.

$\boxed{(B)\text{ voltage controlled capacitor}}$

Q4.94.1 NAT 1994 1M Ans: False ● ○ ○

For depletion mode n-channel MOSFET, if the gate terminal is made more positive then the channel becomes more and more n-type hence the drain current will increase.

$$\boxed{\text{False}}$$

Q4.94.2 MCQ 1994 1M Ans: B ● ○ ○

The threshold voltage including the body effect is given by:

$$V_T = V_{T_0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$$

where γ is the substrate bias coefficient and V_{SB} is the source-to-body voltage, and is defined as:

$$\gamma = \frac{\sqrt{2qN_A\epsilon_S}}{C_{ox}} = \frac{t_{ox}}{\epsilon_{ox}}\sqrt{2qN_A\epsilon_S}$$

N_A is the substrate doping concentration. Increase in N_A causes decrease in channel carrier concentration and increase in V_T.

$$\boxed{(B)\text{ reducing the channel carrier concentration.}}$$

Q4.94.3 NAT 1994 1M Ans: switching ● ○ ○

The transit of a current carriers through the channel of an FET decides its switching characteristics.

$$\boxed{\text{switching}}$$

Q4.90.1 MSQ 1990 2M Ans: B , C ● ○ ○

The collector current I_C is given as:

$$I_C = \beta I_B + (\beta + 1)I_{CO}$$

As temperature increases, I_{CO} increases, so the current (I_C) increases in BJT with rise in temperature.

As temperature increases, mobility μ decreases, thus MOSFET current I_D decreases.

$$\boxed{(B)\text{ Increase in BJT current } I_C}$$

$$\boxed{(C)\text{ Decrease in MOSFET current } I_{DS}}$$

Key Points

- MOSFET current I_D is related to mobility μ as:

$$I_D \propto \frac{\mu C_{ox}}{2}$$

Q4.89.1

MCQ | 1989 | 2M | Ans: D | ● ○ ○

In a MOSFET, the polarity of the inversion layer is the same as that of the majority carriers in the source.

$\boxed{(D)\text{ majority carriers in the source}}$

Key Points

- Taking an example of an nmos: the body/substrate is p-type, and the source/drain regions are n-type. The channel is also n-type, thus having the same polarity as source/drain.

Q4.88.1

MCQ | 1988 | 2M | Ans: B | ● ○ ○

n-channel MOSFET is faster than the p-channel MOSFET because mobility of electrons μ_n is higher than the mobility of holes μ_p. In n-channel MOSFET, the charge carriers are electrons, and in p-channel MOSFET, the charge carriers are holes.

$\boxed{(B)\text{ it is faster.}}$

DEBUG INFO

Chapter IV

MOSCap and MOSFETs

Total Questions : 43

MCQ : 29

MSQ : 1

NAT : 13

SOLUTIONS

V

Optoelectronic Devices

Topics

1. Solar Cell
2. Photodiode
3. LEDs

Q5.26.1 MSQ 2026 2M Ans: A, B, D ● ● ○

A wavelength is absorbed by a semiconductor if the photon energy exceeds the bandgap:

$$\lambda \le \frac{1.24}{E_G}\ \mu\text{m}$$

Substituting $E_G = 1.3\,\text{eV}$:

$$\lambda \le \frac{1.24}{1.3}\ \mu\text{m} = 0.953\ \mu\text{m} = 953\ \text{nm}$$

Since the device is an LED, its emission spectrum is centered around the bandgap wavelength (953 nm) with an approximate bandwidth of ± 20 nm. Therefore, the LED emits light roughly in the range 933 nm to 973 nm .

Among the given options, only 950 ± 20 nm lies within the emission spectrum. Hence, the LED will **not emit** wavelengths corresponding to 1410 $\pm$20 nm, 1090 $\pm$20 nm, and 510 $\pm$20 nm.

$$\boxed{(A)\ 1410 \pm 20\ \text{nm}}\boxed{(B)\ 1090 \pm 20\ \text{nm}}\boxed{(D)\ 510 \pm 20\ \text{nm}}$$

Key Points

- **Absorption** : A wavelength is absorbed by a semiconductor if the photon energy exceeds the bandgap: $\lambda \le \frac{1.24}{E_G}\ \mu\text{m}$. Thus, photons with wavelengths much shorter than the bandgap wavelength (for example, 510 nm in this question) can still be absorbed.
- **Emission** : An LED emits photons due to electron-hole recombination, and the emitted photon energy is approximately equal to the bandgap energy. Therefore, the emission spectrum is centered around the bandgap wavelength with a finite spectral width.

Q5.24.1 NAT 2024 2M Ans: 4 to 4.26 ● ○ ○

The reverse saturation current I_0 is extracted using:

$$I_0 = \frac{I_L}{\left(1 + \frac{V_m}{V_T}\right) e^{V_m/V_T} - 1}$$

Substituting $I_L = 1\,\text{mA}$, $V_m = 0.3\,\text{V}$, and $V_T = 0.03\,\text{V}$:

$$I_0 = \frac{1\ \text{mA}}{\left(1 + \frac{0.3}{0.03}\right) e^{0.3/0.03} - 1} = 4.127 \times 10^{-9}\ \text{A} \approx 4.13\ \text{nA}$$

$$\boxed{4.13}$$

Key Points

- At the maximum power point, the condition $dP/dV = 0$ gives:

$$\left(1 + \frac{V_m}{V_T}\right) e^{V_m/V_T} = \frac{I_L}{I_0} + 1$$

rearranging directly yields the expression for I_0 used above.
- $V_m < V_{OC}$ always; confusing the two voltages is a critical error in solar cell problems.
- $V_T = kT/q$; here $V_T = 0.03\,\text{V}$ implies a slightly elevated temperature ($T \approx 348$ K) or a given ideality factor adjustment.

Mistakes to be avoided

- The voltage given corresponds to he **maximum power point** voltage $V_m = 0.3$ V, not the open-circuit voltage V_{OC}.

Q5.20.1 MCQ 2020 2M Ans: A ● ○ ○

Efficiency of solar cell is given by:

$$\eta = \frac{(FF)V_{OC}I_{SC}}{P_{in}}$$

Substituting given values:

$$0.15 = \frac{0.8 \times 0.7 \times I_{SC}}{100mW}$$

$$I_{SC} = \frac{15}{0.56}mA$$

Optical generation Rate is given by:

$$G_L = \frac{I_{SC}}{q \times \text{Area} \times \text{Thickness}}$$

$$G_L = \frac{15 \times 10^{-3}}{0.56 \times 1.6 \times 10^{-19} \times 1 \times 200 \times 10^{-4}}$$

$$G_L = 0.837 \times 10^{19} \approx 0.84 \times 10^{19}$$

$$\boxed{(A)\, 0.84 \times 10^{19}}$$

Key Points

- Solar cell efficiency is $\eta = (FF \times V_{OC} \times I_{SC})/P_{in}$; rearranging gives $I_{SC} = \eta P_{in}/(FF \times V_{OC})$.
- The optical generation rate $G_L = I_{SC}/(q \times \text{Area} \times \text{Thickness})$; it represents the number of electron-hole pairs generated per unit volume per second.
- Fill Factor $FF = P_{\max}/(V_{OC} \times I_{SC})$; a higher FF (closer to 1) indicates a more ideal solar cell.

Mistakes to be avoided

- Do not confuse I_{SC} (short-circuit current) with I_0 (reverse saturation current); they are fundamentally different – I_{SC} is the photogenerated current while I_0 is the dark diode leakage current.

Q5.19.1 NAT 2019 2M Ans: 0.23 to 0.232 ● ○ ○

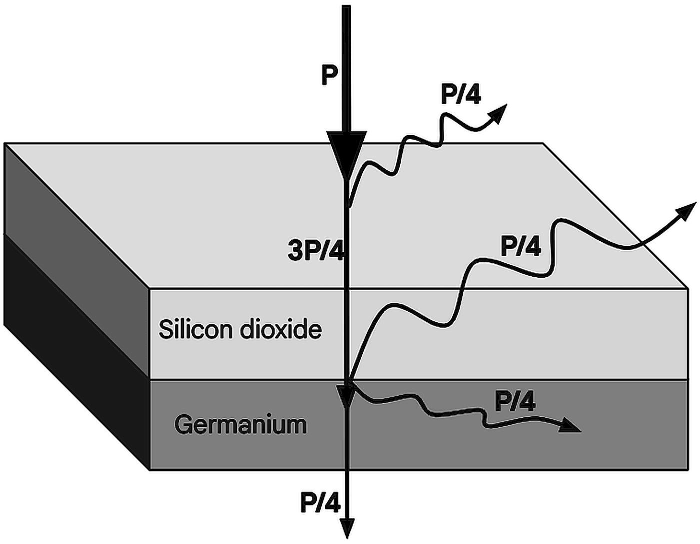

Fig. Solution Q5.19.1

Here, P = $20mW$. It is observed that effective power incident on Germanium is $\frac{3P}{4} - \frac{P}{4} = \frac{P}{2}$, and power $\frac{P}{4}$ is absorbed by Germanium. Thus we have

$$P_{absorbed} = P_{incident}\left(1 - e^{-\alpha T}\right)$$
$$\frac{P}{4} = \frac{P}{2}\left(1 - e^{-\alpha T}\right)$$
$$e^{-\alpha T} = 0.5$$
$$T = 0.231\mu m$$

$$\boxed{0.231}$$

Mistakes to be avoided

- Make sure to account for the reflected power while calculating the incident power.

Q5.19.2 MCQ 2019 2M Ans: A ● ○ ○

The quantum efficiency η of a photodetector relates the output current to input optical power by:

$$\eta = \frac{I_{out}}{q} \times \frac{hf}{P_{in}}$$

The responsivity R is defined as:

$$R = \frac{I_{out}}{P_{in}}$$

Expressing R in terms of η:

$$R = \eta \times \frac{q}{hf} = \eta \times \frac{q\lambda}{hc}$$

If λ is given in μm, substituting λ in μm:

$$R = \eta\lambda \times \frac{q \times 10^{-6}}{hc}$$

Using the standard constant $hc/(q \times 10^{-6}) \approx 1.24\,\mathrm{W/(A \cdot \mu m)}$:

$$\boxed{(A)\ R = \frac{\eta\lambda}{1.24}}$$

Key Points

- Responsivity $R = I_{out}/P_{in}$ in A/W measures how efficiently a photodetector converts optical power to electrical current.
- The relation $R = \eta\lambda/1.24$ (with λ in μm) is a standard and widely used result; the constant 1.24 arises from $hc/q \approx 1.24\,\text{eV}\cdot\mu\text{m}$.

Mistakes to be avoided

- Do not confuse quantum efficiency η (dimensionless, $0 \le \eta \le 1$) with responsivity R (in A/W); η is the fraction of photons converted to electron-hole pairs, while R accounts for the energy per photon.
- Ensure λ is in μm when using $R = \eta\lambda/1.24$; using λ in nm gives a result 10^3 times too small.

Q5.18.1 | NAT | 2018 | 2M | Ans: 0.59 to 0.63 | ● ○ ○

The open circuit voltage V_{OC} for a solar cell is given by:

$$V_{OC} = V_T ln\left(\frac{I_{SC}}{I_0}\right)$$

The difference between the open circuit voltages at different input intensity values is calculated as:

$$V_{OC1} - V_{OC2} = V_T ln\left(\frac{I_{SC1}}{I_{SC2}}\right)$$

Substituting values,

$$V_{OC2} = 0.65 - V_T ln(5) = 0.608V$$

$$\boxed{0.608}$$

Q5.18.2 | MCQ | 2018 | 2M | Ans: | ● ○ ○

It is known that wavelengths of red, blue and green lights are in the following order:

$$\lambda_R > \lambda_G > \lambda_B$$

It is also known that energy is related to wavelength as $E_g \propto \frac{1}{\lambda}$. Therefore

$$E_{gR} < E_{gG} < E_{gB}$$

Materials with higher energy gap will have higher built-in voltages, when doping concentrations are same. Thus, $V_R < V_G < V_B$.

$$\boxed{(B)\; V_R < V_G < V_B}$$

Key Points

- Energy of a photon of wavelength λ is given by $E_g = \frac{hc}{\lambda}$.

Q5.17.1 | NAT | 2017 | 2M | Ans: 0.51 to 0.62 | ● ○ ○

The open circuit voltage V_{OC} for a solar cell is given by:

$$V_{OC} = V_T ln\left(\frac{I_{SC}}{I_0}\right)$$

The difference between the open circuit voltages at different input intensity values is calculated as:

$$V_{OC1} - V_{OC2} = V_T ln\left(\frac{I_{SC1}}{I_{SC2}}\right)$$

Now, we know that current I and current density J are related as $J = \dfrac{I}{Area}$. Also, $J \propto$ Light intensity. Thus the equation can be written as:

$$V_{OC1} - V_{OC2} = V_T ln\left(\frac{J_{L1}}{J_{L2}}\right) = V_T ln\left(\frac{1}{20}\right)$$

$$V_{OC2} = 0.451 + V_T ln(20) = 0.526V$$

$$\boxed{0.526}$$

Q5.16.1 | NAT | 2016 | 1M | Ans: 20.5 to 21.5 | ● ○ ○

Fill Factor FF of a solar cell is given by:

$$FF = \frac{\text{Maximum obtained power}}{V_{OC} I_{SC}} = \frac{P_M}{V_{OC} I_{SC}}$$

$$0.7 = \frac{P_M}{0.5 \times 180 \times 10^{-3}}$$

$$\therefore \text{ Maximum obtained power } P_M = 0.063W$$

$$\text{Input power } P_{in} = 100mW/cm^2 \times 3cm^2 = 0.3W$$

$$\therefore \text{ Efficiency } \eta = \frac{P_M}{P_{in}} = \frac{0.063}{0.3} = 21\%$$

$$\boxed{21}$$

Key Points

- Fill Factor (FF) quantifies the squareness of a solar cell's current-voltage (I-V) characteristic curve, representing the ratio of the maximum obtainable power to the theoretical maximum power.

Mistakes to be avoided

- V_{OC} and I_{SC} do not correspond to maximum obtainable power. These are simply the maximum obtainable voltage and current respectively for the solar cell, and are determined by open circuiting or short circuiting the terminals of the solar cell.
- Power at points V_{OC} and I_{SC} is zero, since V_{OC} corresponds to zero current, and I_{SC} corresponds to zero voltage.

Q5.14.1 | NAT | 2014 | 1M | Ans: 8 to 8 | ● ○ ○

Responsivity R is related to photocurrent I_{ph} and incident power P_{in} as

$$R = \frac{I_{ph}}{P_{in}}$$

Thus photocurrent can be obtained as:

$$I_{ph} = R \times P_{in} = 0.8A/W \times 10\mu W = 8\mu A$$

$$\boxed{8}$$

Q5.07.1 | MCQ | 2007 | 1M | Ans: B | ● ○ ○

Zener diode and avalanche photodiode normally operate in reverse bias to utilize breakdown effects, while a solar cell works like a forward-biased source delivering power. Hence: P–2, Q–1, R–2.

$$\boxed{(B)\text{ P–2, Q–1, R–2}}$$

Q5.03.1 | MCQ | 2003 | 1M | Ans: A | ● ○ ○

Energy of a photon of wavelength λ is given by $E_g = \frac{hc}{\lambda}$

$$E_g = \frac{1242}{\lambda(nm)} = \frac{1242}{549nm} = 2.26eV$$

$$\boxed{(B)\, 2.26\, eV}$$

DEBUG INFO

Chapter V

Optoelectronic Devices

Total Questions : 12

MCQ : 5

MSQ : 1

NAT : 6